

EPSON OPOS ADK MANUAL

User's Manual
(Installer, SetupPOS, TMUSB)

Version 2.80 Dec. 2014

Notes

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Section 1. Introduction

This manual contains an outline of the EPSON OPOS ADK program, and provides installation and setup instructions for the OPOS Setup program.

EPSON OPOS ADK contains software with API functions that can be used easily through ActiveX controls supported by Visual Basic on the using OS platforms. Application developers without any concern for attached peripherals type or protocol, can develop real-time, multitasking, GUI-based, POS applications with lots of functionality.

Section 2, Development Outline, describes the requirements and uses of EPSON OPOS ADK, and lists what is included in the package.

Section 3, Installation, contains directions on how to install the program.

Section 4, Uninstallation, contains directions on how to uninstall the program.

Section 5, SetupPOS Utility, shows you how to use the program to record information about OPOS devices in the system registry.

Section 6, Epson TM USB Driver, provides details on how to install the USB driver and the device names.

Section 7, Warnings, describes the matters that demand attention.

The "UPOS 1.14", created by the OPOS committee, describes API functions in detail. The Application Development Guide describes application development. Please refer to these manuals for information on these subjects.

Section 2. Development Outline

This section describes development outline for the EPSON OPOS ADK.

2.1. Features

EPSON OPOS ADK offers standardized API functions that interface with peripherals used in POS systems through ActiveX controls that can be run in Visual Basic, operating on the using OS platforms. Through these standardized API functions, you can develop powerful, open POS systems easily.

2.2. Operating Environment

The EPSON OPOS ADK operates under the following environment.

- Computer Hardware
 - IBM PC/AT or compatible
 - * Please follow the specifications recommended by your OS for system requirements (CPU, RAM, etc.).
- Operating Systems
 - Windows 8.1
 - Windows 8
 - Windows 7
 - Windows Vista
 - Windows XP
 - Windows 2000
 - Windows Server 2008
 - Windows Server 2008 R2
 - Windows Server 2012
 - Windows Server 2012 R2
 - Windows Embedded Standard
 - Windows Embedded Standard 7
 - Windows Embedded for Point Of Service
 - Windows Embedded POSReady 2009

Windows Embedded POSReady 7

- * For the details of each operation system version, refer to the release note included in the Installation disks.
- * Internet Explorer: Version 5.0 or higher (Recommend)
- * There are unsupported devices in Windows Vista or newer.

For details of contents, please refer to "Accessible Devices" of next page.

- Accessible Serial Ports

COM1, COM2, COM3, COM4 (extended port functions allow use of COM5 through COM10)

- Accessible Parallel Ports

LPT1, LPT2, LPT3 (operation is unverified in LPT3)

- Accessible USB ports

USB equipment must be connected to the IBM PC/AT compatible machines that are equipped with USB connectors or use USB expansion cards. If the USB device cannot be identified when an IBM PC/AT compatible machine equipped with a USB connector is in use (especially for machines put on the market prior to the English version of Windows 98 being available), contact the manufacturer.

- Accessible Network

Ethernet and IEEE802.11 support applies to TCP/IP compliant networks. The setup of a compliant network should be handled by a qualified network administrator.

- Accessible Bluetooth

When using a Bluetooth connection, please use the standard Windows Bluetooth stack.

The printer will not function normally if a different stack is used.

- Supported Languages

Microsoft Visual Basic Version 5.0 or higher

Microsoft Visual C++ Version 5.0 or higher

- Accessible Devices

For details of support devices, please read the release notes included in the Installer disks.

2.3. Necessary System Components

In order to develop an application using OPOS API, the following hardware and software are necessary.

- Computer Body Parts

IBM PC/AT Interchange Machine

* Please follow the specifications recommended by your OS for system requirements (CPU, RAM, etc.).

- Operating System

Windows 8.1

Windows 8

Windows 7

Windows Vista

Windows XP

Windows 2000

Windows Server 2008

Windows Server 2008 R2

Windows Server 2012

Windows Server 2012 R2

Windows Embedded Standard

Windows Embedded Standard 7

Windows Embedded for Point Of Service

Windows Embedded POSReady 2009

Windows Embedded POSReady 7

- Hard Disk, Memory

At least 10M bytes hard disk capacity is needed except for recommended blank capacity of using OS.

- Package

These disks include software, manuals and sample programs.

Please confirm that your package contains all the following items:

Sample Programs:

EPSON OPOS ADK Install Disk contains practical sample program guides using common API for each device. The method for use is described on the sample program guide. Please copy onto the hard disk as necessary.

Manuals:

On-line manuals are included on EPSON OPOS ADK Disk.

USER'S GUIDE

Information on EPSON OPOS ADK installation and setup.

APPLICATION DEVELOPMENT GUIDE

Programming examples and explanations of factors governing of peripherals. The Application Development Guide is divided into common subjects and device subjects. The common subjects portion contains general information about OPOS and device classes. The device subjects' portion contains information about specific devices. Please refer to common subjects and device subjects for the device being used.

SAMPLE PROGRAM GUIDE

Contains outlines, detailed explanations, and notes on the sample program included on the EPSON OPOS ADK.

Section 3. Installation

Running the EPSON OPOS ADK Installer program can perform all the processing required for installation of the EPSON OPOS ADK. To ensure optimal use of the OPOS, various functions are offered during installation. Please read this manual for details.

When installing or reinstalling, first ensure that all running application programs are closed; both those use the OPOS files as well as any other applications.

After installing of EPSON OPOS ADK, it is necessary to confirm the setup, such as port and device settings. For details, please refer to "Section 5. SetupPOS Utility" of this manual.

The EPSON OPOS ADK install program is explained in the following.

3.1. Features of the Installer

This Installer is created using InstallShield 12.

The Installer performs the following operations:

- Specifies the destination directory for installation
- Specifies the destination folder for installation
- Specifies the initial settings for OPOS
- Specifies REG files
- Installation type
 - Full installation ("Common CO" or "EPSON CO")
 - Custom installation
 - (1) Selection of device class units, Selection of CO
 - (2) Selection of the old and new CO (Selection of the CO install)
- Installation of the manual and the Sample Program
- Installation of the driver of EPSON TM USB
- Installation of the driver of parallel
- Uninstallation type
 - Full uninstallation (all)
 - Custom uninstallation
 - (3) Selection of the SO
- Add-in (Service Pack, File update)
- Silent mode installation/uninstallation

3.2. Explanation of Installer Disks

There is only a CD-ROM install disk. The installation from the FD is not supported. Install according to the following procedure.

3.3. Installation

The procedure of install is as follows:

The installation can be done only by authorized users of the PC installation. Please request the authorized users for help.

Installation of the EPSON TM USB Driver (EPSON USB controller for TM printer series) is required when the USB POSPrinter, LineDisplay, MICR, CheckScanner, CashDrawer and ElectronicJournal are to be used. Without the installation in advance, the devices used with USB port cannot be registered normally.

For details of EPSON TM USB Driver, please refer to the "Section 6 EPSON TM USB Drive" of this manual.

3.3.1. First Installation

Without the installation of EPSON OPOS ADK in advance, install according to the following procedure.

<Installation Procedure>

- (1) Run the SETUP.EXE program on disk 1.

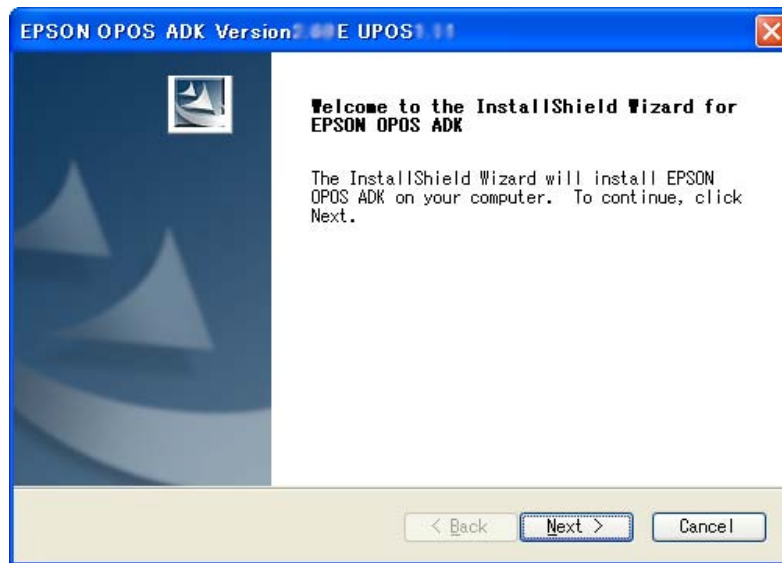
If Setup is executed while other program is running, its operation is not guaranteed. It is suggested to close all programs before executing Setup. Startup of the Setup program may take some time.

The "Choose Setup Language" dialog is displayed.



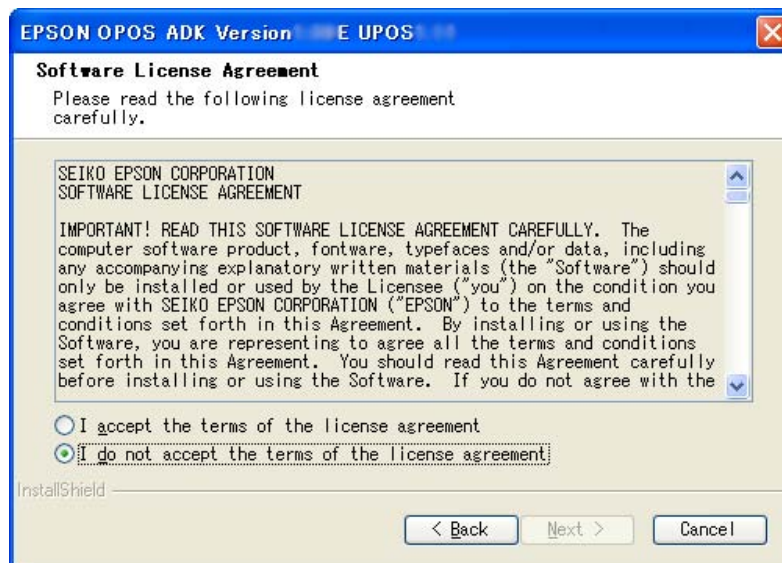
Select the language for the installation. Then, select the [OK] button.

- (2) The “Welcome” dialog is displayed.



Read the contents and then select the [Next] button if agreed with the contents.

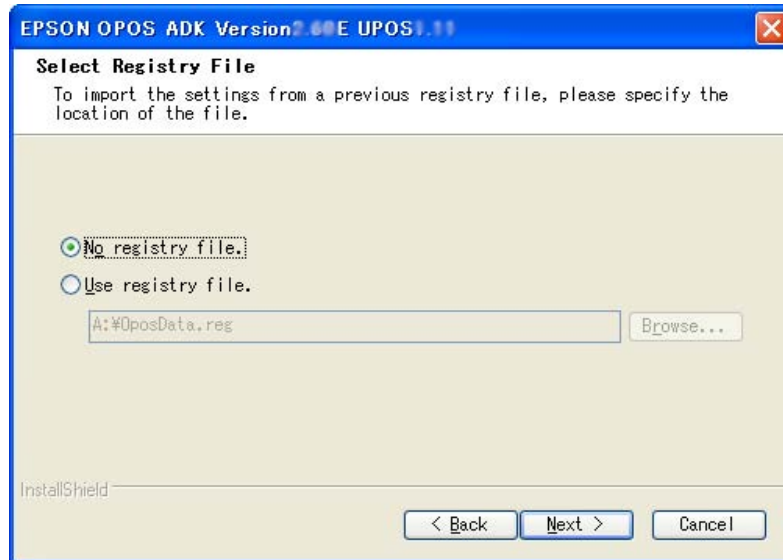
- (3) The “Software License Agreement” dialog box is displayed.



To install the EPSON OPOS ADK, the user must agree to the contents of the displayed agreement. Read the contents. Select “I accept the terms of the license agreement” if agreed with the Software License Agreement. After the selection, the [Next] button becomes effective. Select the [Next] button. If the Software License Agreement is not acceptable, select “I do not accept the terms of the license agreement”. Then, select the [Cancel]

button. Exits installation. Select the [Back] button for returning to the previous dialog box.

- (4) The “Select Registry File” dialog box is displayed.

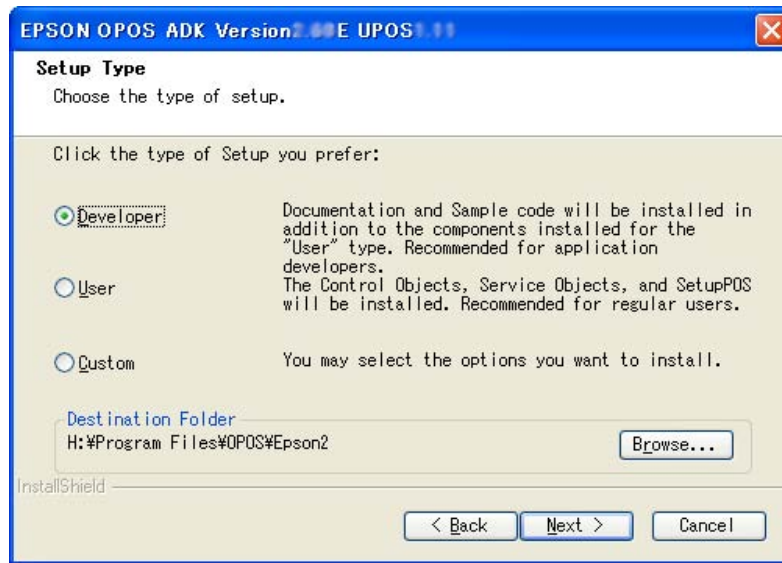


Select “No Registry File” not to use the REG file. Then, select the [Next] button. Select “Use Registry File” to use the REG file. Enter the full path REG file name after the text box is enabled. Then, select the [Next] button. Press the [Browse...] button to specify the folder name that includes the REG file. The REG file is a file for restoring information related to EPSON OPOS ADK. Use the SetupPOS utility to create this file. If an improper Registry File is selected, an error message will be displayed and it will not be possible to move to the next dialog box. Follow the directions of the error message and confirm that the Registry File is correct. When the Registry File is set correctly, subsequent information can be referred from the Registry File.

Select the [Back] button for return to the previous dialog box.

Select the [Cancel] button for quitting.

- (5) The “Setup Type” dialog box is displayed.



Select a setup type to be installed. Selectable types are “Developer”, “User” and “Custom”.

The details of type are as follows:

Developer

Components:

Full installation of programs

Manual

SampleProgram

CO:

<First installation> CCO

User

Components:

Full installation of programs

CO:

<First installation> CCO

Custom

Select components to be installed.

- (6) After selecting a type, press the [Next] button.

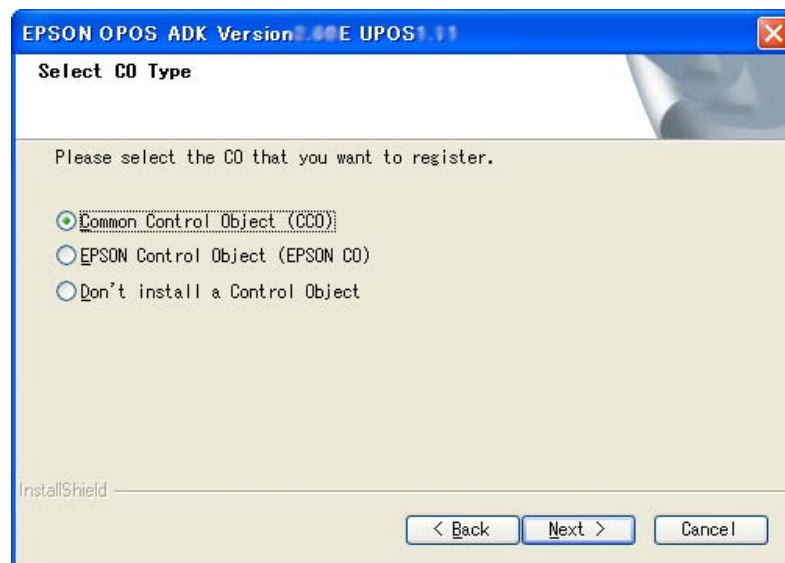
When the "Developer" or the "Users" have been selected, the "Start Copying Files" dialog box is displayed. Confirm the contents and select the [Next] button. See (10).

When the "Custom" has been selected, the "Select CO Type" dialog box is displayed.

Select the [Back] button for return to the previous dialog box.

Select the [Cancel] button for quitting.

- (7) The "Select CO Type" dialog box is displayed.



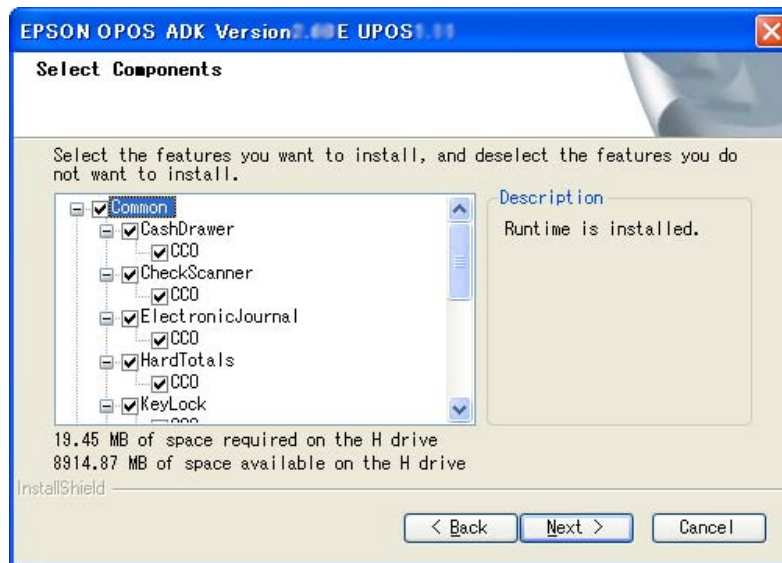
Select the CO that has a priority for registration. The CO registered here will be valid for all devices.

When the type is decided, select the [Next] button.

Select the [Back] button to return to the previous dialog box.

Select the [Cancel] button for quitting.

- (8) The "Select Components" dialog box is displayed.



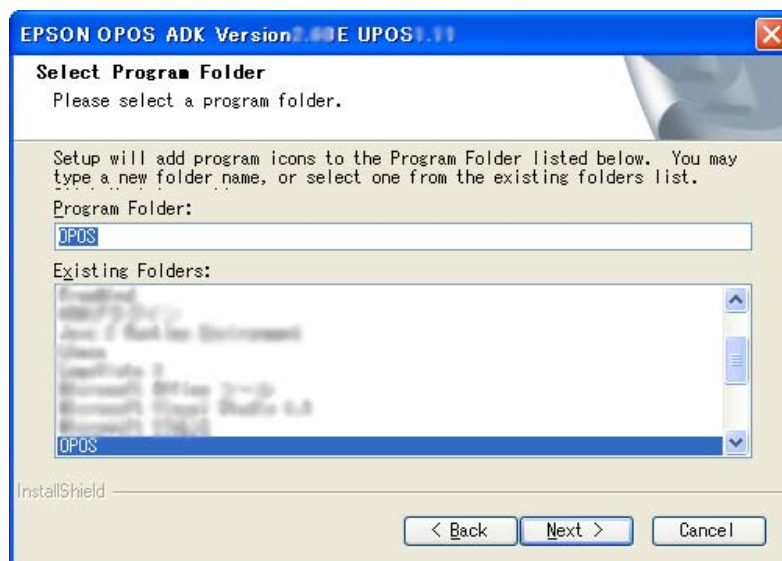
Select the necessary components.

To go to the next dialog box, select the [Next] button.

Select the [Back] button for return to the previous dialog box.

Select the [Cancel] button for quitting.

- (9) The “Select Program Folder” dialog box is displayed.



The previously decided folder name is displayed in “Program Folder”. If the Registry File is specified, this information is referred from the Registry File. Normally, default values should be used. The folder name can be changed, or an existing folder can be selected, but if an existing folder is used, please select it only after obtaining an understanding of the folder’s

attributes. The folder must have COMMON attributes.

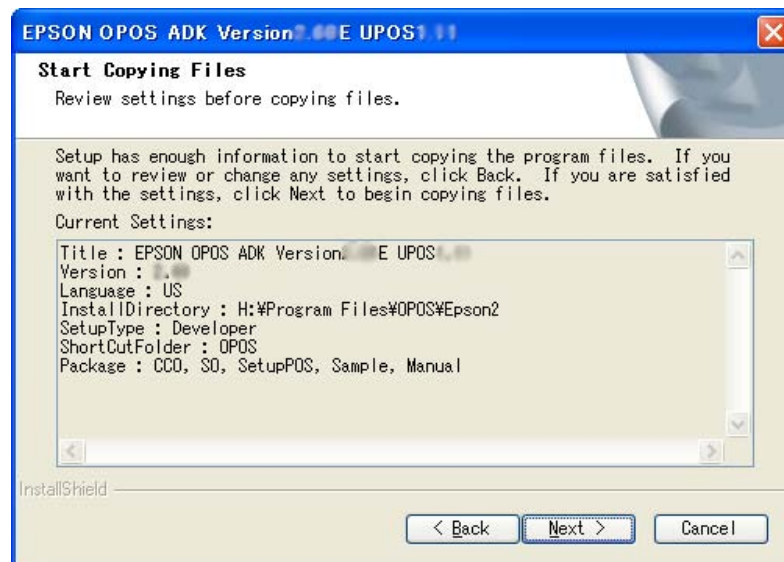
To go to the next dialog box, select the [Next] button.

Select the [Back] button for return to the previous dialog box.

If the [Back] button is selected, the set folder name will be canceled.

Select the [Cancel] button for quitting.

(10) The “Start Copying Files” dialog box is displayed.

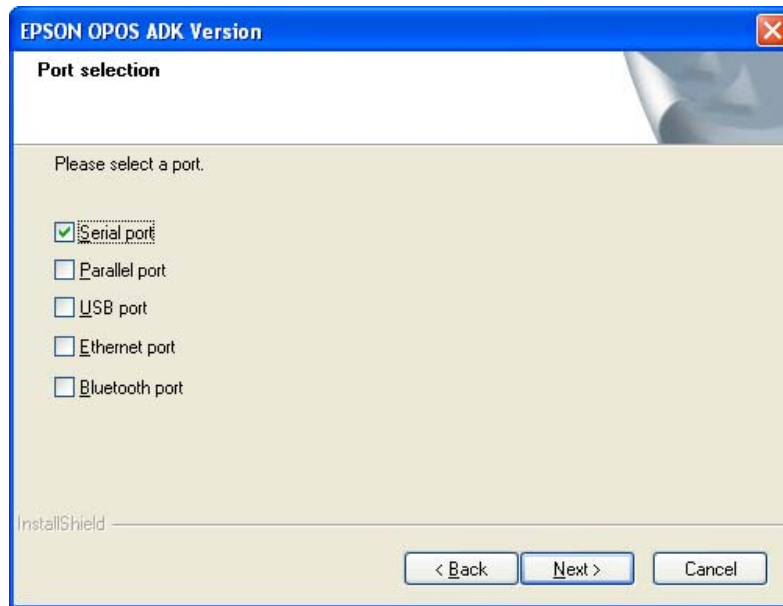


This dialog box shows the information that has already been set. Confirm the contents and select the [Next] button.

Select the [Back] button for return to the previous dialog box.

Select the [Cancel] button for quitting.

(11) The “Port selection” dialog box is displayed.



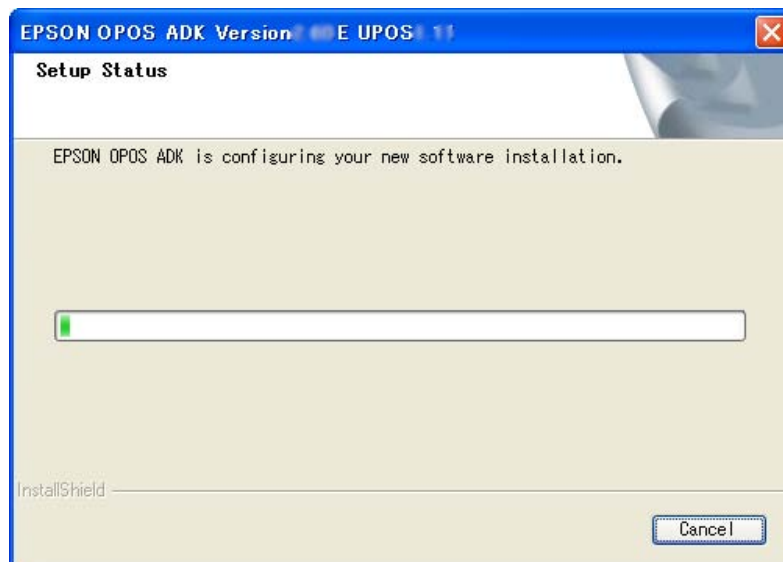
Select the port.

When the Parallel port was selected, the Lilac parallel driver is installed.

When the USB port was selected, the TMUSB driver is installed.

Select the [Cancel] button for quitting.

(12) The “Setup Status” dialog box is displayed.



The progress is shown on the status bar.

The [Cancel] button works at this point, but do not press the [Cancel] button unless there is a particular reason.

Operation after pressing the [Cancel] button is not guaranteed. If pressed,

be sure to perform a reinstallation.

At this point, all the information necessary for the version upgrade shall be updated. It may take a few minutes for all the operations to be completed.

(13) When all the operations are completed, the "Setup Complete" dialog box is displayed. Select in the check box whether or not to display the release note on the installed EPSON OPOS ADK. It is strongly recommended to read the notes.

(14) If system files are renewed because of the installation, rebooting of system may be requested. In this case, please specify whether rebooting right away or later, and execute the rebooting. If later rebooting is specified, make sure to reboot after the completion of the installation. The operation under the un-rebooted environment is not guaranteed at all.

(15) Installation is completed. The SetupPOS utility starts up. Confirm the settings in accordance with the setup described in "Section 5. SetupPOS Utility" of this manual. When system rebooting is executed, SetupPOS utility does not start automatically. Please start SetupPOS utility from start menu.

*In the case of Windows 8/ Windows8.1, start from the shortcut on the Start screen.

3.3.2. Overwriting Installation

With the installation of EPSON OPOS ADK in advance, install according to the following procedure.

3.3.2.1. Overwriting Version 2.xx by this Version

When performing an overwrite installation, user information, specified installation destination, and specified program folder in the previous installation will remain effective. This information cannot be changed.

If Version 2.xx of EPSON OPOS ADK already has been installed, when you try to install the new version, Add-in Installation starts. In this case, only the installed devices and files are overwritten.

When installing this version over an existing install of version 2.67 or older and the following devices are registered, the devices must be re-registered after installation is complete. Re-register the devices according to the specifications in "Section 5 SetupPOS Utility" of the "EPSON OPOS ADK MANUAL User's Manual (Installer/ SetupPOS/ TMUSB)" manual.

TM-H6000II(M), TM-H6000III(M), TM-T88III(M), TM-T88IV(M), TM-T70(M),

TM-T90(M), TM-L90(M), TM-U220A(M), TM-U220AM_BM, TM-U220B(M),
TM-U220BM_BM, TM-U220D(M), TM-U220DM_BM

<Installation Procedure>

- (1) Run the SETUP.EXE program on disk 1.

Before running SETUP.EXE, close the other programs already running. Otherwise, installation may fail. It takes some time for Setup to start. Wait a while.

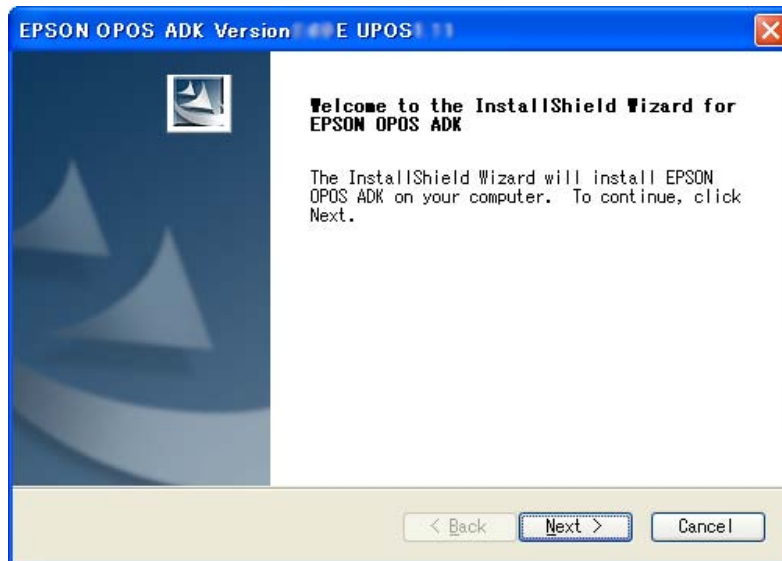
Compares the module already installed with the module contained in the package. If there is a possibility that problem occurs, the “Question” dialog box is displayed and a summary is notified.

The “Choose Setup Language” dialog is displayed.



Select the language for the installation. Then, select the [OK] button.

- (2) The “Welcome” dialog box is displayed.



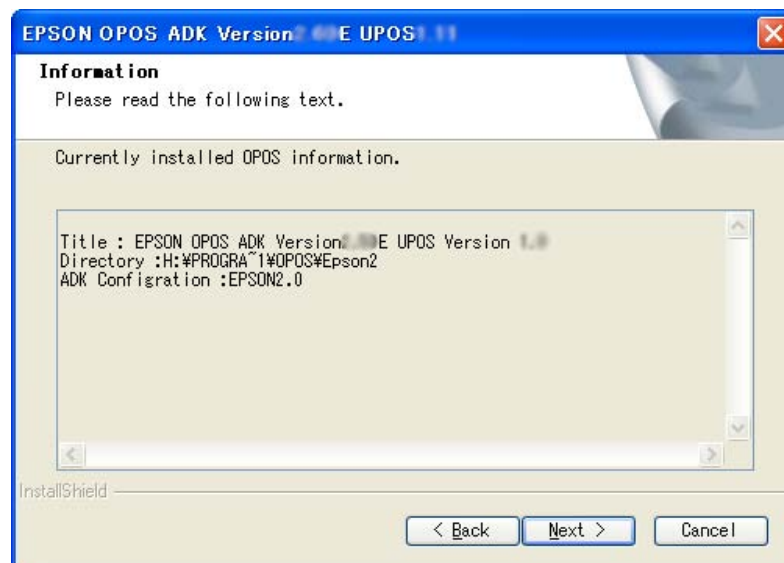
Read the contents and then select the [Next] button if agreed with the contents.

Select the [Cancel] button for quitting.

Note:

Depending on the environment, the "Select Language" dialog box will be displayed. In this case, please select the language that you want.

- (3) The "Information" dialog box is displayed.

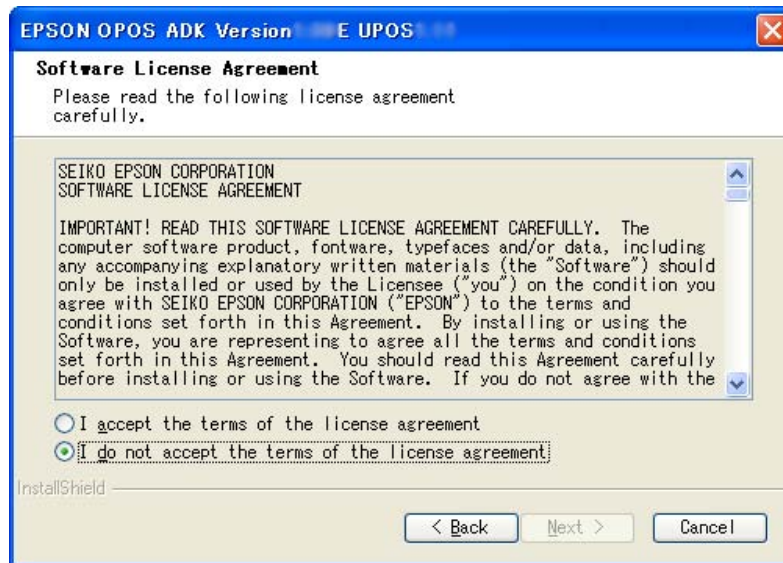


This dialog box displays information on the settings of the already installed EPSON OPOS ADK. Confirm the contents of the list and select the [Next] button if the contents are satisfactory.

Select the [Back] button to return to the previous dialog box.

Select the [Cancel] button for quitting.

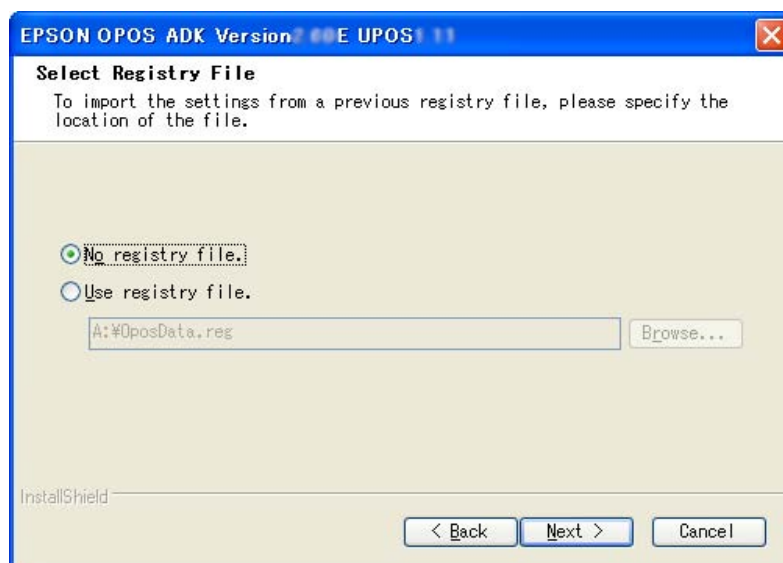
- (4) The “Software License Agreement” dialog box is displayed.



To install the EPSON OPOS ADK, the user must agree to the contents of the displayed agreement. Read the contents. In addition, select “I accept the terms in the License” if agreed with the Software License Agreement. After the selection, the [Next] button becomes effective. Select the [Next] button. If the Software License Agreement is not acceptable, select “I do not accept the terms in the License Agreement”. Then, select the [Cancel] button. Exits installation.

Select the [Back] button for returning to the previous dialog box.

- (5) The “Select Registry File” dialog box is displayed.

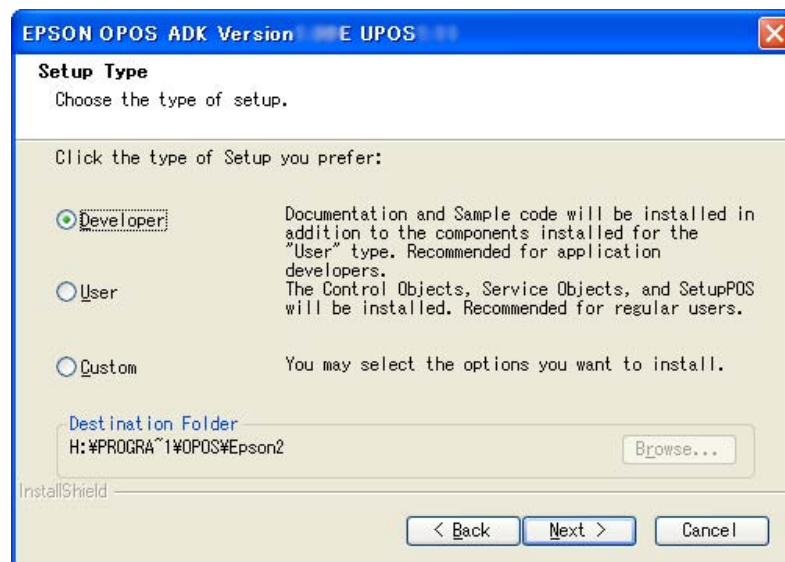


Select “No Registry File” not to use the REG file. Then, select the [Next] button. Select “Use Registry File” to use the REG file. Enter the full path REG file name after the text box is enabled. Then, select the [Next] button. Press the [Browse] button to specify the folder name that includes the REG file. The REG file is a file for restoring information related to EPSON OPOS ADK. Use the SetupPOS utility to create this file. If an improper Registry File is selected, an error message will be displayed and it will not be possible to move to the next dialog box. Follow the directions of the error message and confirm that the Registry File is correct. When the Registry File is set correctly, subsequent information can be referred from the Registry File.

Select the [Back] button to return to the previous dialog box.

Select the [Cancel] button for quitting.

- (6) The “Setup Type” dialog box is displayed.



Select a setup type to be installed. Selectable types are “Developer”, “User” and “Custom”.

The details of type are as follows:

Developer

Components:

Full installation of programs

Manual

SampleProgram

CO:

<Overwrite installation> Complies with CO/CCO that has

already been installed.

User

Components:

Full installation of programs

CO:

<Overwrite installation> Complies with CO/CCO that has already been installed.

Custom

Select components to be installed.

When you chose the "Developer" or "User":

Proceeds to the [Start Copying Files] dialog box. See (9).

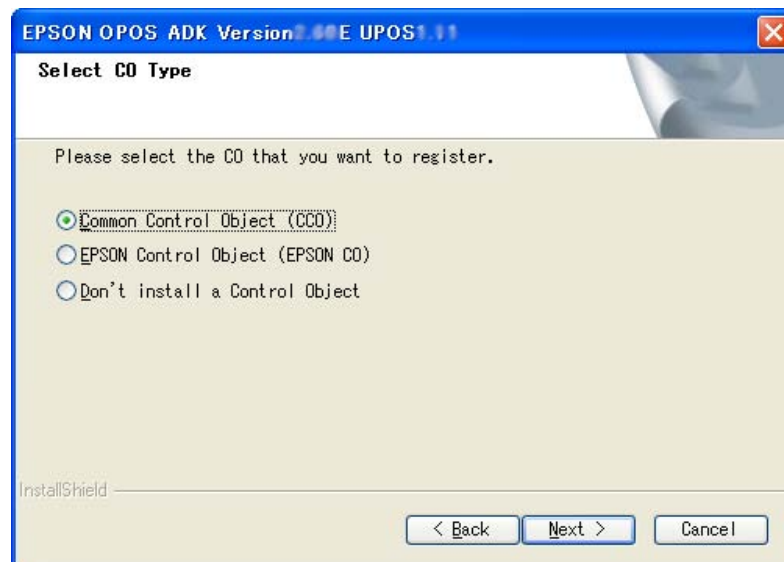
When you chose the "Custom":

If the CO/CCO is already registered, go to the "Select Components" screen in order to overwrite the base and minor versions. See (8).

If the CO/CCO is already registered, go to the "Start Copying Files" screen in order to overwrite the SP version. See (9).

If the CO/CCO is not already registered, go to the "Select CO Type" screen. See (7).

- (7) The "Select CO Type" dialog box is displayed.



Select the CO that has a priority for registration. The CO registered here will be valid for all devices.

When the type is decided, select the [Next] button.

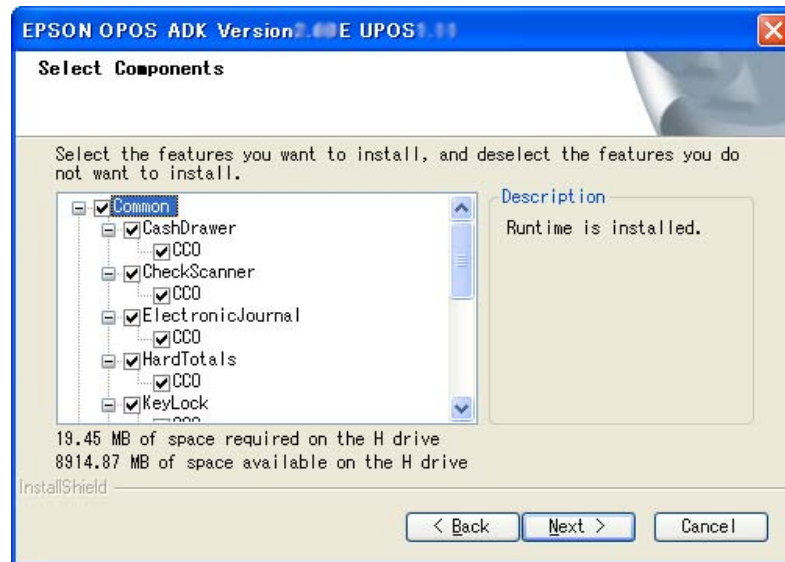
Select the [Back] button to return to the previous dialog box.

Select the [Cancel] button for quitting.

To overwrite the base and minor versions, go to the "Select Components" screen. See (8).

To overwrite the SP version, go to the "Start Copying Files" screen. See (9).

- (8) The "Select Components" dialog box is displayed.



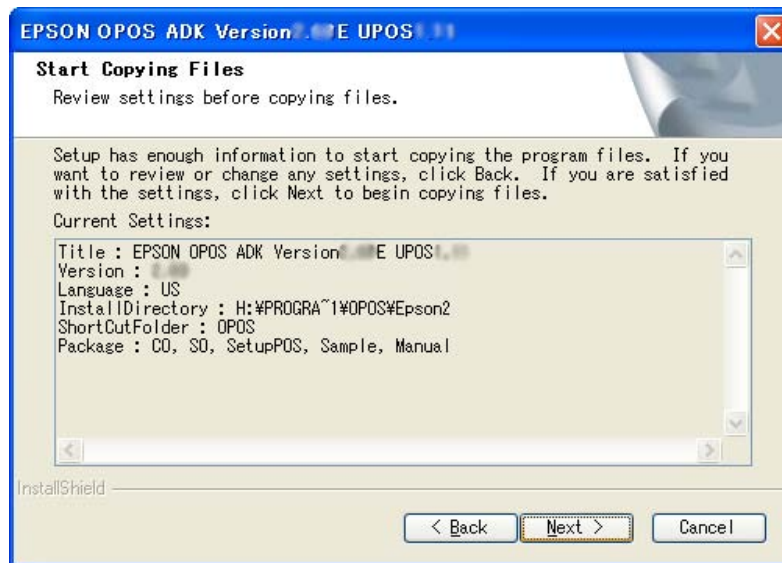
Select the necessary components.

To go to the next dialog box, select the [Next] button.

Select the [Back] button for return to the previous dialog box.

Select the [Cancel] button for quitting.

- (9) The "Start Copying Files" dialog box is displayed.

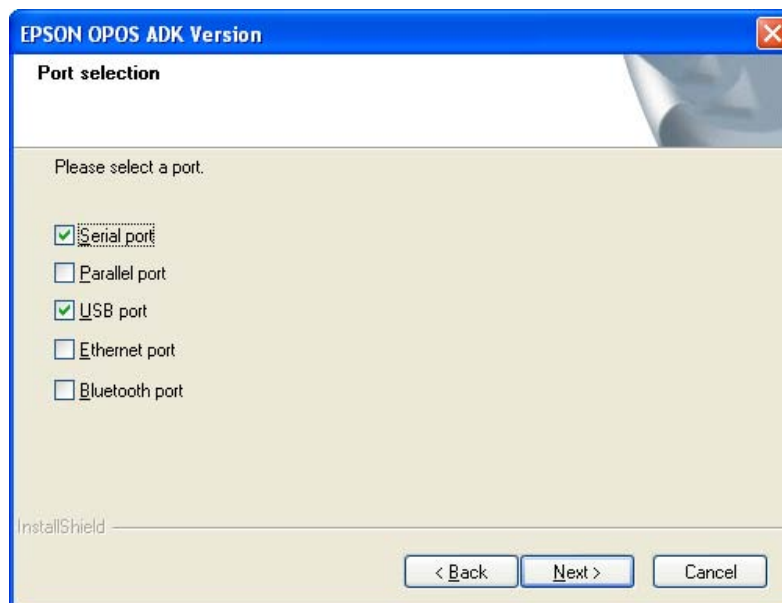


This dialog box shows the information that has already been set. Confirm the contents and select the [Next] button.

Select the [Back] button to return to the previous dialog box.

Select the [Cancel] button for quitting.

(10) The “Port selection” dialog box is displayed.



Select the method of connection used by device, select the [Next] button.

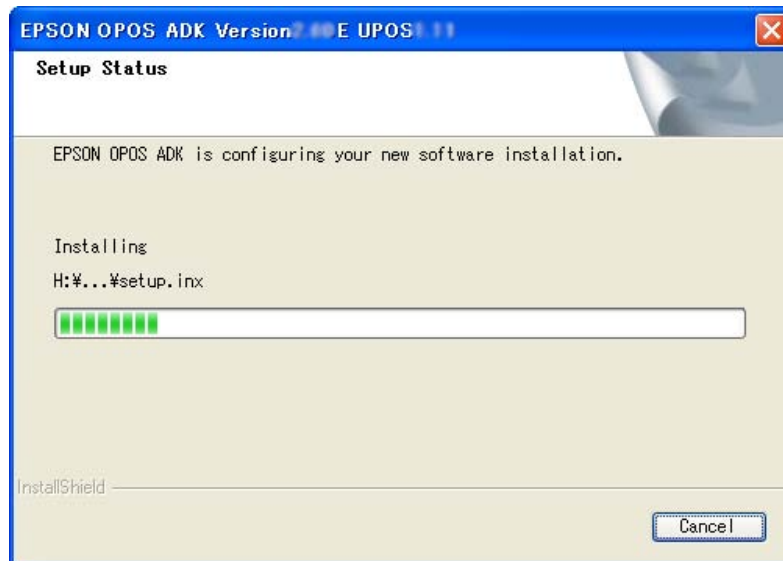
When the Parallel port is selected, the Lilac parallel driver is installed.

When the USB port is selected, the TMUSB driver is installed.

Select the [Back] button to return to the previous dialog box.

Select the [Cancel] button for quitting.

- (11) The “Setup Status” dialog box is displayed.



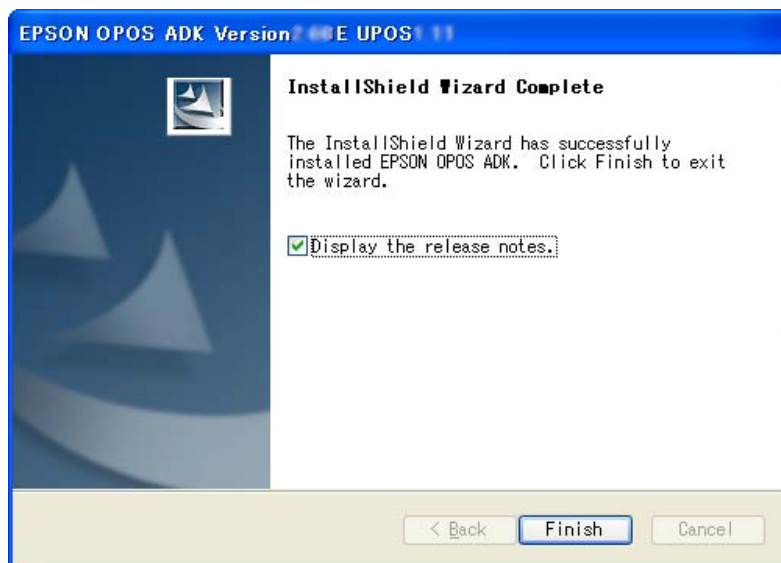
The progress is shown on the status bar.

The [Cancel] button works at this point, but do not press the [Cancel] button unless there is a particular reason.

Operation after pressing the [Cancel] button is not guaranteed. If pressed, be sure to perform a reinstallation.

At this point, all the information necessary for the version upgrade shall be updated. It may take a few minutes for all the operations to be completed.

- (12) When all the operations are completed, the “InstallShield Wizard Complete” dialog box is displayed.



Select in the check box whether or not to display the release note on the installed EPSON OPOS ADK. It is strongly recommended to read the notes.

- (13) If system files are renewed because of the installation, rebooting of system may be requested. In this case, please specify whether rebooting right away or later, and execute the rebooting.
If later rebooting is specified, make sure to reboot after the completion of the installation. The operation under the un-rebooted environment is not guaranteed at all.

- (14) Installation is completed. The SetupPOS utility starts up. Confirm the settings in accordance with the setup described in "Section 5. SetupPOS Utility" of this manual.

When system rebooting is executed, SetupPOS utility does not start automatically. Please start SetupPOS utility from start menu.

*If "TM-T88V" is selected from the "Select Device Name" combo box,

3.3.2.2. Overwriting Version 1.xx by Version 2.xx

EPSON OPOS ADK Version 1.xx can be overwritten only by Version 2.xx. The Version 2.xx overwrites all the components of the Version 1.xx.

When performing an overwrite installation, user information, specified installation destination, and specified program folder in the previous installation will remain effective. This information cannot be changed.

<Installation Procedure>

- (1) Run the SETUP.EXE program on disk 1.

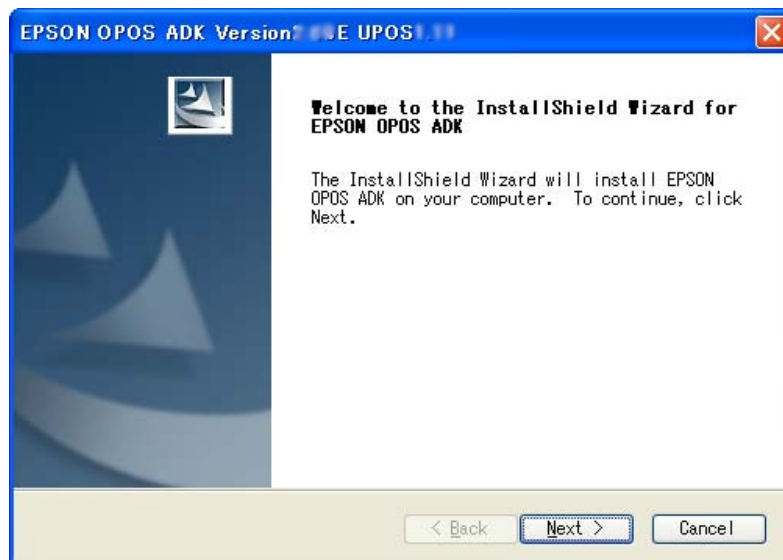
Before running SETUP.EXE, close the other programs already running. Otherwise, installation may fail. It takes some time for Setup to start. Wait a while.

The “Choose Setup Language” dialog is displayed.



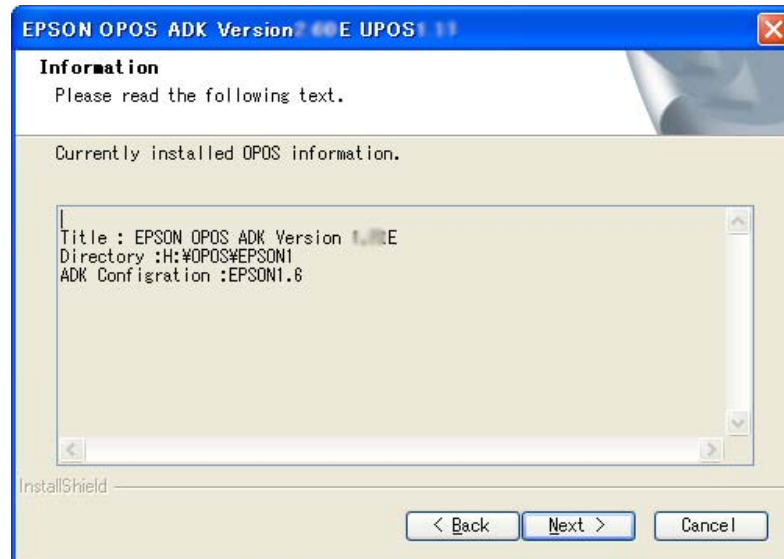
Select the language for the installation. Then, select the [OK] button.

- (2) The “Welcome” dialog box is displayed.



Read the contents and then select the [Next] button if agreed with the contents. Select the [Cancel] button for quitting.

- (3) The “Information” dialog box is displayed.



This dialog box displays information on the settings of the already installed EPSON OPOS ADK. Confirm the contents of the list and select the [Next] button if the contents are satisfactory.

Select the [Back] button to return to the previous dialog box.

Select the [Cancel] button for quitting.

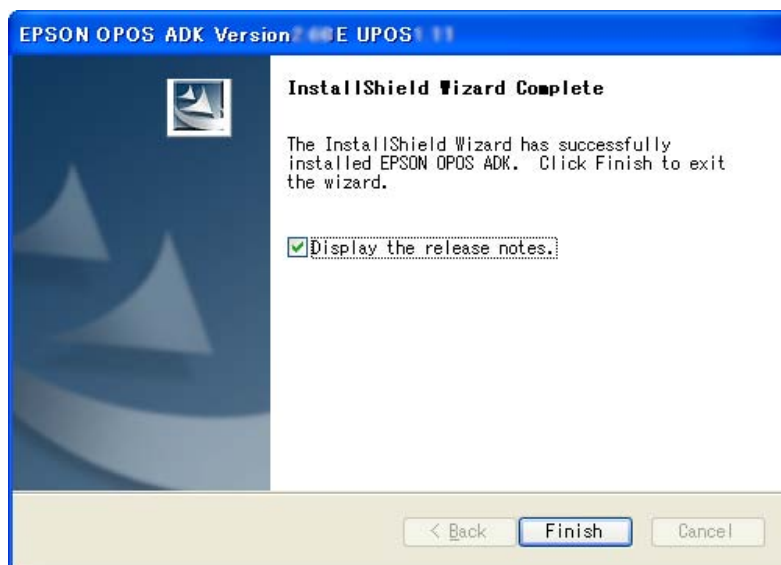
From (4) to (11) steps are the same as those described in “3.3.1 First Installation”. Please refer to “3.3.1 First Installation” of this manual.

- (12) The “Question” dialog box is displayed.

Select the [Yes] button for updating to the information of the OPOS ADK version that coexisted.

Select the [No] button for not updating to the information of the OPOS ADK version that coexisted

- (13) When all the operations are completed, the "InstallShield Wizard Complete" dialog box is displayed.



Select in the check box whether or not to display the release note on the installed EPSON OPOS ADK. It is strongly recommended to read the notes.

- (14) If system files are renewed because of the installation, rebooting of system may be requested. In this case, please specify whether rebooting right away or later, and execute the rebooting.

If later rebooting is specified, make sure to reboot after the completion of the installation. The operation under the un-rebooted environment is not guaranteed at all.

- (15) Installation is completed. The SetupPOS utility starts up. Confirm the settings in accordance with the setup described in "Section 5. SetupPOS Utility" of this manual.

When system rebooting is executed, SetupPOS utility does not start automatically. Please start SetupPOS utility from start menu.

3.3.2.3. Version 1.xx and Version 2.xx Coexisting Installation

Coexistence implies that the devices of Version 1.xx and the devices of Version 2.xx operate simultaneously. The above versions are able to coexist, but it is recommended to use Version 2.xx alone since there are some restrictions as follows:

<Restrictions on coexistence>

Hydra settings devices on the same port cannot coexist and be shared between versions.

(Example) Version 1.xx : Printer, Display, Drawer are hydra settings.
Version 2.xx : When printer alone is intended installed.

In this case, the whole settings cannot be succeeded as they are unless Printer, Display and Drawer are also installed in Version 2.xx.

Only the printer information is succeeded, and the device settings for Display and Drawer are deleted.

It is not possible for devices with shared keyboard interface to coexist and be shared by different versions.

(Example) Version 1.xx : MSR, Keylock.
Version 2.xx : When Keylock alone is intended installed.

In this case, the whole settings cannot be succeeded as they are unless both Keylock and MSR are also installed in Version 2.xx. Only the Keylock information is succeeded, and the device settings for MSR are deleted from both versions.

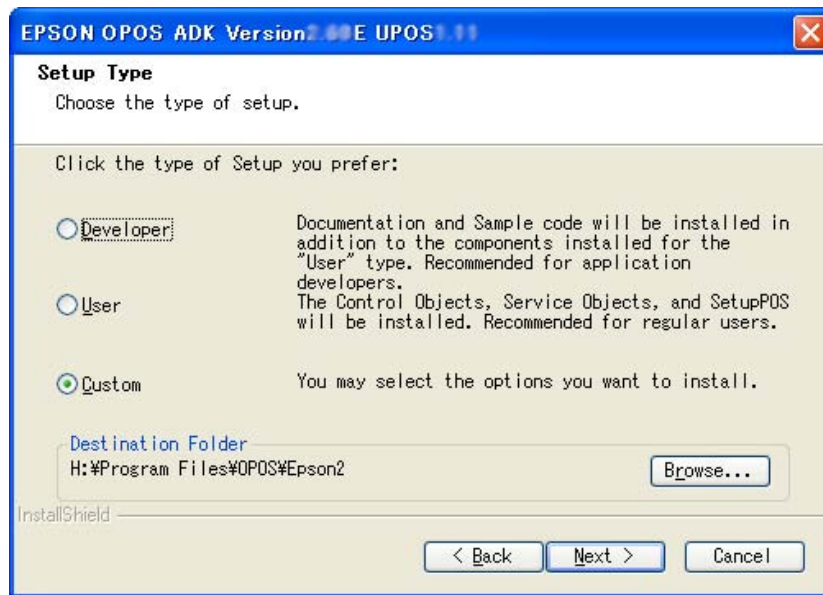
The restrictions apply to the following devices.

COM hydra : Printer, Display, Drawer, MICR, CheckScanner
Keyboard hydra : MSR, Keylock

<Installation Procedure>

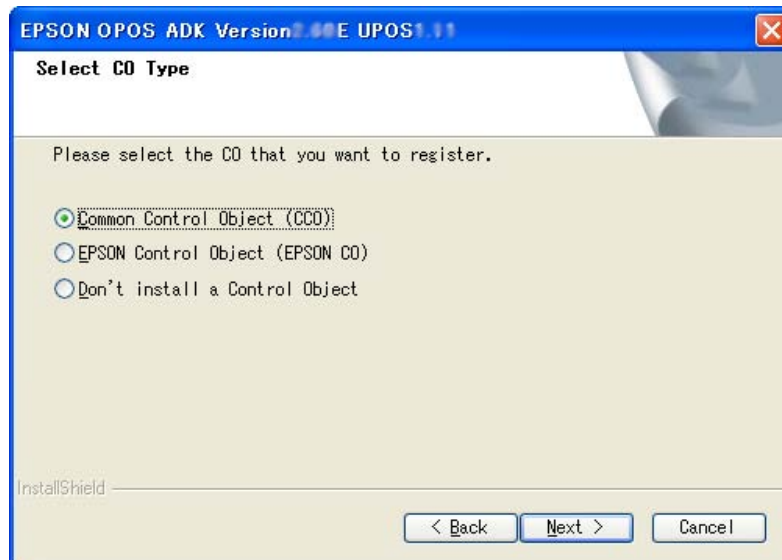
From (1) to (4) steps are the same as those described in "3.3.1 First Installation". Please refer to "3.3.1 First Installation" of this manual.

- (5) The “Setup Type” dialog box is displayed.



Select the [Custom] button, and then select the [Next] button.
 Select the [Back] button to return to the previous dialog box.
 Select the [Cancel] button for quitting.

- (6) The “Select CO Type” dialog box is displayed.



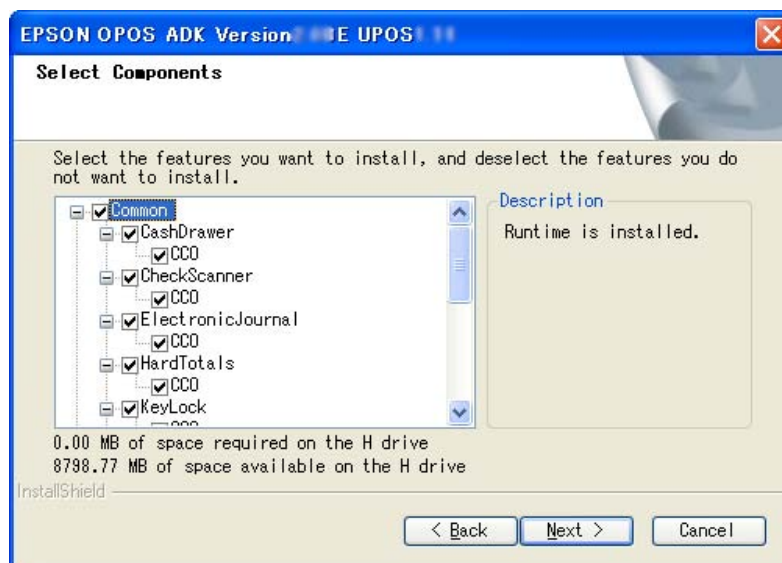
Select the CO that has a priority for registration. The CO registered here will be valid for all devices.

When the type is decided, select the [Next] button.

Select the [Back] button to return to the previous dialog box.

Select the [Cancel] button for quitting.

- (7) The “Select Components” dialog box is displayed.



Select the necessary components.

Select the [Next] button to go to the next dialog box.

Select the [Back] button to return to the previous dialog box.

Select the [Cancel] button for quitting.

- (8) The “Start Copying Files” dialog box is displayed.

This dialog box shows the information that has already been set. Confirm the contents and select the [Next] button.

Select the [Back] button to return to the previous dialog box.

Select the [Cancel] button for quitting.

- (9) Copying of files, registration of DLL and OCX, and setting the registry begins.

During the copying, the confirmation message “Inherit the previous version information?” is displayed. When overwriting, be sure to select yes. This ensures that the device information set in Version 1.xx can be used by Version 2.xx. It may take a few minutes for all the operations to be completed. When the [No] button is selected, information on Version 1.xx remains as it is.

At this point, all the information necessary for the version upgrade shall be updated.

It may take a few minutes for all the operations to be completed.

The [Cancel] button works at this point, but do not press the [Cancel] button unless there is a particular reason. Operation after pressing the [Cancel] button is not guaranteed. If pressed, be sure to perform a reinstallation.

- (10) When all the operations are completed, the “Setup Complete” dialog box is displayed.

Select in the check box whether or not to display the release note on the installed EPSON OPOS ADK. It is strongly recommended to read the notes.

When the installation of the parallel driver is successfully done, system files are renewed and a message to query rebooting method is indicated. Please specify whether rebooting right away or later, and execute the rebooting. If later rebooting is specified, make sure to reboot after the completion of the installation. The operation under the un-rebooted environment is not guaranteed at all.

- (11) Installation is completed. The SetupPOS utility starts up. Confirm the settings in accordance with the setup described in “Section 5. SetupPOS Utility” of this manual.

When system rebooting is executed, SetupPOS utility does not start automatically. Please start SetupPOS utility from start menu.

3.3.2.4. Others

When both Version 1.xx and Version 2.xx prior to Version 2.60 are already installed and in addition, Version 2.xx is upgraded to this version, the method described in Section 3. “3.3.2.1. Overwriting Version 2.xx by this version” of this manual applies.

3.4. Files to be installed

The following files are installed when the installation is normally completed.

- Program files:

These files are installed in the directory where the EPSON OPOS ADK is installed. Software files such as CO, SO, utilities, and device information files are included in this directory. To install these files, select "Program Files" in the "Select Install Module" dialog box at installation.

- Include files:

These files are installed in the Include folder under the folder where the EPSON OPOS ADK is installed. Files necessary for the OPOS such as C++ header file, VB header file, and TLB file for CO are included in this directory. To install these files, select "Program Files" in the "Select Install Module" dialog box at installation.

- Sample:

These files are installed in the Sample folder under the folder where the EPSON OPOS ADK is installed. Sample programs for VC++ and VB are included in this directory. To install these files, select "Sample" in the "Select Install Module" dialog box at installation.

When you use the sample program in the Windows XP and Windows Vista or newer, the Logon ID needs to be the Administrator account group.

- Manuals:

These files are installed in the Manual folder under the folder where the EPSON OPOS ADK is installed. To install these files, select "Manuals" in the "Select Install Module" dialog box at installation.

3.5. Types of Installation

There are two types of installation: full and custom.

Select the type in the "Install Type" dialog box.

- Full installation

Full installation installs all the supported devices. For supported devices, read the release note included in the Installer disks.

- Custom installation

Custom installation installs selected devices from the supported devices. Selectable units are device classes. If selecting custom installation in the "Install Type" dialog box, the "Select Components" dialog box appears. The displayed list shows the devices that can be installed. Check only the devices that you want to install.

3.6. CO Registration

Both CCO and EPSON CO can be installed and registered in EPSON OPOS ADK. In addition, you can also choose to register neither of the two, and thus it becomes possible to use the COs of a previous version or from another company. When using multiple COs in this way, please note the following precautions.

Please note the following points when the multiple CO are used.

(When application is created with VB)

(1) Pasting control on form by adding VB component

In this case, character strings applicable to the CO type library is displayed in the "Component dialog" box.

As this character strings is an individual string, it is possible to select the CO to be actually used. If multiple COs is registered, select the CO that you want to use from the list of components.

(2) Creating CO by specifying Program ID using Create Object

In this case, specifies the program ID of the CO.

The program ID is character strings determined by the OPOS specifications.

Ex: the CO of a POSPrinter is "OPOS.POSPrinter".

If multiple COs of the same device class has been registered, the latest one is created.

If this method is used, it is also necessary to pay attention to the order of the registrations of COs.

(3) Creating CO using WithEvents

In this case, it is not possible to create "as Object"; it is necessary to create it "as Class Name".

The CO Class Name may be the same or different.

Before declaring a variable using WithEvents, components are selected using method (1), and the components selected in the "Component" dialog box can be used.

3.6.1. The method for updating the CCO

The CCO that is bundled with this product is the latest version at release time. Please use the following procedure to update to a later version.

<Installation Procedure>

- (1) Download the CCO installer from the following web site.
<http://www.monroecs.com/>
- (2) Installs by the downloaded CCO installer.

<Warnings>

- If you install EPSON OPOS after updating the CCO, the CCO will become the older version.
Follow the above steps, then update the CCO again.
- If you uninstall EPSON OPOS, the updated CCO will also be uninstalled.
Follow the above steps, then update the CCO again.

3.7. INI Files

The INSTALL.INI file contains information copied to the directory that is specified by the installation program. Also, SETUP.INI is a file referred by SETUP.EXE. Make sure these file are always in the same directory as the SETUP.EXE file. Please do not edit these files. If modified, the operation is not guaranteed.

3.8. System Setup Files

There is no need for the user to edit any system files (AUTOEXEC.BAT, CONFIG.SYS, SYSTEM.INI).

3.9. Installation Using the Registry File

The registry file is used when a user wants to restore an environment created on one PC to another PC. When the various settings are made with the SetupPOS utility, this information is retained in the registry. With the SetupPOS utility, only the information necessary for the registry is stored in a file. By specifying this file

with the Installer, it is possible to restore the same environment.

To use this function, Please follow either procedure 1 or procedure 2 below.

In this section, the originally created environment is called the "main PC", and the environment that you want to reconstruct is called "target PC".

<Procedure 1>

- (1) Install the EPSON OPOS ADK in the main PC, and set the necessary information and the various devices using the SetupPOS utility.
Use the SetupPOS utility to save the registry. At this point, pay attention to the following and create the files.

- Specify "Register All".

- (2) Port information files (pcs.properties) will be retrieved from the following folder and saved to the same location as the registry file.

Windows 2000/XP:

(System Drive):\Documents and Settings\All Users\Application
Data\EPSON\portcommunicationservice

Windows Vista or newer:

(System Drive):\ProgramData\EPSON\portcommunicationservice

- (3) Perform installation on the target PC.

At this point, the following two methods are available. Select the method that suits the environment.

- Installation using the normal installation method.
- Installation using the Silent Installation method.

For the normal installation method, take the procedures described in "[3.3 Installation](#)". For installation using the Silent Installation method, see Section 3. "[3.11. Silent Installation](#)".

When installing on the target PC, specify the registry file saved in procedure (1).

- (4) At the time of installation, specify the registry file saved in step (1).

<Procedure 2>

- (1) After installing Epson OPOS ADK on the main PC, use the SetupPOS utility to configure the various devices and information.

- (2) Run the information collection tool.

This tool is in the same location as the program file.

Use the following command to run the tool. Run the information collection tool with administrative rights.

Ex.) "C:\Program Files\OPOS\Epson2\CollectCopyInstallModules.bat"

When you run the tool, an "EpsonCopyInstallModules" folder is created in

the same location and the following files are stored in it.

- Registry saved by the SetupPOS utility
- Port information file (pcs.properties)

(3) Perform installation on the target PC.

At this point, the following two methods are available. Select the method that suits the environment.

- Installation using the normal installation method.
- Installation using the Silent Installation method.

For the normal installation method, take the procedures described in "[3.3 Installation](#)". For installation using the Silent Installation method, see Section 3. "[3.11. Silent Installation](#)".

When installing on the target PC, specify the registry file saved in procedure (2).

(4) At the time of installation, specify the registry file saved in step (2).

<Restrictions>

- Be sure to create the REG file with the SetupPOS utility or acquire it with the information collection tool.
- It is a prerequisite that the Registry File contains the same version information as the version to be installed, or the same information as the installed environment.

Installation can not be done successfully in the following cases:

- If system drive is different:

(e.g. The original drive is C and the destination drive is D)

- If the physical port configuration is different:

(e.g. The original has COM1 and COM2, and the destination has only COM1)

- If the OS is different:

(e.g. The original is 32bit and the destination is 64bit)

- When a registry file is specified using the installer User Interface, install components are not determined automatically. Desired components should be selected using the installer User Interface.
- With the installation using the Registry File, the old version's information is not succeeded.
- Please close all applications and install OPOS when migrating the port configuration.

3.10. Add-in Functions (Service Pack)

The add-in function is a function installing the package containing update

requirement only. A service pack is created based on the version exist. (This existed version is called the base version, in this section.) A service pack may be made available when devices are added or some kind of update becomes necessary after the release of the base version. For the latest service pack, please consult the manufacturer.

When you install the service pack, the registry information before the service pack has been installed is saved as a file in the destination folder for installation (file name: "OposReg.reg"). Use this registry file for Silent Installation of the base version to the target PC.

This function is also used when you overwrite an old version (Version 2.xx) of EPSON OPOS ADK with a new version. In this case, the installed devices or the files only are all updated.

<Procedure>

- (1) Run the SETUP.EXE file on the service pack disk.
If Setup is executed with other programs' running, its operation is not guaranteed. It is suggested to close all programs before executing Setup. Startup of the Setup program may take some time.
- (2) Displayed message before the installation.
Before the installation, a message about the installed devices or files appears.
For service pack:
If a device included in the service pack is not installed, a warning message appears. This is because devices cannot be added after installing the service pack. (Refer to <Restrictions> below for details.)
To overwrite Version 2.xx with a new version:
If the devices and files included in the new version are not installed, a prompt message appears. All devices or files not listed in the message will be overwritten. To install other files, run the installation again after the update.
Select [Next] to continue. Select [Cancel] to cancel the installation.
- (3) The "Welcome" dialog box is displayed.
Read the contents and then select the [Next] button if you agree with the contents.
- (4) The "Software License Agreement" dialog box is displayed.
To install the EPSON OPOS ADK, you must agree to the contents of the

displayed agreement. Read the contents and select "I accept the terms in the License" if agreed with the Software License Agreement. After the selection, the [Next] button becomes effective. Then, select the [Next] button. If the Software License Agreement is not acceptable, select "I do not accept the terms in the License Agreement". Then, select the [Cancel] button. Select the [Back] button to return to the previous dialog box.

- (5) The "Information List" dialog box is displayed.
Information of the installed EPSON OPOS ADK is displayed. Confirm the contents, and then select the [Next] button if the contents are acceptable.
- (6) The "Select Registry File" dialog box is displayed.
Select "Not Use Registry File" when the REG file is not used. Then select the [Next] button. Select "Use Registry File" to use the REG file. Enter the full path REG file name after the text box is enabled. The default value is "A:\OPOSDATA.reg". Then, select the [Next] button. Press the [Browse] button to specify the folder name that includes the REG file.
The REG file is a file for restoring information related to EPSON OPOS ADK. Use the SetupPOS utility to create this file.
If an improper Registry File is selected, an error message will be displayed and it will not be possible to go to the next dialog box. Follow the directions of the error message and confirm that the Registry File is correct. When the Registry File is set correctly, subsequent information can be referred from the Registry File. When installing by overwriting, the Registry File information will have priority over the installation information of the previous installation.
Select the [Back] button to return to the previous dialog box.
- (7) The "Select Installation Module" dialog box is displayed.
Select a module to be installed. Selectable modules are "Program Files", "Samples", and "Manuals".
Installed modules cannot be selected.
After selecting a module, press the [Next] button. Press the [Back] button to go back to the previous dialog box.
- (8) The "Start Copying Files" dialog box is displayed.
The dialog box shows the information, which has already been set.
Confirm the contents and select the [Next] button.
Select the [Back] button to return to the previous dialog box.

- (9) Copying of files, registration of DLL and OCX, and setting the registry begins.

The [Cancel] button works at this point, but do not press the [Cancel] button unless there is a particular reason. Operation after pressing the [Cancel] button is not guaranteed. If pressed, be sure to perform a reinstallation.

At this point, all the information necessary for the version upgrade shall be updated.

It may take a few minutes for all the operations to be completed.

- (10) When all the operations are completed, the "Setup Complete" dialog box is displayed. Select in the check box whether or not to display the release note on the installed EPSON OPOS ADK. It is strongly recommended to read the notes.

When the installation of the parallel driver is successfully done, system files are renewed and a message to query rebooting method is indicated. Please specify whether rebooting right away or later, and execute the rebooting. If later rebooting is specified, make sure to reboot after the completion of the installation. The operation under the un-rebooted environment is not guaranteed at all.

- (11) Installation is completed. The SetupPOS utility starts up. Confirm the settings in accordance with the setup described in "Section 5. SetupPOS Utility" of this manual.

When system rebooting is executed, SetupPOS utility does not start automatically. Please start SetupPOS utility from start menu.

<Restrictions>

- Be sure to install the base version before installing the service pack.
- As long as the service pack version number is higher than already installed service packs it can be installed without installing intermediate service packs. For instance, it is possible to add-in Service Pack Version 3 directly to the base version. It is not necessary to first add-in Service Pack Version 1 or 2.
- It is not possible to install a service pack older version than the one currently installed. For instances; after Service Pack Version 3 has been installed, it is not possible to install Service Pack Version 1 or 2.

- The base version cannot be overwritten after installing the service pack. Also, the service pack alone cannot be uninstalled. In other words, after the service pack is installed, later installations of additional devices are not possible. Be sure to reinstall or overwrite the base version and the desired devices before installing the service pack.

3.11. Silent Installation

Installation can be performed without using the user interface. This is called "Silent Installation". This function uses the Registry File explained in the previous section. For details, please refer to the "[3.9 Installation Using the Registry File](#)" of this manual. This function is mainly used for installation on the target PC. Please use another installation function using normal user interface on the main PC. For description of target PC and main PC, please refer to the "[3.9 Installation Using the Registry File](#)" of this manual. When executing a Silent Installation, the registry information from the main PC is succeeded. OPOS ADK Manuals and the Sample Programs cannot be installed when using the Silent Installation function. To use this function, Please follow either procedure 1 or procedure 2 below.

<Procedure 1>

Install the EPSON OPOS ADK on the main PC, and set the necessary information and the various devices using the SetupPOS utility. Then, save the registry using the SetupPOS utility.

(1) Make a recording file.

Make a file for recording the operation of install. To install the EPSON OPOS ADK, perform the following commands.

(Example) "Full Path\Setup.exe" /r

When the installation finished, the install operation is recorded to the "Setup.iss" file that is made in the Windows folder.

(2) Use the "Export Registry" in SetupPOS utility to save the registry information. Select the [All] button

(3) Port information files (pcs.properties) will be retrieved from the following folder and saved to the same location as the registry file.

Windows 2000/XP:

(System Drive):\Documents and Settings\All Users\Application
Data\EPSON\portcommunicationservice

Windows Vista or newer:

(System Drive):\ProgramData\EPSON\portcommunicationservice

- (4) Copy the record file created in (1) to the same folder as "Setup.exe" and begin silent installation.

(Example)

"Full Path\Setup.exe" /s /a"a:\OposData.reg"

To specify the path of the recording file, use the "/f1" option.

(Example)

"Full Path\Setup.exe" /s /f1"c:\Temp\setup.iss" /a"a:\OposData.reg"

<Procedure 2>

Install the EPSON OPOS ADK on the main PC, and set the necessary information and the various devices using the SetupPOS utility.

- (1) Make a recording file.

Make a file for recording the operation of install. To install the EPSON OPOS ADK, perform the following commands.

(Example) "Full Path\Setup.exe" /r

When the installation finished, the install operation is recorded to the "Setup.iss" file that is made in the Windows folder.

- (2) Run the information collection tool.

This tool is in the same location as the program file.

Use the following command to run the tool. Run the information collection tool with administrative rights.

Ex.) "C:\Program Files\OPOS\Epson2\CollectCopyInstallModules.bat"

When you run the tool, an "EpsonCopyInstallModules" folder is created in the same location and the following files are stored in it.

- Registry saved by the SetupPOS utility
- Port information file (pcs.properties)
- Record file (setup.iss)

- (3) Copy the record file created in (2) to the same folder as "Setup.exe" and begin silent installation.

(Example)

"Full Path\Setup.exe" /s /a"a:\OposData.reg"

To specify the path of the recording file, use the "/f1" option.

(Example)

"Full Path\Setup.exe" /s /f1"c:\Temp\setup.iss"

/a"a:\OposData.reg"

The functions are added to Silent Installation by adding the following parameters to command lines.

The following parameters can be used at the same time. If you wish to use them at once, input the parameter you want to use after the parameter added to the command line.

Parameters do not distinguish capital and small letters. It is not necessary specifying only with small letters as the following examples show.

<Making a recording file>

Adding Parameter: "/r"

Example: "Full Path\Setup.exe" /r

<Indicating the Silent Installation>

Adding Parameter: "/s"

Example: "Full Path\Setup.exe" /s /a"Full Path\OposData.reg"

<Specify the substitute recording file name and the path name>

Adding Parameter: "/f1"

Example: "Full Path\Setup.exe" /s /f1"C:\Temp\Setup.iss" /a"Full Path\OposData.reg"

<Specify the substitute log file name and the path name>

Adding Parameter: "/f2"

Example: "Full Path\Setup.exe" /s /f2"C:\Temp\Setup.iss" /a"Full Path\OposData.reg"

<Indicating that adds the registry >

Adding Parameter: "/a"

Example: "Full Path\Setup.exe" /s /a"Full Path\OposData.reg"

<Rebooting after the Silent Installation>

Adding Parameter: "/n"

Example: "Full Path\Setup.exe" /s /a"Full Path\OposData.reg" /n

- The result is stored in the log file.
The log file is created with the name "Silence.log" in the target directory. All errors can be referred here. However, before the completion of the installation,

the file is created in the root where Windows is (e.g. "C:\") since the target directory is not decided until the completion of the installation. Please ensure no errors exist in the log file after the Silent Installation.

<How to view Silence.log>

[SilenceMode Log]

[Silence Mode Log Start]: Silence mode starts

[SilenceMode Log End] : Silence mode ends

ERROR : Error message

WARNING : Warning message

INFORMATION : Information

MESSAGE : Message

Messages are logged in the same way as the messages listed in Section 3. "[3.12. Messages for Installation](#)" of this manual.

Once the Silence.log file is created, results are added to this file. For the latest result, please see the line, which indicates the start of the latest Silent Installation.

<Restrictions>

- The registry file is necessary for the Silent Installation.
- The registry file saved by the base version can be used only for the Silent Installation of the base version. It cannot be used to install the service pack.
- When installing a service pack with Silent Installation function, registry files have to be made under the condition that the service pack is being installed in advance. When the environment in which the service pack has been installed is imported to the target PC with the Silent Installation, first, execute the Silent Installation with the base version and the registry file for the base version (created automatically in the destination folder for installation during the service pack installation), then execute the Silent Installation with the service pack and the registry file for the service pack.

EX: Procedures for service pack Silent Installation

2. First Silent Installation of the OPOS Version 2.xx by using the following Example:
"Full Path\Setup.exe" /s /a"Full Path\OposReg.reg"
3. Next, Silent Installation of the service pack by using the following Example:

"Full Path\Setup.exe" /s /a"Full Path\OposData.reg"

Notes: The registry files such as "OposReg.reg" or "OposData.reg" are the example, which are already saved.

- When the environment (Main PC) for creating the registry files to be used is installing EPSON TM USB Driver or parallel driver with this installer, the driver being installed is installed with Silent Installation regardless of parameter's presence. If you wish not to install those drivers to the target PC, please make a registry file with a PC that is not installing respective driver.
- When a parallel driver is installed with Silent Installation, the system rebooting is not executed upon success of installation. A reboot may be required when system files are updated. When installing a parallel driver with Silent Installation or updating a system file, we recommend appending the reboot parameter to restart the system after installation is complete.
- If you specify the "/f1" parameter when creating the record file, the record file cannot be acquired with the information collection tool.

3.12. Messages for Installation

The program shows a user message when necessary after the execution or setting of an item selected in a dialog box. Follow the instructions in them. Conditions in which messages appear and the messages themselves are listed below.

3.12.1. Messages at the Time of Startup

Condition	Error Message
A user without administrator authority is attempting to perform installation. (Common to all of supported OS)	The user doesn't have Administrator authority.
The "Install.ini" file is not found.	The install.ini file cannot be found in the current directory. Aborting installation.
The "Relnote.txt" file is not found.	The Relnote.txt file cannot be found in the current directory. Aborting installation.
The contents of the "Install.ini" file are illegal and the contents are changed.	The setting files required for installation can either not be found, or they are corrupt.
The registry information is corrupted.	Registry information couldn't be read.

Attempting to install a previous version over a new version.	A newer version of OPOS is already installed.
Attempting to install a special version by overwriting but the special version does not allow overwriting installation.	This version cannot be overwritten.
Attempting to install a special version by coexistence but the special version does not allow coexisting installation.	This version cannot be overwritten.
Attempting to install a version that allows overwriting in the condition where a special version that does not allow overwriting installation is already installed.	This OPOS version cannot be overwritten.
Attempting to install a version that allows coexistence in the condition where a special version that does not allow coexisting installation is already installed.	This OPOS version cannot be overwritten.
Attempting to install by overwriting/coexistence when the version allows neither overwriting nor coexisting installation.	The OPOS information in the registry file is neither compatibly with this version, nor can it be overwritten.
A version in different language is already installed.	The language information of the installed OPOS is different. Installation cannot continue.
When an installed version of Epson software is incompatible this version.	This product cannot be used on the same PC as the products above. If you wish to use the products above after installing this product, please upgrade them to the latest version. Do you want to continue installation?

3.12.2. Messages during Installation

Condition	Error Message
There is not enough free space in the directory in which EPSON OPOS ADK to be installed.	The necessary free space cannot be secured to install OPOS.
Another application is using a file to be installed with EPSON OPOS ADK.	Cannot overwrite the subjected file. It may be in use. Please close all applications and restart installation.
Not a single component has been checked for custom installation.	The component is not checked.
The registry information contains illegal data.	An invalid registry information value was returned.
When there was a failure to load an installer internal DLL.	The specified file cannot be found. Please prepare a valid disk and restart installation.
When the selected CO type is different from the installed CO type.	The selected control objects type is not compatible with the one already installed. If you continue and install with this control objects type, the existing objects will be automatically removed and invalidated.
The registry recordation of uninstall information is failed.	Failed to record uninstall information. The uninstaller may not work correctly.

When the information for registered devices that is not supported by the current version is contained in the specified registry file.	Information for registered devices that is not supported by the current version is contained in the specified registry file. Since the use of this file as it is could lead to unexpected results, it is necessary to delete this device information. Do you wish to delete this device information? Click "Yes" to continue the install and delete this device information.
---	--

3.12.3. Message during Driver Installation after EPSON OPOS ADK Installation

Condition	Error Message
There is no driver supporting the OS currently used.	This OS is not supported.
The Installation transaction of the parallel driver is failed.	When parallel driver installation fails
An error is returned during the installation of the parallel driver.	The Installer of the Parallel driver returned an error.
An unexpected error is returned during the installation of the parallel driver.	An unexpected error is detected.
The update transaction of the parallel driver is success.	To use the Parallel interface, the PC must be restarted. To restart now, remove the disk and click "Yes". To restart later, click "No".

3.12.4. Messages When the Silent Mode is specified

Condition	Error Message
An illegal value is specified for the Silent Installation command line.	An illegal command line parameter was input. Please define as follows: setup /a"RegData.reg"
An inexistent registry file is specified in the command line.	Registry File specified in the command line parameter can't be found.
The Silent mode's log file cannot be created.	Failed to create the silent installer log file.
The Silent Installer log file cannot be opened.	Failed to open the silent installer log file.
An unsupported registry file is specified in the parameter.	The registry file specified in the command line parameter cannot be found.

3.12.5. Messages Related to Registry File

Condition	Error Message
The registry file, which is not being installed, is specified for registry import or the parameter specification at starting up.	The version stored in the registry file does not match the version of this installation.

The coexistence registry file of a version that is not installed is specified at the time of coexistence installation	The shared version in the registry file, and the version of the currently installer OPOS are different.
A registry file that contains coexistence information is specified at the time of sole installation	A shared version in the registry file is not installed in the target computer.
A registry file created with a different language version is specified at the time the parameter is specified during registry file selection or startup	The OPOS language stored in the registry file does not match the language of this installation.
A registry file created with a version that is not supported is specified.	The specified registry file contains version information that is not supported by this installation. This file cannot be used.
An internal processing of the registry file isn't performed smoothly due to some reasons.	Could not create a copy of the selected registry file. Registry file selection is disabled.
A registry file that was made by different version OS is specified when registry file import or the parameter is specified at startup.	The OPOS language stored in the registry file does not match the language of this installation.

3.12.6. Messages during Coexisting Installation

Condition	Error Message
Coexisting with a current version, a registry file containing only information on Version 2.xx is specified when registry file is selected or the parameter is specified at startup.	This registry file only has information of this version. The file can not be used when a shared installation is performed.
Coexisting installation with a version that does not support coexistence is executed.	A version of OPOS that is incompatible with this version is already installed. Please uninstall the existing version of OPOS prior to installing this version.
Overwriting Ver1.9x by Ver2.xx.	Do you want to update the shared OPOS information?

3.12.7. Add-in Installation Messages

Condition	Error Message
Installation of an add-in service pack whose language is different from that of the base version is executed.	The language information of the installed OPOS is different. Installation cannot continue.
Installation of an add-in service pack over a wrong base version is executed.	The installed version is not supported.
Installing the base version in the condition where the add-in service pack has been installed.	This service pack is already installed for OPOS.
Installing a service pack alone without its base version is being installed.	The installed OPOS version does not support service packs.

Devices and files included in the add-in service pack have not been installed in the base version.	The following devices and files cannot be installed after the installation of service pack x. ---- ---- These files cannot be installed after service pack x is installed. If necessary, after installing the above files please reinstall the Service Pack. Do you wish to continue?
The devices and files included in the full package are not installed yet	The following devices and files cannot be installed after the installation of Ver2.xx. ---- ---- Only the files already installed will be updated. To install other files of EPSON OPOS ADK, please select "Custom" in the Setup Type dialog box and then add the necessary components. Do you wish to continue?
The standard package has already been installed and you have tried to install either the same version or an older version.	The files of this version have already been installed. Installation is not necessary.

3.13. Precautions for Installation

- Overwriting installation is not possible when an EPSON OPOS ADK of different version or language (US version, Japanese version) has already been installed.
- It is not possible to read registry information files stored in an EPSON OPOS ADK of different version (including the service pack) or language (US version, Japanese version).
- Be sure to install the devices stated in the REG file information that is specified in the registry file import (that is, the devices that had already been installed or registered when the Setup POS utility stores to the REG file). If these do not match, operation cannot be guaranteed.

REG file information : POSPrinter, LineDisplay, MSR

Installed device : MSR

If such a setting is made as shown above, a conflict is generated, as it is only the settings of the POSPrinter and LineDisplay that will become valid whereas the actual files do not exist.

- If the installation fails for some reason, or it is canceled during the execution, be sure to perform a reinstallation. In the condition where installation has failed or been canceled, the settings, etc. may not be completely removed even if

uninstallation is performed. EPSON OPOS ADK may malfunction due to the failure of the first installation even after the reinstallation. It is recommended to uninstall EPSON OPS ADK and install it again.

- When using USB POSPrinter, MICR, LineDisplay, CheckScanner, CashDrawer and ElectronicJournal, install the Epson TM USB Driver (Epson USB controller for TM printer series) in advance. It is not possible to install the USB OPOS information unless the Epson TM USB driver is installed. Users of the EPSON TM USB Driver should refer to "Section 6. EPSON TM USB Driver" of this manual.
- When Version 1.xx and Version 2.xx are installed coexisting, the operation of Version 2.xx will not be normal if the uninstaller of Version 1.xx is used. Be sure to uninstall Version 2.xx first, or use the uninstaller of Version 2.xx to uninstall Version 1.xx.
- If Version 2.xx is uninstalled under the coexisting environment of Version 1.xx and Version 2.xx, the subsequent behavior of Version 1 is not guaranteed. In such a case reinstall Version 1.xx and make the prescribed settings.
- You can install and uninstall CCO using the SetupPOS utility; subsequently, however, the CCO registration information and the Installer cannot be synchronized.

When CCO is registered by the SetupPOS utility while CCO is not installed by the Installer, or when CCO is released by the SetupPOS utility while CCO is installed by the Installer, the Installer message may not be appropriate.

- When overwriting Version 1.xx by Version 2.xx, the settings of the trace function on Version 1.xx may not be succeeded. They are as follows.
 - CashDrawer: "Open Status Phase" (Device Specific Settings)
 - POSPrinter: "Reverse feed removal" (Device Specific Settings)
 - All devices: When the Trace file name is setting to default value by the OPOS API Trace Setting.

Set the trace function again before using it after the information was succeeded. When overwriting Version 1.xx by Version 2.xx in the above cases, please redefine the setting by the SetupPOS Utility after installing.

If both Version 1.xx and Version 2.xx exist at the same time and "Trace All" is selected for each version, only trace of either one of the versions can be taken.

- When the reboot setting after the Silent Installation is being specified, the system reboots automatically after the installation. The data being edited should be saved or terminated before the installation.
- When installation of parallel driver with this installer is succeeded, please make sure to reboot the system. The rebooting enables the use of recommended parallel driver.

Please note that if the system is not rebooted, the operation is not guaranteed at all.

- Upon completion of the installation, a Tmusb folder that stores the installer of EPSON TM USB Driver and a Lilac folder that stores the installer of recommended parallel driver are created at the target folder.
Even if these drivers are not installed with this installer, they can be installed by execution of their installers later on.
- Certain Epson software packages are incompatible with this version. A warning message will appear when the installer is launched if one of these versions is already installed. Choose to abort or continue the installation according to the message. If you choose to continue, be sure to upgrade the affected software after installation is complete.

Section 4. Uninstallation

The provided Uninstaller must be used to uninstall an installed EPSON OPOS ADK program. The following methods are available for uninstallation.

- Choose the "Add/Remove Programs" (on Windows 2000/XP) or the "Uninstall a program" (on Windows Vista or newer) in the Control Panel, and then select the "EPSON OPOS ADK Version 2.xx".

When uninstalling, first make sure that all running applications are closed; both those used by OPOS as well as any other applications.

Only privileged users can use install programs. Please consult a user who has privilege for the machine.

4.1. Uninstallation Procedures

Perform uninstallation using the methods prescribed.

- (1) The "Information..." dialog box is displayed.
This dialog box displays the version(s) compatible with the Uninstaller. Confirm that the version to uninstall using the Uninstaller is listed. To continue, select Yes. To cancel the operation here, select No.
- (2) The "ADK Selection" dialog box is displayed.
All the currently installed EPSON OPOS ADK versions are displayed. Select the version to uninstall. Multiple versions cannot be selected. If you check the Complete uninstall of selected OPOS ADK check box, full uninstallation is performed. If the check mark is removed, it is possible to select the device(s) to be removed. When the selection has been made, select the [Next] button. If checked, go to step (4).
To return to the previous dialog box, select the [Back] button.
- (3) The "Selection of Devices to Delete" dialog box appears.
The "Selectable devices" window shows the currently installed devices. Select a device to be deleted, and then press the [Remove] button. The device is moved to the "Devices to delete" window. To return to the "Selectable devices" window, press the [Add] button. Move devices from the "Selectable devices" window to "Devices to delete" the window. To move all the devices shown in the "Selectable devices" window to the "Devices to delete" window, press the [All] button. All the devices are

moved. When the selection is done, select the [Next] button.

To return to the previous dialog box, select the [Back] button.

(4) The “Confirm Deletion” dialog box appears.

Information related to the Uninstallation is displayed. Confirm the contents.

Uninstallation starts when the [Start] button is selected. To return to the previous dialog box without starting deleting, select the [Back] button if you do not want to start deleting.

- The Uninstallation process starts. A progress bar indicates the status of the Uninstallation.
- A message indicating completion is displayed on the progress bar when the Uninstallation is completed.

4.2. Types of Uninstallation

There are two types of Uninstallation: full and custom. The types of the Uninstallation can be selected by “Completely remove the selected ADK configuration” check box in the “Select ADK Version” dialog box.

- Full uninstall

Deletes all the devices registered in the selected version.

- Custom uninstall

Select the devices to be uninstalled from all the devices registered in the selected the devices using the “Selection of Devices to Delete” dialog box.

The device can be specified by device classes.

4.3. Silent Uninstallation

It is possible to perform uninstallation without using the user interface. This is called Silent Uninstallation. When Silent Uninstallation is used, only full uninstallation is performed. Specify in the command line as follows:

The way to specify depends on the installing version.

- Silent Uninstallation of Version 1.xx

Example: “FullPath\Uninstaller.exe” /s”Version”

- Silent Uninstallation of from Version 2.00 to 2.50 (including service pack version)

Example: “FullPath\Uninstaller.exe” /s

When only /s is specified as the parameter, the currently installed program is uninstalled. This parameter can only be used when only one version is currently installed. If multiple versions are installed, an error message is displayed. When multiple versions are installed, specify as follows.

- Silent Uninstallation of from Version 2.60 to 2.66 (including service pack version)

<Procedure>

- (1) Make a recording file.

Make a file for recording the operation of uninstall. To uninstall the EPSON OPOS ADK, perform the following commands.

(Example) "Full Path\Setup.exe" /r

When the uninstallation finished, the uninstall operation is recorded to the "Setup.iss" file that is made in the Windows folder.

- (2) Using the (1) recording file file for Silent Uninstall. Copy the (1) recording file to the same folder as the "Setup.exe".

(Example) "Full Path\Setup.exe" /s

To specify the path of the recording file, use the "/f1" option.

(Example) "Full Path\Setup.exe" /s /f1"c:\Temp\Setup.iss"

- Silent Uninstallation of Version 2.67 or newer

<Procedure>

- (1) Make a recording file.

Make a file for recording the operation of uninstall. To uninstall the EPSON OPOS ADK, perform the following commands.

(Example) "Full Path\Setup.exe" /r /removeonly

When the uninstallation finished, the uninstall operation is recorded to the "Setup.iss" file that is made in the Windows folder.

- (2) Using the (1) recording file file for Silent Uninstall. Copy the (1) recording file to the same folder as the "Setup.exe".

(Example) "Full Path\Setup.exe" /s

To specify the path of the recording file, use the "/f1" option.

(Example) "Full Path\Setup.exe" /s /f1"c:\Temp\Setup.iss"

The functions are added to Silent Installation by adding the following parameters to command lines.

The following parameters can be used at the same time. If you wish to use them at once, input the parameter you want to use after the parameter added to the command line.

Parameters do not distinguish capital and small letters. It is not necessary specifying only with small letters as the following examples show.

<Indicating the Silent Uninstallation>

Adding Parameter: /s

Example: "FullPath\Setup.exe" /s

<When the Silent Uninstallation finished, not to display the dialog box>

Adding Parameter: /m

Example: "FullPath\Setup.exe" /s /z"/m"

To specify the "/m" option, use the "/z" option.

<Not to delete the file that is made and used by ElectronicJournal>

Adding Parameter: /e

Example: "FullPath\Setup.exe" /s /z"/e"

To specify the "/e" option, use the "/z" option.

< When the Silent Uninstallation finished, not to display the dialog box, and not to delete the file that is made and used by ElectronicJournal>

Adding Parameter: /m, /e

Example: "FullPath\Setup.exe" /s /z"/m /e"

To specify the "/m" option and the "/e" option, use the "/z" option.

<Forced uninstall>

In some cases it may not be possible to reinstall the EPSON OPOS ADK Driver. The forced uninstall option can be used to resolve this issue.

Adding Parameter: /z"uninstall"

Example: "FullPath\setup.exe" /z"uninstall"

When this is complete, run the installer again without this parameter to reinstall the EPSON OPOS ADK Driver.

Please note that the parameter "/s" is not added.

4.4. Messages for Uninstallation

The Uninstaller program shows user messages when necessary after the

execution of dialog boxes or making of settings. Follow the instructions in them. Conditions in which messages appear and the messages themselves are listed below.

Condition	Error Message
A user who does not have Administrator authority is attempting to perform Uninstallation.	Administrator Access Required. Execute this program from an account with administrative access.
The Uninstaller is started while OPOS is not installed.	OPOS is not installed.
The Uninstaller of a version in a different language was executed.	The target language of OPOS is not installed.
The Uninstaller does not have the required information.	Internal Error Occurred. Premature program was preliminarily terminated. Some component may not have been deleted. Please manually delete remaining components.
The registry contains illegal information.	Unable to open registry. Program terminating.
The required information was not found in the registry.	Unable to get required information from registry. Program terminating.
Other files have been copied to the install directory.	Unable to remove directory. Remove directory manually.
An illegal value has been entered on the command line.	Invalid Parameter. Required format is: Uninstaller /s"X.xx"
The version that is not installed is specified as the command line parameter.	Version not installed or invalid command parameter.
The file to be uninstalled is being used by another application.	X.xx is being used by other application. To delete this file, exit the application using the file and push the "Retry" button. To leave this file, push the "Cancel" button.
Only the /S parameter is specified on the command line when multiple versions of OPOS are installed.	There is more than one installed version of OPOS. Use the command line option Uninstaller /s"X.xx" to uninstall a specific version or execute the user interface version.
The forced uninstall option has been used to resolve the issue.	Are you sure you want to completely remove the EPSON OPOS ADK and all of its features?

4.5. Precautions for Uninstallation

- Uninstallation is not possible when an EPSON OPOS ADK of different version or language (US version, Japanese version) is installed.
- The Uninstallation is executed on the condition that OPOS-related files are not in use. If OPOS-related files are in use, a message that files cannot be deleted is displayed. When this message is displayed, close the application that is using the OPOS-related files. Then try again to continue the Uninstallation process. If the Uninstallation is not retried, files alone will remain in the directory.

- Close Explorer for executing Uninstaller.
- This Uninstaller does not uninstall the EPSON TM USB Driver and the recommended parallel driver that are provided by the installer. However, the system operation is not affected by those drivers.
- If the parameter "/m" is specified for the Silent Installer, it operates in Silent mode. For this reason, after the Silent Uninstaller is finished, do not execute the installer.

Section 5. SetupPOS Utility

SetupPOS is a setup utility for registering, modifying, or deleting devices and port environments of EPSON OPOS ADK to the system registry.

Depending on the permissions of the user, the user may only be able to view the settings but does not have the ability to register or modify the settings without a few exceptions.

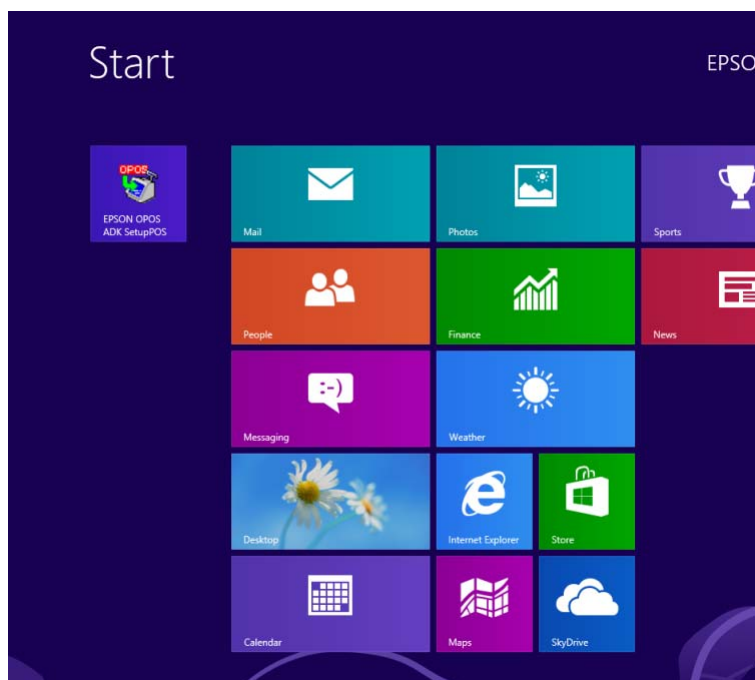
5.1. Starting and Exiting

5.1.1. Starting SetupPOS

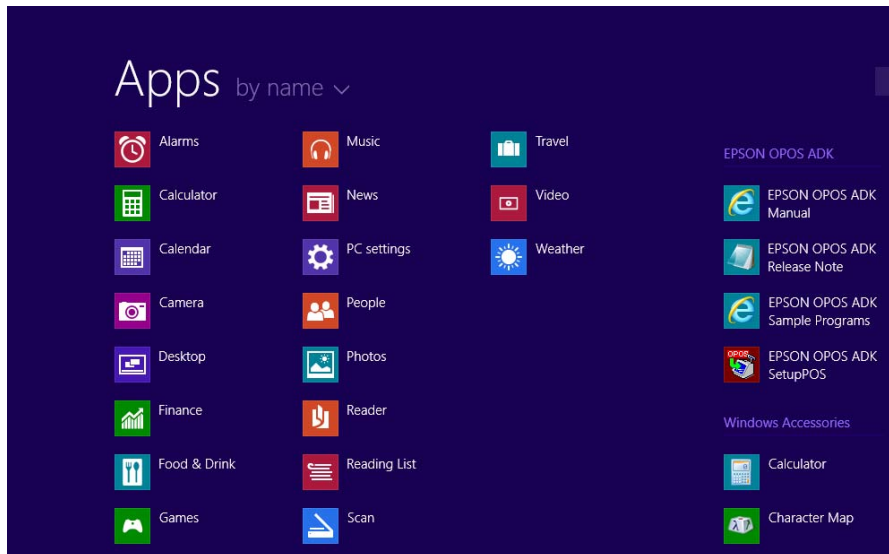
At the end of the installation, continue and use this utility.

Or, after installation, click the [Start] button, point to Programs, and then select [OPOS]. Finally, select [SetupPOS Ver2.00] to start the utility.

* In the case of Windows 8, start from the shortcut on the Start screen.



* In the case of Windows 8.1, start from the shortcut on the Start screen.



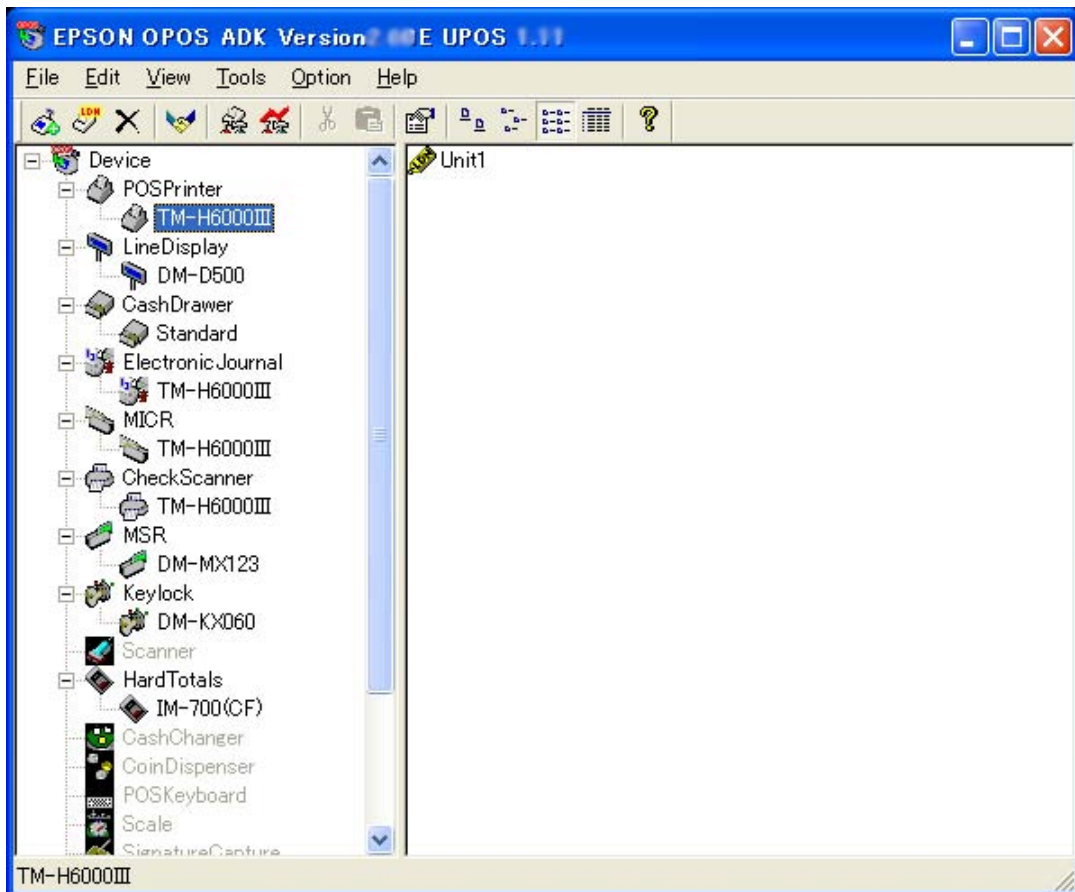
5.1.2. Exiting SetupPOS

Use one of the following three methods to exit the program.

1. In the SetupPOS main window, select Exit from the File menu.
2. Click the [Close] button in the upper right corner of the window.
3. Click the  mark in the upper left corner of the window. From the shortcut menu, select Close.

5.2. About the Basic Operations (MAIN WINDOW)

When SetupPOS runs, its main window appears as shown below. In this window, it is possible to add or delete devices, and make the settings by the explorer-like operations. This window consists of a menu bar, tool bars, and device settings panes.



The following sections describe the various components of the SetupPOS main window.

5.2.1. Menu

1. File

Use the File menu to display properties and exit SetupPOS.

i. Property

Use this menu command to display properties of the item selected in the

left or right pane. The properties can only be viewed but not modified. Depending on the item selected, the properties that can be displayed are as follows:

Device class/device class name

This gives information about EPSON OPOS ADK, CO, or version of SetupPOS.

Device name key

This gives information about the device's detail, communication settings^{*1}, trace settings, message handling, sleep time settings and DeviceStatistics settings.

^{*1} It becomes key hook driver in case of the MSR and Keylock.

Port connection

This gives information about the range of port.

Serial port (COM) or parallel port (LPT)

This gives information about the range of port.

USB/KeyBoard/Memory/Network/Wireless/Bluetooth/(None)

No property is displayed.

ii. Exit

This menu command exits SetupPOS.

2. Edit

This menu contains commands for setting and editing devices.

i. Cut

This menu command prepares a selected device for moving, and is valid only when "Port Connection View" is selected. Use this command, for example, when moving a device to another port. Usually, this command is used with the Paste command. Cutting and pasting the other device, or drag-and-drop of the other device cancel the Cut command to the device which has been prepared for moving (or cut). After cutting the device, using the Paste command appropriately moves a port.

Example: Moving device from COM1 to COM2, from USB2 to USB1, or from LTP2 to "None".

This operation becomes disabled if at "Port Connection View" the device

name key is not selected.

To cut a device

Select the device name key and select Cut from the Edit menu, in tool bar, or from a pop-up menu, which is brought up by right-click.

The user must have the administrative permissions in order to use this function.

ii. Copy

This menu command copies character strings. Note that a device cannot be copied.

To copy character strings

Select the character strings, then select Copy from the Edit menu.

iii. Paste

This menu command pastes character strings that has been copied earlier, or moves a device that has been cut earlier to a valid port.

Note that pasting a device (which has been cut earlier) backs to its original port or to another type of port (for example from COM1 to LPT3) is not allowed. Also, this command does not work if nothing has been cut or copied earlier. This command is valid only when "Port Connection View" is selected.

To paste a device that has been cut earlier

After copying the device, the [Paste] icon becomes enabled as well.

Select a valid port, and then select Paste from the Edit menu. The user must have the administrative permissions in order to use this function.

iv. Select All

Use this menu command to select all items displayed in the right pane.

v. Invert Selection

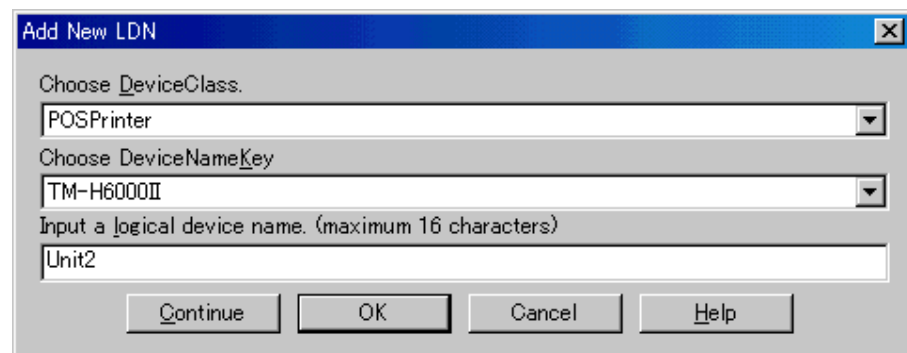
Use this menu command to invert the state of the selection of all items displayed in the right pane.

vi. Add New Device

Use this menu command to display the Add New Device wizard and add a new device to the device class. The device class displayed here refers to devices prepared by the install program. For details, see the Section 5. "5.3.1. Using the Add New Device wizard" of this manual. The user must have the administrative permissions in order to use this function.

vii. Add New LDN

Use this menu command to display the "Add New LDN" dialog box and add a logical device name to a device.



A logical device name that has already been used by the same device class cannot be specified. When re-specifying a logical device name that has been used and then deleted the corresponded device, a confirmation message appears that is whether to replace the old logical device name with the new one. The user must have the administrative permissions in order to use this function.

<About the dialog box for adding logical device name>

"Choose DeviceClass." combo box:

Select a device class for adding the logical device name. If the device class has already been selected in the window, then it appears in the combo box by default; otherwise the device classes appear in standard way or by name. Also, only device classes which the control objects (CO) supporting the device is installed are listed in the combo box.

"Choose DeviceNameKey" combo box:

Select a device name key for adding the logical device name. If the device name key has already been selected in the window, then it appears in the combo box by default; otherwise the device names appear by ascending order. If no device exists for the device class, then

the combo box becomes empty.

"Input a logical device name. (maximum 16 characters)" edit field

Enter a logical device name to be added. The name can be up to 16 characters, but it cannot contain the "\" character. If the logical device name is invalid or is already registered, then the Continue or the [OK] button becomes disabled.

[Continue] button

Use this button to set up the logical device name and then continue to add more logical device names.

[OK] button

Use this button to set up the logical device name.

[Cancel] button

Use this button to cancel the setting and close the dialog box.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

Adding a logical device name

In the window, select a device name key for adding the logical device. From the Edit menu, select "Add New LDN" and then enter the logical device name. Another method is to select a device name key for adding the logical device and right-click it. From the pop-up menu, select "Add New LDN" and then enter the logical device name.

viii. Delete

Use this menu command to delete a device selected in the current window. Before deleting, a confirmation message box appears; when clicking the [Yes] button, the device is deleted.

When deleting a master device of Hydra connection, a confirmation message box appears that is whether to delete the slave devices as well. It is possible to select to delete the master device only or all including the slave devices.

Delete of ElectronicJournal device

If an Electronic Journal device is deleted, a data file confirmation message is displayed. If "Yes" is selected, the Electronic Journal data files are deleted. If "No" is selected and the data files are not deleted, if

is possible that the files will not be able to be deleted using EPSON OPOS ADK.

If no device name key or logical device name is selected, then this operation becomes disabled. The user must have the administrative permissions in order to use this function.

To delete a device, use one of the following methods:

- 1) In the main window, select a device name key or the LDN to be deleted. From the Edit menu, select Delete.
Or, in the main window, select a device name key or the LDN to be deleted. Right-click to bring up a pop-up menu and select Delete.
- 2) In the main window, select a device or the LDN to be deleted, and then press the Delete key.
- 3) In the main window, select a device or the LDN to be deleted, and then click the [Delete] icon in the tool bar.

ix. Rename LDN

This menu command renames the logical device currently selected in the window. The name can be up to 16 characters, but it cannot contain the "\" character. Also, the same logical device name cannot be used again in the same device class.

A confirmation message box appears before the name change takes effect. If no logical device name is selected, then this operation becomes disabled. The user must have the administrative permissions in order to use this function.

To rename a logical device:

While in the main window, select a device name. From the Edit menu, select "Rename LDN". The name is ready for editing now. Change the device's name and then press the Enter key. When a confirmation message appears, click the [Yes] button and the logical device name is changed.

x. Communication Settings

Use this menu command to display the "Communication Settings" dialog

box and change the communication settings of the selected device. This command works only on the following devices: COM, LPT, USB, Ethernet, Wireless(IEEE802.11), or devices using the keyboard interface (such as Printer, LineDisplay, CashDrawer, MSR, Keylock, MICR, CheckScanner, ElectronicJournal).

Communication Settings of slave devices with Hydra connection cannot be modified. Also, Communication Settings of device with Hydra connection can only be performed from the master device. When the master device is changed, settings of all of its slave devices are changed as well.

The user must have the administrative permissions in order to use this function.

<Communication Settings dialog box>

COM/LPT/USB/Network/Wireless interface:

The screenshot shows the 'Communication Settings' dialog box. It has a title bar with a close button. The main area is divided into several sections. The first section is 'USB Port Setting', which includes a dropdown menu currently set to 'USB1' and an 'Auto Setting' button. To the right of this is a label 'Port is being used by' followed by a dropdown menu set to '(None)'. Below this is a section titled 'Setup of details of the port.' which contains several fields: 'BaudRate', 'BitLength', 'Parity', 'StopBit', and 'Handshake', each with a dropdown menu. To the right of these are 'InputBuf', 'OutputBuf', 'Output Interval Time' (set to 2500 msec), and 'Input SleepTime' (set to msec). Below this is a section 'Master device to:' with a dropdown menu set to '(None)'. At the bottom of the dialog are two buttons: 'CheckHealth Interactive' and 'Device Specific Settings'. At the very bottom are three buttons: 'OK', 'Cancel', and 'Help'.

Port settings

"Port number" combo box

Use this combo box to display the port name of the selected device, as follows:

- For serial port: "COM Port settings" combo box
- For parallel port: "LPT Port settings" combo box
- For USB port: "USB port settings" combo box
- For network: "Network Setting" edit box
- For wireless: "Network Setting" edit box

Use this combo box to select a serial, parallel, or USB port number. For network and wireless, set the IP address and the host name. The network settings set the input value by pushing the [Update] button. When selecting "None", the registry's port information "(Port)" is deleted. The setting ranges for serial, parallel port number can modify in the "Port Range Setting" dialog box. For USB port, the ports, which can be connected by the USB driver when the "Communication Settings" dialog box appears, become the selectable range. If the device, which has

been set is on the port to be set, depending on the capability of the device and the selected device, a message not to set the device to the port appears. In this case, the port setting is invalid. Please select other available port.

[Auto Setting] button

This command becomes disabled if a non-USB port is selected. When clicking this button, the "USB Auto Setting" wizard is displayed and it is possible to retrieve a USB port number automatically. For details, refer to the Section 5. "5.3.2. Using the USB Auto Setting wizard" of this manual.

[Update] button

Use this button to update the IP address or host name. The [update] button works only when the IP address or host name entered into the edit field is different from the existing one. If a device already exists at the IP address or host name entered into the edit field, then depending on the device or the capability of the selected device, a message appears to indicate that the IP address or host name cannot be used. When continuing and adding the device, the IP address or host name of that device becomes invalid. Please select other IP address or host name.

"Port is being used by" combo box

This combo box contains a list of devices already set up for the port number selected.

Setup of Details of the port

The settings from "BaudRate" to "Handshake" apply to the serial port. Change these settings to match the printer's settings. If the detailed settings of a master device in Hydra connection are changed, then the settings of its slave devices are also changed automatically. Also, as mentioned above, the detailed settings of the slave devices cannot be changed directly.

"BaudRate" combo box

Use this combo box to display or set the communication speed of the serial port. The values that can be used depend on the device information file (INF file). For information on the communication speed,

refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the speed is displayed but cannot be modified.

"BitLength" combo box

Use this combo box to display or set the data bit length of the serial port. The values that can be used depend on the device information file (INF file). For information on data bit length of the device, refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the value is displayed but cannot be modified.

"Parity" combo box

Use this combo box to display or set the parity bit length of the serial port. The values that can be used depend on the device information file (INF file). For information on parity bit of the device, refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the value is displayed but cannot be modified.

"StopBit" combo box

Use this combo box to display or set the stop bit of the serial port. The values that can be used depend on the device information file (INF file). For information on stop bit of the device, refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the value is displayed but cannot be modified.

"Handshake" combo box

Use this combo box to display or set the handshake of the serial port. The values that can be used depend on the device information file (INF file). For information on handshake of the device, refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, or "None", then this setting cannot be set or changed. For a slave device in Hydra

connection, the value is displayed but cannot be modified.

"InputBuf" edit field

This edit field supports only Version 1.xx devices.

Use this edit field to display or set the input buffer size of the serial port.

The range of acceptable values is 32 to 8192.

If the selected port is parallel, USB, network, wireless port, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the information is displayed but cannot be modified. Also, the default value has been optimized to allow for normal operation.

"OutputBuf" edit field

Use this edit field to display or set the size of the output buffer size of the serial, parallel, network, or wireless port. The range of acceptable values is usually 32 to 1024. For some devices, the range is modified by settings of the device information file (INF).

If the selected port is USB or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the value is displayed but cannot be modified. In addition, the default value has been optimized to allow for normal operation.

"Output Interval Time" edit field

Use this edit field for set the retry time to use when output fails. For details of output interval time, please refer to the ["5.8 Glossary"](#) of this manual. The range of acceptable values is 0 to 9999. This setting is valid for COM, LPT, USB, network, and wireless port.

If the selected port is "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the value is displayed but cannot be modified. Also, the default value has been optimized to allow for normal operation.

"Input SleepTime" edit field

This edit field supports only Version 1.xx devices.

Use this edit field to display or set the sleep time at the location where API is processed, during input from the port. The range of acceptable values is 0 to 99.

If the selected port is USB or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the value is displayed but cannot be modified. Also, the default value has been optimized to allow for normal operation.

"Master device of the following device(Master device to:)" edit field

If the selected device is a master device connected to Hydra, then the slave device is displayed; if the device is slave, then the master device is displayed.

[CheckHealth Interactive] button

Use this button to execute the CheckHealth method using Level = OPOS_CH_INTERACTIVE for the device. If the port setting is "None", then this button is disabled. For details, refer to the manual of the device.

[Device Specific Settings] button

Use this button to display the "Device Specific Settings" dialog box and change the device specific settings for the selected device. If the selected device does not support device specific settings, then this button is disabled. For details, refer to the manual of the device.

[OK] button

Use this button to close the dialog box and save the device's communication settings to the registry.

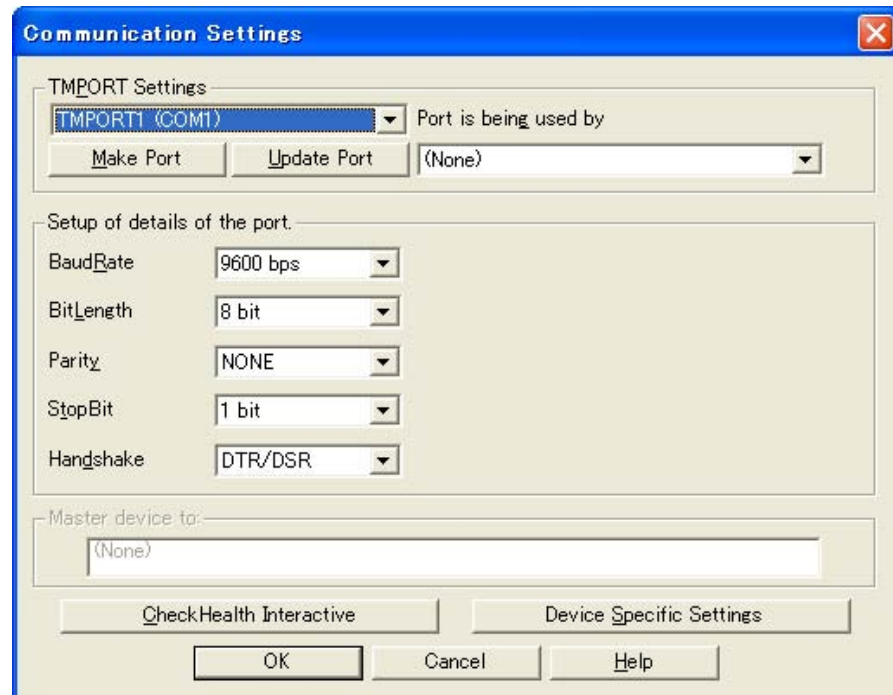
[Cancel] button

Use this button to cancel the changes and close the dialog box. Any changes in the settings will not be saved to the registry.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

TMPORT interface:



The Communication Settings dialog box enables to modify the TMPORT settings.

"TMPORT Settings" combo box

Selects a port of the TMPORT for connect.

[Make Port] button

Displays the [TMPORT making] dialog box.

[Update Port] button

The connectable port is updated.

"Port is being used by" combo box

This combo box contains a list of devices already set up for the port number selected.

Setup of Details of the port

The settings from "BaudRate" to "Handshake" apply to the serial port. Change these settings to match the printer's settings. If the detailed settings of a master device in Hydra connection are changed, then the settings of its slave devices are also changed automatically. Also, as mentioned above, the detailed settings of the slave devices cannot be changed directly.

"BaudRate" combo box

Use this combo box to display or set the communication speed of the serial port. The values that can be used depend on the device information file (INF file). For information on the communication speed, refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, Bluetooth, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the speed is displayed but cannot be modified.

"BitLength" combo box

Use this combo box to display or set the data bit length of the serial port. The values that can be used depend on the device information file (INF file). For information on data bit length of the device, refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, Bluetooth, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the value is displayed but cannot be modified.

"Parity" combo box

Use this combo box to display or set the parity bit length of the serial port. The values that can be used depend on the device information file (INF file). For information on parity bit of the device, refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, Bluetooth, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the value is displayed but cannot be modified.

"StopBit" combo box

Use this combo box to display or set the stop bit of the serial port. The values that can be used depend on the device information file (INF file). For information on stop bit of the device, refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, Bluetooth, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the value is displayed but cannot be modified.

"Handshake" combo box

Use this combo box to display or set the handshake of the serial port. The values that can be used depend on the device information file (INF file). For information on handshake of the device, refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, Bluetooth, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the value is displayed but cannot be modified.

"Master device of the following device(Master device to:)" edit field

If the selected device is a master device connected to Hydra, then the slave device is displayed; if the device is slave, then the master device is displayed.

[CheckHealth Interactive] button

Use this button to execute the CheckHealth method using Level = OPOS_CH_INTERACTIVE for the device. If the port setting is "None", then this button is disabled. For details, refer to the manual of the device.

[Device Specific Settings] button

Use this button to display the "Device Specific Settings" dialog box and change the device specific settings for the selected device. If the selected device does not support device specific settings, then this button is disabled. For details, refer to the manual of the device.

[OK] button

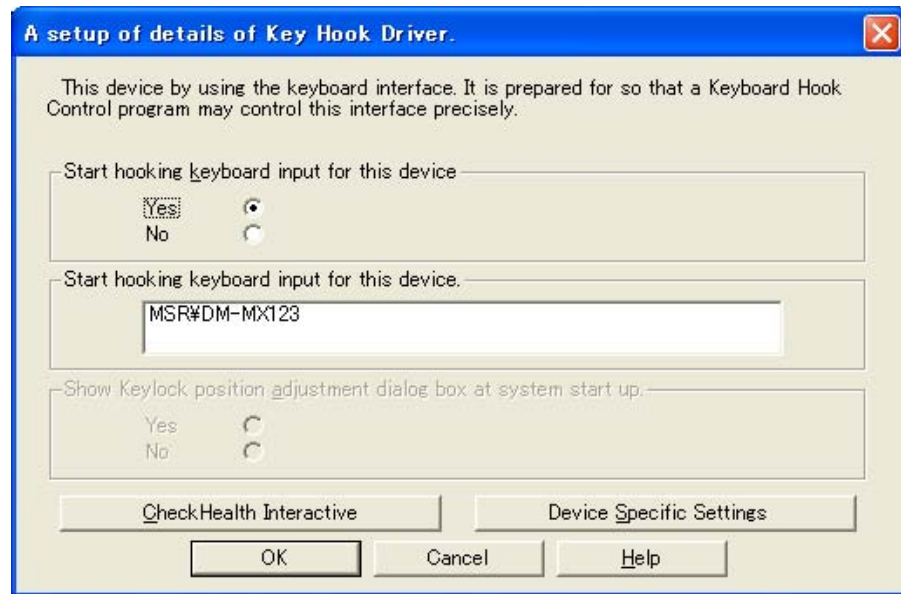
Use this button to close the dialog box and save the device's communication settings to the registry.

[Cancel] button

Use this button to cancel the changes and close the dialog box. Any changes in the settings will not be saved to the registry.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

Keyboard interface:

Use this screen to set the details of Keyboard Hook Control.

"Start hooking keyboard input for this device" check button

If the [Yes] button is selected, then the EPSON Keyboard Hook Control utility (hereafter referred to as "Hook Control") is placed in the Windows Startup group and when the system runs, monitoring of keyboard input to this device also starts.

If the [No] button is selected, the hook control utility is not always loaded or placed in the Windows Startup group.

In this case, however, it is possible to use this device in the OPOS application.

If an input event occurs at this device then the OPOS control is not opened yet, then unanticipated keyboard input may result.

To avoid this problem, it is necessary to register the hook control utility as the target of this device.

If no keyboard-monitoring driver is used, then these buttons become disabled.

Note that when version 1.xx and version 2.xx are installed simultaneously, it is impossible to use MSR and Keylock devices of these versions at a time. Use the combination of MSR and Keylock of either version (only version 1.xx or only version 2.xx). MSR and Keylock use the keyboard interface and keyboard monitoring driver

(EPSONHKO). However, this driver differs from version 1.xx and version 2.xx, so it is impossible to use MSR and Keylock devices of both versions at a time.

"Start hooking keyboard input for this device" list box

This list box displays the list of devices which are monitored by the Hook Control.

If the setting of the "Start hooking keyboard input for this device" check buttons is changed, the setting is valid.

"Show Keylock position adjustment dialog box at system start up" check buttons

This setting is available only when selecting the Keylock.

If the [Yes] button is selected, when the system runs, a dialog box appears and prompts for change of the position.

Then follow the instructions and modify the position so that the key position can be correctly adjusted. Usually, select the [Yes] button.

If the [No] button is selected, then the dialog box will not appear.

However, when starting the system, the Keylock position must at least be modified once. To know the initial state of the Keylock correctly, select the [Yes] button.

[CheckHealth Interactive] button

Use this button to call the CheckHealth method for the selected device using Level = OPOS_CH_INTERACTIVE.

For details, see the manual of each device.

[Device Specific Settings] button

Use this button to display the "Device specific Setting" dialog box and change the device specific settings for the selected device.

If the selected device does not support device specific settings, then this button becomes disabled. For details, see the manual of each device.

[OK] button

Use this button to close the dialog box and save the setting information to the registry.

[Cancel] button

Use this button to cancel the settings and close the dialog box.

The modifying setting does not be set to the registry.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

3. View

Use this menu to control display of the main window.

i. Tool Bar

Use this menu command to hide or display the tool bar.

If the command comes with a check mark : the tool bar is displayed.

If the command comes with no check mark : the tool bar is hidden.

Once changed, the setting remains effective even when restarting the system. As the default, the tool bar is displayed.

What is the tool bar?

The tool bar appears below the menu bar of the main window and contains a list of frequently used menu commands. Clicking a button easily run a menu command associated with the button.

ii. Status Bar

Use this menu command to hide or display the status bar.

If the command comes with a check mark: the status bar is displayed.

If the command comes with no check mark: the status bar is hidden.

Once changed, the setting remains effective even when restarting the system. As the default, the status bar is displayed.

What is the status bar?

The status bar displays various information of the main window. For example, when a menu bar item is displayed, its description appears on the status bar.

iii. Large Icon

Use this menu command to display items in the right pane in large icons.

It is possible to display information using one of the following menu commands: [Large Icon], [Small Icon], [List], or [Details]. When selecting

[Large Icon], a check mark appears together with the command. Once selected, the setting remains effective even when restarting the system.

iv. Small Icon

Use this menu command to display items in the right pane in small icons. It is possible to display information using one of the following menu commands: [Large Icon], [Small Icon], [List], or [Details]. When selecting [Small Icon], a check mark appears together with the command. Once selected, the setting remains effective even when restarting the system.

v. List

Use this menu command to display items in the right pane as a list. It is possible to display information using one of the following menu commands: [Large Icon], [Small Icon], [List], or [Details]. When selecting [List], a check mark appears together with the command. Once selected, the setting remains effective even when restarting the system. As the default, items in the right pane are displayed as a list.

vi. Details

Use this menu command to display items in the right pane as a detailed list. It is possible to display information using one of the following menu commands: [Large Icon], [Small Icon], [List], or [Details]. When selecting [Details], a check mark appears together with the command. Once selected, the setting remains effective even when restarting the system. In the Details view, "Device Description" and "Connection condition" are displayed. For "Connection condition", it can be Stand alone, Hydra, master, or slave. Once set, the setting remains effective even when restarting the system.

vii. Device Class View

This view displays devices in the left pane for each device class. "Device Class View" and "Port Connection View" form a group. When selecting "Device Class View", a check mark appears together with the command. Once set, the setting remains effective even when restarting the system. As the default, "Device Class View" is selected.

viii. Port Connection View

This view displays devices in the left pane for each port. "Device Class View" and "Port Connection View" form a group. When selecting "Port Connection View", a check mark appears together with the command. Once set, the setting remains effective even when restarting the system.

ix. Device Class Ordering

Use this menu command to change the sort order of the device classes in "Device Class View", either "by Standard" or "by Name". If "by Standard" is selected, then device classes supported by EPSON have the priority in the display. If "by Name" is selected, then device classes are sorted alphabetically. Once set, the setting remains effective even when restarting the system.

What is "by Standard"?

With this view, device classes are displayed in the following order.

1. POSPrinter
2. LineDisplay
3. CashDrawer
4. ElectronicJournal
5. MICR
6. CheckScanner
7. MSR
8. Keylock
9. Scanner
10. HardTotals
11. CashChanger
12. CoinDispenser
13. POSKeyboard
14. Scale
15. SignatureCapture
16. ToneIndicator
17. BumpBar
18. FiscalPrinter
19. PINPad
20. RemoteOrderDisplay

What is "by Name"?

With this view, device classes are displayed in the alphabetical order.

x. All View

Use this menu command to expand the view of the left pane. After selecting "All View", all device information is not always displayed, but right after "All View" is selected. Using this menu command enables to check the details of the device currently set.

4. Tools

Use this menu to display items for supplementary tools of devices.

i. CheckHealth Interactive

This menu command calls the CheckHealth method using Level = OPOS_CH_INTERACTIVE for the device (key of the device name) currently selected in the window. If the call succeeds, the result is displayed in CheckHealthText. If error occurs during the call, then an error API is called, and ResultCode and ResultCodeExtended are displayed.

This menu command becomes disabled if no device name key is selected in the window.

What is "CheckHealth Interactive (OPOS_CH_INTERACTIVE)"?

CheckHealth Interactive executes an interactive test with device. The supported service object usually provides a dialog box to display the test options as well as test results.

ii. Device Specific Settings

Use this menu command to perform device specific settings for the device currently selected in the window. If no device name key is selected in the window, then this operation becomes disabled. Also, if the selected device does not support device specific settings, then this operation becomes disabled.

The user must have the administrative permissions in order to use this function.

iii. Validate

Use this menu command to validate the registry settings. When a check mark appears on the menu command, the validation is carried out. When the validation is valid, after device settings are changed, the OPOS registry settings is checked if they are correctly set. If conflict exists in the registry settings, a "Warning" dialog box appears to indicate the conflict. We recommend turning on this function (that is, command with the check mark).

If automatic validation is not required, turn off the command (check mark cleared). However, in this case the user must take the responsibility to make sure the OPOS registry settings are correctly adjusted. After turning off the function, if the function is turned on again, registry validation starts from that moment. Once set, the setting remains effective even when restarting the system. For details on this function, see the description of "Warning" dialog boxes.

iv. Export Registry

Use this menu command to display the "Export Registry" dialog box and export the current OPOS registry settings to a registry data file. The saved file can then be used to recover, copy, or install the OPOS environment.

In the "Save As" dialog box, specify a file name to export the registry. If the file already exists, then a message box appears to confirm to overwrite the existing file. Select the [Yes] button to overwrite, or the [No] button to append. If the existing file is overwritten, then all existing registry data is deleted; if the existing file is appended, then the new settings are appended to the end of the file. It is possible to specify the extent of the export from the following options: Register All, Selecting Object, and Hydra.

<Export registry dialog box>

[All] button

Use this button to export all OPOS registry settings to a registry data file. All settings below the ADK configuration key of the registry are always saved.

[Selecting] button

Use this button to export only the OPOS registry settings for the item

currently selected in the window, to a registry data file. If multiple items are selected, then multiple registry settings are exported. This button becomes disabled if no item with registry settings is selected in the window.

[Hydra] button

This button becomes enabled if the device currently selected in the window is a Hydra device, and then the Hydra-related settings of the device are exported. This button becomes disabled if the device currently selected in the window is not a Hydra device.

[Cancel] button

Use this button to cancel the changes and close the dialog box. Any changes in the settings will not be saved to the registry.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

5. Option

This menu contains commands for maintenance.

i. Confirm the Hydra Settings closely

When a check mark appears, this menu command is turned on and a stringent check will be carried out to confirm the validity of the registry in version 1.xx. If the command is turned off, then only the minimum check will be carried out. With this command, it is possible to add devices of the same device class to the same port, or create multiple Hydra configurations.

The user must have the administrative permissions in order to use this function.

ii. Port range setting

Use this menu command to display the "Port Range Settings" dialog box and set the maximum numbers of the serial or parallel ports according to the hardware configuration. Usually, as long as the port configuration is not changed, it is not necessary to use this dialog box. The default range depends on the PC, but the acceptable range are as follows:

COM port : 10

LPT port : 3

The user must have the administrative permissions in order to use this function.

<Port range dialog box>

"Maximum COM ports" edit field with spin buttons

Set the largest number of serial (COM) ports that can be used. The range that can be chosen is 1 to 10. However, it is impossible to select a number which is smaller than the maximum number of COM ports currently in use.

"System default" button

Use this button to set the suitable value for the current system as the largest number of serial ports (COM). Before using this function, use the control panel of the OS to make sure the system has been configured correctly.

"Maximum LPT ports" edit field

Set the largest number of parallel (LPT) ports that can be used. The range that can be chosen is 1 to 3. However, it is impossible to select a number which is smaller than the maximum number of LPT ports currently in use.

[System default] button

Use this button to set the suitable value for the current system as the largest number of serial ports (LPT). Before using this function, use the control panel of the OS to make sure the system has been configured correctly.

[OK] button

Close the dialog box and set the port range to the registry.

[Cancel] button

Use this button to cancel the setting and close the dialog box.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

<Port property dialog box>

"Port Name" edit field (read only)

Displays the name of port.

"Port Type" edit field (read only)

Displays the type of port.

The strings that are displayed are as follows:

COMn (n= 1, 2 ...)

LPTn (n= 1, 2 ...)

USB

NET:XXX.XXX.XXX.XXX (XXX.XXX.XXX.XXX = IP address)

"Connected Device" edit field (read only)

The name of the currently or previously physically connected device is displayed.

[OK] button

Closes the dialog box.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

iii. OPOS API Trace Setting

Use this menu command to display the "OPOS API trace settings" dialog box and set the trace settings of OPOS ADK. Note that for version 1.xx, the trace function can be used only if an EPSON OPOS ADK supporting trace is installed.

The trace (log) function provides support for application development. Usually this function is used only for debugging OPOS applications; it is not used for other purposes.

The user must have the administrative permissions in order to use this function.

When EPSON CO is used, the log of all devices can be taken. When CCO is used, only the log of POSPrinter, LineDisplay, CashDrawer, MICR, CheckScanner and ElectronicJournal can be taken.

<OPOS API Trace setting dialog box>

"DeviceName" combo box

Use this combo box to select a device name for enabling the OPOS API trace function.

The device is the target to use when "Private Trace" is selected.

If a device name is currently selected in the window, the device appears as the default name in the combo box.

If the currently selected device's settings have been changed, changing the settings of another device displays a message box to confirm to save the settings. Select the [Yes] button to save the settings to the registry. Even if the [Cancel] button is selected to close the dialog box, the settings are not abandoned.

"Tracing mode" radio buttons

Select a tracing mode from the following options:

"No Trace" : Disable the OPOS API trace function.

This option is selected as the default.

"Global Trace" : Trace all OPOS devices (EPSON devices only).

"Private Trace" : Trace only the specified valid device.

Usually, select "No Trace".

When the OPOS API trace is turned on, all OPOS API calls (all property access and all method calls) are traced, with the trace result recorded in the trace log file. Note that the "Global Trace" option takes priority over the "Private Trace" option, but when the "Private Trace" option is used, the "Global Trace" settings are deleted. For example, if "Global Trace" has been set to the POS printer and the user uses the line display to set trace to "Private Trace", the POS printer will be excluded from the target of trace.

To avoid this problem, use the "Private Trace" option for the device to trace.

When CCO is registered, the device that can be traced is limited. Even when [Global Trace] is selected, only the device that can be traced is traced.

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"Use specific trace log file" check box

Select this check box when using a specific trace log file.

When selected, it is possible to change the trace log file name. When cleared, the trace log file name is automatically set.

If the name of log file is set the same name as the other log file (even error log file of POSPrinter), log information of the following device don't be recorded. To avoid this problem, please don't use the same name as the other log file.

If a special trace log file is specified, please be careful of the access privileges for both the specified file, and for the save location.

If there is an insufficient access privilege for the specified location, the trace log file will not be created.

"Notify when trace log file becomes full. " check box

In order to receive notification when the trace log file has exceeded the maximum size limit, check this checkbox.

[OK] button

Close the dialog box and save the current settings.

[Cancel] button

Use this button to cancel the setting and close the dialog box.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

iv. Message Handling

Use this menu command to display the "Message handling" dialog box and set up message handling for the registered devices.

The user must have the administrative permissions in order to use this function.

What is message handling?

Some application needs to extract a message while a method requiring much time is being processed. In this case, use this command. If "Message is processed during synchronous" is selected, then OPOS methods or properties are not executed while an OPOS method requiring much time is being processed.

(For example, during synchronization, if the Close method is executed, this may result in fatal errors such as causing the PC to hang.)

<Message handling dialog box>

"Message is not processed during synchronous" radio button

Use this radio button to disable message handling.

"Message is processed during synchronous" radio button

Use this radio button to enable message handling.

When this button is selected, a warning message box appears;

when choosing the [Yes] button, the "Message is processed during

synchronous" radio button is then selected and message handling will be performed.

[OK] button

Close the dialog box and save the current settings.

[Cancel] button

Use this button to cancel the setting and close the dialog box.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

v. Setting Sleep Time

Use this menu command to display the "Setting Sleep Time" dialog box and set the Sleep time of Win32 API.

In the programs for waiting for input from device inside the driver or long processing programs, to keep the CPU time use exclusively, EPSON OPOS ADK calls Sleep (a WIN32 API) and specifies the time to pass control to other programs.

By using this dialog box, it is possible to set the length of the Sleep time for processing long programs.

By modifying the sleep time, it is possible to change the ratio of the CPU time used by a particular device. For the sleep time of input-waiting programs, refer to the port details of the device to learn to how to change it. Because the settings have been optimized, usually it is not necessary to modify them.

The user must have the administrative permissions in order to use this function.

<Setting Sleep Time dialog box>

"DeviceName" combo box

Use this combo box to select a device for setting the sleep time.

If a device name is currently selected in the window, the device appears as the default name in the combo box.

To change the sleep time of the other device, select the device name from the list.

If the currently selected device's settings have been changed, changing the settings of another device display a message box to confirm to save the settings.

If the [Yes] button is selected, then the even if the [Cancel] button is selected to close the dialog box, the settings are not abandoned.

"Setting Individual Device" check box

To set the sleep time for each device, select this check box.

Usually, this check box is selected so that the sleep time value applies only to the device selected in the "DeviceName" combo box.

If this check box is cleared, then the sleep time value applies to all devices and the value set to each device becomes ignored.

However, the individual sleep time values are stored in the OPOS registry even if the check mark is cleared, when selecting the check box again, the individual sleep time values will be used again.

"Setting Value (msec)" edit field

Use the edit field to specify the sleep time for the Sleep API. The acceptable range is 0 to 999.

[OK] button

Close the dialog box and save the sleep time to the registry.

[Cancel] button

Use this button to cancel the setting and close the dialog box.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

vi. DeviceStatistics Settings

The "DeviceStatistics Settings" dialog box is displayed.

The device statistics information file is the file that SO uses to provide the DeviceStatistics functions.

<DeviceStatistics Settings dialog box>

"DeviceStatistics information files preservation folder." edit box

Shows the path to a folder where DeviceStatistics information is to be saved.

[OK] button

Close the dialog box.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

vii. Version Compatibility

Use this menu command to display the "Version compatibility" dialog box and ensure version compatibility in terms of the differing specifications among versions of OPOS ADK.

The user must have the administrative permissions in order to use this function.

<Version compatibility dialog box>**"Common" radio button**

Use the radio buttons to specify the sending conditions of DataEvent when AutoDisable = TRUE.

This button is available only when using EPSON OPOS ADK version 1.xx.

For "Open, Claim&Enable" (default):

Immediately after SO receives the data notified as DataEvent, DeviceEnabled is set to FALSE.

At this moment, DataEvent is not sent. Afterwards, when DeviceEnabled is set to TRUE, DataEvent is sent.

(AutoDisable UPOS specifications)

For "Open, Claim":

Immediately after SO receives the data notified as DataEvent, DeviceEnabled is set to FALSE.

At this moment, DataEvent is sent (EPSON specific specifications).

Usually, select "Open, Claim&Enable" (recommended).

To upgrade OPOS ADK from version 1.5x or earlier and enable the existing applications to run in the new version without any modification, use "Open, Claim".

The specifications for operations are different in pre-Ver2.60 and Ver2.60. Select which specification. These are settable only for the EPSON OPOS ADK of the starting from Ver2.60.

(3) When the specification starting from OPOS Ver2.60:

The specifications for the initialization, devices with hydra connection, and Error Event are operated in compliance with UPOS1.11. (UPOS Specification)

(4) When the specifications prior to OPOS Ver2.60:

The specifications for the initialization, devices with hydra connection, and Error Event are operated in compliance with the specifications Pre-Ver2.60. (EPSON Specification)

The default setting is different by the condition.

Condition	Default setting
Without the installation of EPSON OPOS ADK in advance	Ver2.60 specification
When the prior to Version 2.60 of EPSON OPOS ADK already has been installed	Pre-Ver2.60 specification

The target categories are as follows:

<Initialization processing>

POSPrinter
 LineDisplay
 CashDrawer
 (ElectronicJournal)

<Switch-over of Hydra error code>

LineDisplay
 CashDrawer

<Error Event processing>

Scanner
 MSR

"POS Printer [Prior to Ver2.0]" check box

This check box is available only when using EPSON OPOS ADK version 1.xx.

Use this check box to specify the operation of BeginInsertion/EndInsertion of slip.

If the check box is selected (default):

BeginInsertion only waits for insertion of Slip, and the printer has not yet entered into the slip selection state. When EndInsertion selects slip after the slip insertion is confirmed. The printer goes into the slip selection state and slip printing becomes possible.

If the check box is cleared:

BeginInsertion/EndInsertion of OPOS ADK version 1.5x or earlier works.

Usually, select this check box (recommended).

To upgrade OPOS ADK from version 1.5x or earlier and enable the existing applications to run in the new version without any modification, clear this check box.

"MSR" check box

Use this check box to specify the operation of the data set by TrackXData.

If the check box is selected (default):

Header and footer are not added to the data of the TrackXData properties.

If the check box is cleared:

Operation follows the old specifications. Header and footer are added to the data of the TrackXData properties.

Usually, select this check box (recommended).

To upgrade OPOS ADK from version 1.5x or earlier and enable the existing applications to run in the new version without any modification, clear this check box. For version 2.xx, because the equivalent properties are added with UPOS 1.5, this setting cannot be changed; this is possible only when version 1.xx of EPSON OPOS ADK is used.

Specify a track for storing the JIS-II data.

Set track 3 to the JIS-II data.

If the check box is cleared (default): Store JIS-II data to track 4.

If the check box is selected: Store JIS-II data to track 3.

Usually, do not select this check box.

It is possible to operate OPOS ADK in the same way as version 1.xx.

[OK] button

Close the dialog box and save the version compatibility settings to the registry.

[Cancel] button

Use this button to cancel the setting and close the dialog box.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

viii. Communication compatibility

The communication compatibility dialog box is displayed to allow enabling or disabling of compatibility between different communication types.

Only users with administrator authority can use this function.

<Communication compatibility dialog box>

"Use operations in serial connection for USB connection." **check box**
Enables/disables compatibility of USB connection and serial connection communication. If this checkbox is selected when a device is connected by USB, operation will be equivalent to a serial connection. When a device that has been used in a serial connection is changed to a USB connection, this allows you to use the device without changing the application.

Checked: Enables USB and serial compatibility.
Operation is equivalent to a serial connection.
Unchecked: Disables USB and serial compatibility.
Normal USB operation.

Default is "Unchecked".

[OK] button

Close the dialog box and save the version compatibility settings to the registry.

[Cancel] button

Use this button to cancel the setting and close the dialog box.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

ix. Change of component

Use this menu command to start the installer to register or delete EPSON OPOS ADK/CCO.

Change EPSON OPOS ADK:

Start the installer (SetupPOS.exe).

After the installer starts, SetupPOS ends.

Change CCO:

Start the CCO Installer/Uninstaller.

Even if the Installer/Uninstaller starts, SetupPOS does not end.
The user must have the administrative permissions in order to use this function.

<Change of component dialog box>

To change EPSON OPOS ADK:

"The path of the install file" edit field

The path of the install file of EPSON OPOS ADK appears in the edit field.

The default path displayed:

If only version 1.xx is installed: (installed folder)\Setup.exe

If version 2.xx is installed: A:\ Setup.exe

[Browse] button

Use this button to open the file open common dialog box and select the install file. After selecting the install file, the path of the file appears in the "The path of the install file " edit field.

[OK] button

Use this button to start the install of the file specified in the " The path of the install file " edit field. If the install file does not exist, a message appears and the dialog box closes.

[Cancel] button

Use this button to cancel the setting and close the dialog box.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

To register or delete CCO:

"CCO registration" radio button

To register CCO, select this radio button.

"CCO deletion" radio button

To delete CCO, select this radio button.

This button is disabled if CCO is not installed.

"The path of the install file" edit field

The path to where EPSON OPOS ADK is installed, as well as the installer file appears in the edit field.

The path displayed:

If the "CCO registration" radio button is selected: (EPSON OPOS ADK's installed folder)\OposCCOs.exe

If the "CCO deletion" radio button is selected: the path to where CCO is installed, as well as the Uninstaller file.

Note that EPSON does not provide CCO installer.

Obtain it separately. CCO is available from the website of Microsoft.

[Browse] button

Use this button to open the file open common dialog box and select the Install or Uninstall file of CCO. After selecting the file, the information appears in the " The path of the install file " edit field.

[OK] button

Use this button to start the Installer or Uninstaller of the file specified in the "The path of the install file " edit field.

Even if the Installer/Uninstaller starts, SetupPOS does not end.

If the install file does not exist, a message appears and the dialog box closes.

[Cancel] button

Use this button to cancel the setting and close the dialog box.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

6. Help

This menu contains commands on help.

i. Help Topics

Use this menu command to search help by topic and keyword.

ii. About EPSON SetupPOS

Use this menu command to display the version information dialog box and view the EPSON OPOS ADK, control object information (CO) as well as the version information of SetupPOS.

<About EPSON SetupPOS dialog box>

*** EPSON OPOS ADK Version Information**

"EPSON OPOS ADK Version" edit field

The version of EPSON OPOS ADK appears in the edit field.

If a Service Pack is installed, then "SPxx" appears after the version information.

*** Control Object Information**

If multiple CO exists, then the version information of the CO found first is displayed.

"Device Class" combo box

Use this combo box to select a device class for display of the version information of a control object.

"Control Object Version" edit field

The version information of a control object appears in the format of "X.Y.Z".

If no appropriate control object is installed in the system, then "(Not available)" appears instead.

"Control Object Description" edit field

Description of the control object appears in the edit field.

If no appropriate control object is installed in the system, then "(Not available)" appears instead.

***"SetupPOS Version Information"**

It displays the version of the SetupPOS and copyright information.

[OK] button

Close the dialog box and return to the main window.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

5.2.2. Tool Bar

It is possible to operate the main function of the menu bar by clicking a button.

The following tool bar buttons are available:

For the function, refer to the appropriate explanation of the menu bar.

1.  Add New Device
2.  Add New LDN
3.  Delete
4.  CheckHealth Interactive
5.  Device Specific Settings
6.  Communication Settings
7.  Cut

8.  Paste
9.  Properties
10.  Large Icons
11.  Small Icons
12.  List
13.  Details
14.  Help

5.2.3. Left Pane

The left pane of the window displays a tree structure of devices.

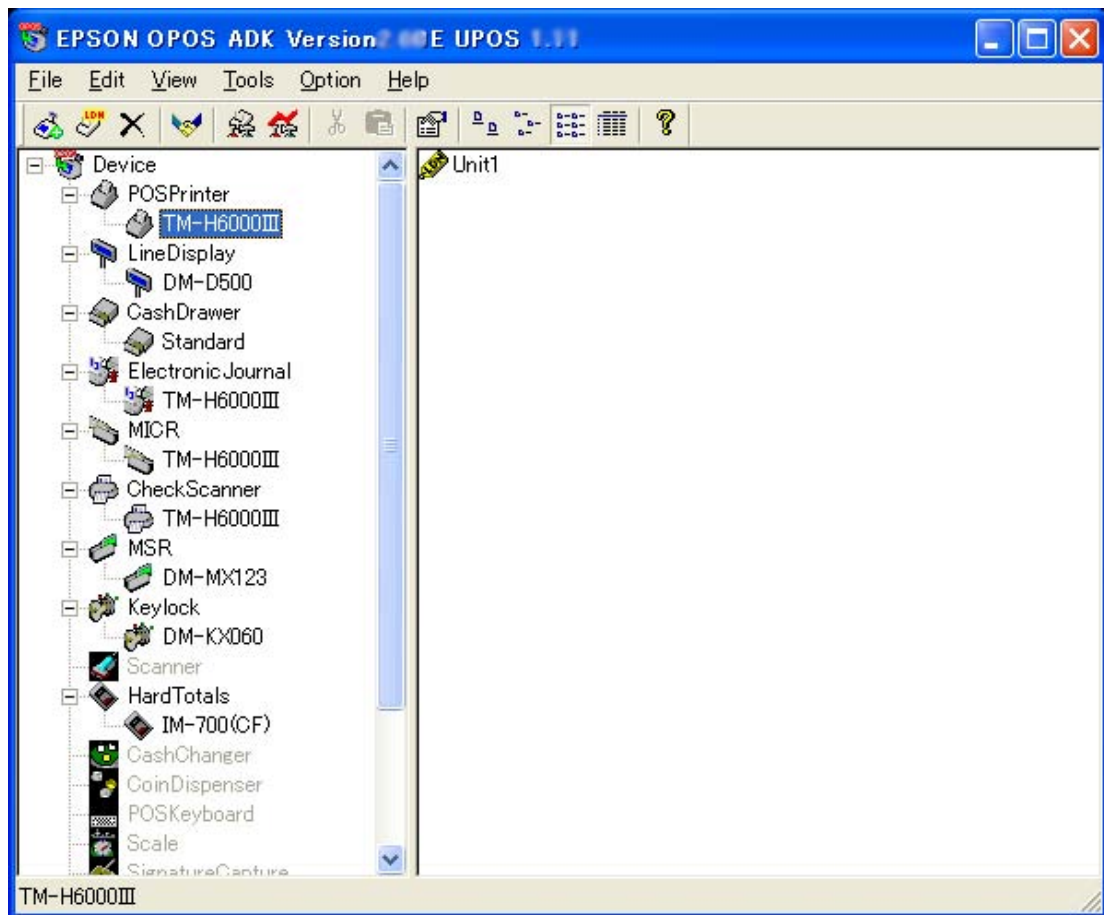
- From the View menu, it is possible to select "Device Class View" or "Port Connection View" to change the way devices are displayed.
- Menu commands can be applied to a selected item on the tree structure.
- For a device, its port can be modified using the drag-and-drop method.
- The pop-up menu can be used with each item.
- The user must have the administrative permissions in order to modify the settings.

5.2.4. Right Pane

The right window displays the detailed information on the selected item in the left window.

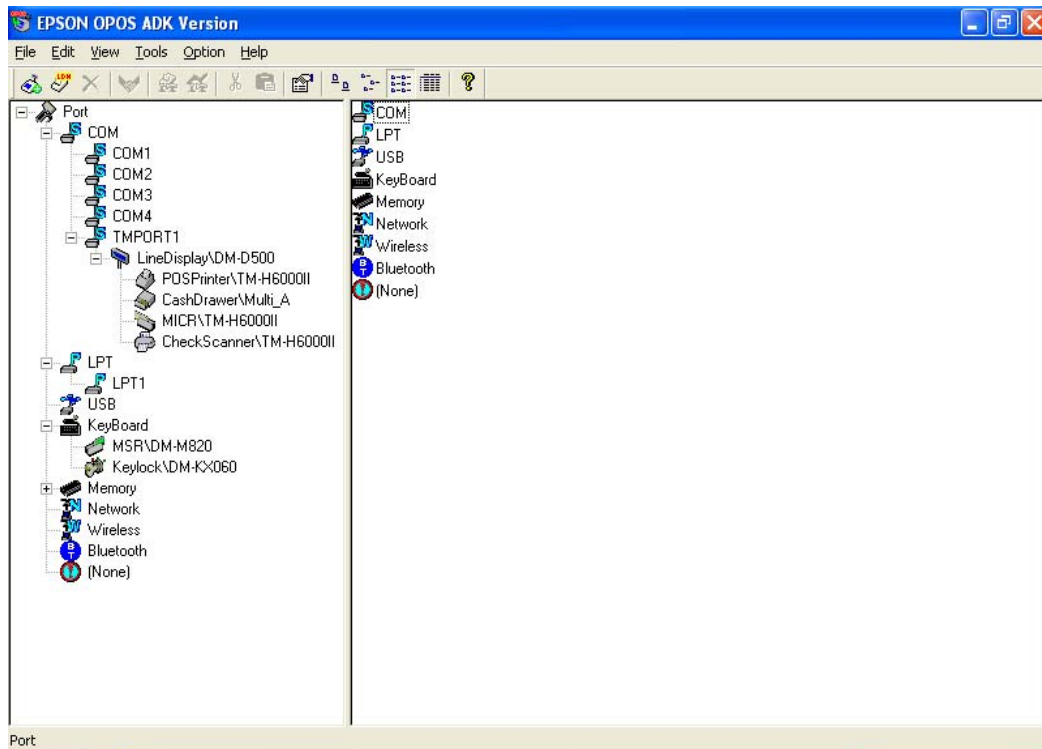
(1). "Device Class View"

- Device class exists below "Device".
- Device name key for each device exists beneath the device.
- When selecting a device name key in the left pane, the logical device name which belongs to that device appears in the right pane. Note that logical device names always appear in the right pane; they never appear in the left pane.
- A device class is also displayed even if any corresponding device is registered.
However, icons of installed devices not from EPSON are dimmed.



(2). "Port Connection View"

- The port type appears below "Port".
- Below the port type, port names (only those which can be set) are displayed.
- Below the port name, the corresponding "Device Class" and device name key are displayed.
For Hydra connection, the displayed device indicates the master device.
- For Hydra connection, slave devices appear below the master device.



What is COM?

It indicates a serial port.

Devices appearing below this port are serial port devices only.

What is LPT?

It indicates a parallel port.

Devices appearing below this port are parallel port devices only.

What is USB?

It indicates a USB port.

Devices appearing below this port are USB port devices only.

What is KeyBoard?

It indicates a standard keyboard interface.

Devices appearing below it are MSR or Keylock device that uses the keyboard interface.

What is Memory?

It indicates a memory interface.

Devices appearing below it are HardTotals devices that use the memory interface.

What is Network?

It indicates a Ethernet.

Devices appearing below this port are network devices only.

What is Wireless?

It indicates a IEEE802.11.

Devices appearing below this port are wireless devices only.

What is Bluetooth?

It indicates a Bluetooth.

Devices appearing below this port are Bluetooth devices only.

What is "(None)"

It indicates that the port settings are not defined yet.

Devices appearing below it are devices with their ports not defined yet.

- COM and LPT appear according to the range set in "Port Range Setting".
- USB appears according to the port range where the USB driver is created.
- Network appears according to the network devices which are connected.
- Wireless appears according to the wireless devices which are connected.
- Bluetooth appears according to the Bluetooth devices which are connected.

5.2.5. Pop-up Menus

Items in the left and right panes of the window support pop-up menus.

The user must have the administrative permissions in order to use this function.

What is a pop-up menu?

When clicking an item and then right-clicking the mouse, a menu appears. This menu is a pop-up menu.

The following pop-up menus are available to items selected in the window.

Selected Item	Pop-up Menu	Description
Device	Add New Device	Use this menu command to display the "Add New Device" wizard.
	Export Registry	Use this menu command to display the "Export registry" dialog box.
	Property	Use this menu command to display "About EPSON SetupPOS" dialog box.
	Help	Use this menu command to display appropriate help to a device.
Device class name	Add New Device	Use this menu command to display the "Add New Device" wizard.

	Export Registry	Use this menu command to display the "Export registry" dialog box.
	Property	Use this menu command to display "About EPSON SetupPOS" dialog box.
	Help	Use this menu command to display appropriate help to a device class.

Selected Item	Pop-up Menu	Description
Device name key	CheckHealth Interactive	Use this menu command to start interactive CheckHealth.
	Device Specific Settings	Use this menu command to display the "Device Specific Settings" dialog box. If the selected device does not support device specific settings, then this command is disabled.
	Add New LDN	Use this menu command to display the "Add new logical device name" dialog box. The key of the selected device name appears as the default.
	Communication Settings	Use this menu command to display the "Communication Settings" dialog box. If the selected device is not from EPSON, then this command is disabled.
	Delete	Use this menu command to delete a device.
	Export Registry	Use this menu command to display the "Export registry" dialog box.
	Property	Use this menu command to display the following setting tabs: "Details of device", "Communication Settings" ^{*1} , "Trace Settings", "Message Handling", "Sleep Time Settings" and "DeviceStatistics settings". ^{*1} It becomes key hook driver in case of the MSR and Keylock.
	Help	Use this menu command to display appropriate help to a device name key.
Logical device name	Delete	Use this menu command to delete the logical device name.
	Rename LDN	Use this menu command to change the mode to the edit mode for modifying the logical device name.
	Help	Use this menu command to display appropriate help to a logical device name.

Selected Item	Pop-up Menu	Description
Port	Add New Device	Use this menu command to display the "Add New Device" wizard.
	Export Registry	Use this menu command to display the "Export registry" dialog box.
	Property	Use this menu command to display the "Port range" dialog box.
	Help	Use this menu command to display appropriate help to the connected port.
COM/LPT	Add New Device	Use this menu command to display the "Add New Device" wizard.
	Export Registry	Use this menu command to display the "Export registry" dialog box.
	Property	Use this menu command to display the "Port range" dialog box.
	Help	Use this menu command to display appropriate help to the connected port (COM or LPT).
COM1/LPT1/ TMPORT1	Add New Device	Use this menu command to display the "Add New Device" wizard.
	Paste	Use this menu command to paste a device that has been prepared for moving (or cut) to a port specified. This command is disabled if there is no port that can be set, or no device has been prepared for moving yet.
	Export Registry	Use this menu command to display the "Export registry" dialog box.
	Property	Others: Use this menu command to display the "Port range" dialog box. TMPORT: Use this menu command to display the "Port property" dialog box.
	Help	Use this menu command to display appropriate help to the connected port (COM or LPT).
USB/ Network/ Wireless	Add New Device	Use this menu command to display the "Add New Device" wizard.
	Export Registry	Use this menu command to display the "Export registry" dialog box.
	Help	Use this menu command to display appropriate help to the connected port (USB/ NETWORK/ WIRELESS).

Selected Item	Pop-up Menu	Description
USB1/Network/ Wireless	Add New Device	Use this menu command to display the "Add New Device" wizard.
	Paste	Use this menu command to paste a device that has been prepared for moving (or cut) to a port specified. This command is disabled if there is no port that can be set, or no device has been prepared for moving yet.
	Export Registry	Use this menu command to display the "Export registry" dialog box.
	Help	Use this menu command to display appropriate help to the connected port (USB).
KeyBoard/Memory	Add New Device	Use this menu command to display the "Add New Device" wizard.
	Export Registry	Use this menu command to display the "Export registry" dialog box.
	Help	Use this menu command to display appropriate help to the connected port (KBD or HT).
IP address or Host name	Add New Device	Use this menu command to display the "Add New Device" wizard.
	Paste	Use this menu command to paste a device that has been prepared for moving (or cut) to a port specified. This command is disabled if there is no port that can be set, or no device has been prepared for moving yet.
	Export Registry	Use this menu command to display the "Export registry" dialog box.
	Help	Use this menu command to display appropriate help to the connected port.
(None)	Paste	Use this menu command to paste a device that has been prepared for moving (or cut) to a port specified. This command is disabled if there is no port that can be set, or no device has been prepared for moving yet.
	Help	Use this menu command to display appropriate help to the connected port.

Selected Item	Pop-up Menu	Description
Device name key (display for each port only)	Add New Device	Use this menu command to display the "Add New Device" wizard.
	CheckHealth Interactive	Use this menu command to start CheckHealth Interactive.
	Device Specific Settings	Use this menu command to display the "Device Specific Settings" dialog box. If the selected device does not support device specific settings, then this command is disabled.
	Add New LDN	Use this menu command to display the "Add new logical device" dialog box. The key of the selected device name appears as the default.
	Communication Setting	Use this menu command to display the "Communication Setting" dialog box. If the selected device is not from EPSON, then this command is disabled.
	Delete	Use this menu command to cut and then delete the device.
	Cut	Use this menu command to cut the device to prepare it for moving.
	Export Registry	Use this menu command to display the "Export registry" dialog box.
	Property	Use this menu command to display the following setting tabs: "Details of device", " Communication Settings", "Trace Settings", "Message Handling", and "Sleep Time Settings" and "DeviceStatistics Settings". *1 It becomes key hook driver in case of the MSR and Keylock.
	Help	Use this menu command to display appropriate help to the device name key.

5.2.6. Drag-and-drop

The drag-and-drop method moves a device to another port.

The user must have the administrative permissions in order to use this function.

What is drag-and-drop?

Drag-and-drop is a mouse operation. Place the mouse pointer over an object, hold down the mouse button and move it (drag) to a desired location, then release the mouse (drop) and the object is "dropped" to the new location.

Hydra connection and drag-and-drop

When the master device of a port with Hydra connection is moved:

Both the master and the slave devices are all moved.

When a slave device of a port with Hydra connection is moved:

Only the slave devices are moved.

In addition, by dragging an individual device that allows Hydra connection to a port connected to a device that also allows Hydra connection, it is possible to perform the Hydra connection.

If the drag operation results in having the device added to the port with Hydra connection set up, a conformation message appears. Follow the instructions in the message.

- It is possible to use the drag-and-drop operation between the right and left panes.
- It is impossible to drag an item which does not allow the drag operation.
- The items which can be dragged are the device name key displayed in each port (COM, LPT, USB, Network, Wireless or Bluetooth) only. It is impossible to drop an item onto an object which does not allow the drop operation.
For drag-and-drop, the following operations are available.
- Moving from COM1 to LPT is not allowed (drop).
- It is impossible to drag-and-drop devices between the physical port and the TMPORT port.

(Example)

- Moving the port settings of the LineDisplay on COM1 to (none) is allowed.
- Moving a drawer with its port setting as (None) to COM3 is allowed.
- Moving a series of devices with Hydra established on COM1 to COM2 is allowed.

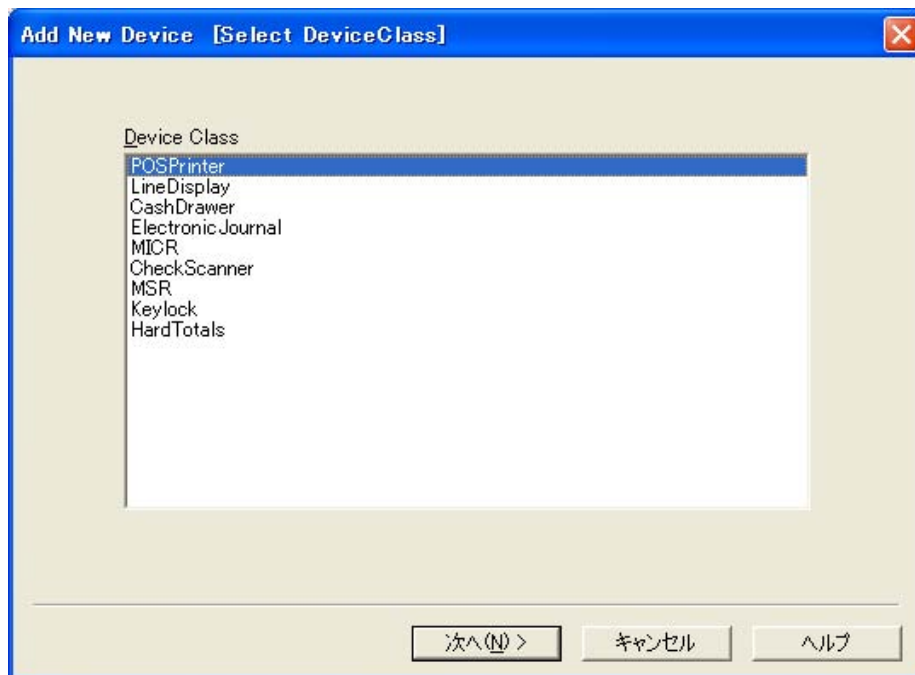
5.3. Using Wizards

5.3.1. Using the Add New Device wizard

This wizard adds devices and performs communication settings. Follow the instructions on the screen to perform the operations.

5.3.1.1. Device Class Selection Screen

In this screen, select a device class to be added.



If a device class has been already selected in the main window before starting this wizard, this screen does not appear; instead the next screen will appear.

"Device Class" list box

Use this list box to display a list of device classes to be selected. Click to select the device class of a device to be added.

Note that the list displays only device classes that are installed control object (CO) that supports the device.

If the necessary device classes are not listed in the list, re-install OPOS ADK, or update it to the appropriate version.

[Next] button

Use this button to move to the device selection screen.

[Cancel] button

Use this button to cancel the settings and return to the main window.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

5.3.1.2. Device Selection Screen

In this screen, set the details of the device to add.

"Select Device Name" combo box

Use this combo box to display a list of selectable devices and select a device from the list.

If there are multiple types of devices (type of ports, etc) for the selected device, use the "Select detailed model." combo box on the right to specify the type of the device.

If no appropriate devices are provided, then this list becomes empty.

The devices are displayed according to the INF file located in the folder where EPSON OPOS ADK is installed.

"Display Ver1.xx" check box

This check box is checked when there is the device of version 1.xx in the list of selectable devices by default. If there is not the device of version 1.xx, this check box is gray out and available.

Clearing the check box delete the device of version 1.xx from the list and display the list. Rechecking the check box add the device of version 1.xx in the list and display the list.

Usually, use the check box to display either device when using both devices of the version 1.xx and version 2.xx.

"Display Ver2.xx" check box

This check box is checked when there is the device of version 2.xx in the list of selectable devices by default. If there is not the device of version 2.xx, this check box is gray out and available.

Clearing the check box delete the device of version 2.xx from the list and display the list. Rechecking the check box add the device of version 2.xx in the list and display the list.

Usually, use the check box to display either device when using both devices of the version 1.xx and version 2.xx.

"Select detailed model." combo box

Use this combo box to specify a type for the device, if the device has multiple types.

While the choices vary according to the information file (INF file) of the selected device, if the device comes with multiple ports, select one from the following: serial, parallel, USB, network, or wireless port.

If the selected device does not have multiple device types, then the combo box display only that port.

When the TMPort is displayed in "Used Port" edit field:

*If "TM-T88V" is selected from the "Select Device Name" combo box, the "Select detailed model" combo box displays TM-T88V only.

When the ports other than TMPort are displayed in "Used Port" edit field:

*If "TM-T88IV" is selected from the "Select Device Name" combo box, the "Select detailed model." combo box displays the following: TM-T88IV, TM-T88IVP, TM-T88IVU and TM-T88IVE Select one of them.

"Used Port" edit field

This edit field displays the port for the device selected in the "Select detailed model." combo box.

Character Strings to be indicated and their meanings are as follows:

"COM": Serial Port.
 "LPT": Parallel Port.
 "USB": USB Port.
 "NET": Ethernet Port.
 "WIR": IEEE802.11 Port.
 "TMPort": TMPort Port.

"Device Description" edit field

This edit field displays description of the device selected in the "Select Device Name" combo box.

If the device comes with multiple types, then description of the selected port appears.

"INF File Name." edit field

This edit field displays the INF file name of the device selected in the "Select Device Name" combo box.

If the same device name has different INF files, then each is treated as a different device.

(This happens when, for example, the same device exists in version 1.xx and version 2.xx.)

[Browse] button

Use this button to open the file open common dialog box and select an INF file from a folder other than the one specified.

If an invalid INF file is selected or if the INF file does not belong to the device class of the selected device, then an error message appears and the selection becomes not possible.

"Add New LDN" edit field

Enter a logical device name for the selected device.

The name can be up to 16 characters, but it cannot contain the "\" character.

In addition, the same logical device name cannot be used again in the same device class.

If the logical device name is invalid, then the [Next] button appears dimmed and it is impossible to proceed to the next step.

"CheckHealth Interactive" button

Use this button to call the CheckHealth method for the selected device using Level = OPOS_CH_INTERACTIVE.

If the selected device is not HardTotals, then it is not displayed.

"Device Specific Settings" button

Use this button to display the "Device Specific Settings" dialog box and change the device specific settings for the selected device.

If the selected device is not HardTotals, then it is not displayed.

Even for HardTotals device, if it does not support device specific settings, it appears dimmed.

[Back] button

Use this button to go back to the previous screen.

[Next] button

Use this button to go to the next screen, the communication setting screen. If the selected device is HardTotals, then this button becomes the [Finish] button.

[Finish] button

Use this button to save the current settings to the registry and close the “Add New Device” wizard.

If the selected device is not HardTotals, then this button becomes the [Next] button.

[Cancel] button

Use this button to cancel the settings and return to the main window.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

5.3.1.3. Communication Settings Screen (for serial, parallel, or USB)

In this screen, set a port to connect as well as its details.

The addition of device. [Port Details]

COM Port Setting

COM1 Port used by (None)

A setup of details of the port.

BaudRate 9600 bps InputBuf 1024

BitLength 8 bit OutputBuf 1024

Parity NONE Output Interval Time 2500 msec

StopBit 1 bit Input SleepTime 10 msec

Handshake DTR/DSR

Interactive CheckHealth Device Specific Settings

< Back Finish Cancel Help

This screen appears if a device of the serial, parallel, or USB port is selected in the add device screen.

"A setup of the port" combo box

Depending on the type of port of the selected device, the following combo boxes are available.

For serial port : "COM Port Settings" combo box

For parallel port: "LPT Port Settings" combo box

For USB port : "USB Port Settings" combo box

From the combo box, select a port number for the serial, parallel, or USB port.

If "None" is selected, then the port settings will be deleted.

For serial and parallel ports, it is possible to use the port range dialog box to change their port ranges.

For USB port, when the communication settings screen appears, all of the USB driver's connectable ports become the selectable range.

"Port is being used by" combo box

Use this combo box to display a list of devices that has been already set up for the selected port number.

"Auto Setting" button

Use this button to start the setting wizard to search and set the USB port numbers automatically.

If a port that is not USB is selected, then this button does not appear.

"Baudrate" combo box

Use this combo box to display and select the communication speed for the serial port.

The values that can be used depend on the device information file (INF file).

For details, refer to the hardware manual of the device.

If the port selection is parallel, USB, or "None", then this combo box cannot be used to change the setting.

"BitLength" combo box

Use this combo box to display and select the data bit length of the serial port.

The values that can be used depend on the device information file (INF file).

For details, refer to the hardware manual of the device.

If the port selection is parallel, USB, or "None", then this combo box cannot be used to change the setting.

"Parity" combo box

Use this combo box to display and select the parity bit of the serial port.

The values that can be used depend on the device information file (INF file).

For details, refer to the hardware manual of the device.

If the port selection is parallel, USB, or "None", then this combo box cannot be used to change the setting.

"StopBit" combo box

Use this combo box to display and select the stop bit of the serial port.

The values that can be used depend on the device information file (INF file).

For details, refer to the hardware manual of the device.

If the port selection is parallel, USB, or "None", then this combo box cannot be used to change the setting.

"Handshake" combo box

Use this combo box to display and select the handshake type of the serial port.

The values that can be used depend on the device information file (INF file).

For details, refer to the hardware manual of the device.

If the port selection is parallel, USB, or "None", then this combo box cannot be used to change the setting.

"InputBuf" edit field

This edit field supports only Version 1.xx devices.

Use this edit field to display and set the input buffer size of the serial port.

The acceptable range is 32 to 8192.

For details, refer to the hardware manual of the device.

If the port selection is parallel, USB, or "None", then this edit field cannot be used to change the setting.

"OutputBuf" edit field

Use this edit field to display and set the output buffer size of the serial or parallel port.

The acceptable range is 32 to 1024, but restrictions apply to some devices depending on the device information file (INF).

This setting is valid for parallel port, and it is possible to specify the size of data to output.

For details, refer to the hardware manual of the device.

If the port selection is USB or "None", then this edit field cannot be used to change the setting.

"Output Interval Time" edit field

Use this edit field for set the retry time to use when output fails. For details of output interval time, please refer to the "[5.8 Glossary](#)" of this manual. The

acceptable range is 0 to 9999. For details, refer to the hardware manual of the device.

If the port setting is "None", this field cannot be used to change the setting.

"Input SleepTime" edit field

This edit field supports only Version 1.xx devices.

Use this edit field to display and set the sleep time to use by the Sleep API in input.

The acceptable range is 0 to 99.

For details, refer to the hardware manual of the device.

If the port selection is USB or "None", then this edit field cannot be used to change the setting.

[CheckHealth Interactive] button

Use this button to call the CheckHealth method for the selected device using Level = OPOS_CH_INTERACTIVE.

If the port setting is "None", then this button is disabled and afterwards interactive CheckHealth is not executed.

[Device Specific Settings] button

Use this button to display the "Device Specific Settings" dialog box and change the device specific settings for the selected device.

If the port setting is "None" or the selected device does not support device specific settings, then this button becomes dimmed and disabled.

[Back] button

Use this button to go back to the device selection screen.

[Finish] button

Use this button to save the current settings to the registry and close the "Add New Device" wizard.

[Cancel] button

Use this button to cancel the settings and return to the main window.

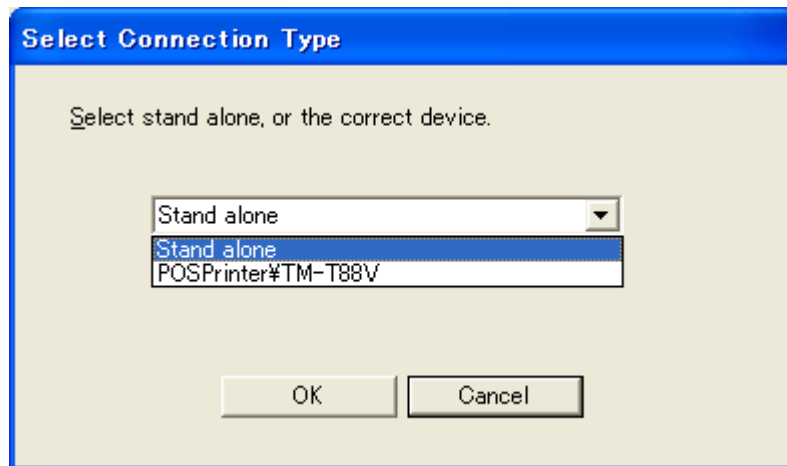
[Help] button

Use this button to display the help on the dialog box by clicking this button.

5.3.1.4. Select Connection Type Screen (for LineDisplay and CashDrawer device)

In this screen, select the connection condition and connection device.

The connection type and/or connected device must be selected when registering a LineDisplay or Cashdrawer.



“Select Stand alone, or the correct Master device.” combo box

Available connection types or devices are displayed for the device selected on the "Add New Device [Select DeviceNameKey]" screen.

[OK] button

Use this button to select the connection condition or connection device and proceed to the next screen.

[Cancel] button

Use this button to cancel the operation and return to the previous window.

5.3.1.5. Communication Settings Screen (for network or wireless device)

In this screen, set a port to connect as well as its details.

This screen appears if the network or wireless device is selected in the add device screen.

“IP address/Host name” edit box

Use the edit field to specify a network or a wireless device's IP address or host name.

If an IP address or host name is already set up, the address or name appears in the edit field.

The network settings set the input value by pushing the [Update] button.

If the device that has been set is on the port to be set, depending on the capability of the device and selected device, a message not to set the device to the port appears. In this case, the port setting is invalid.

[Update] button

Use this button to save the entered IP address or host name to the registry.

If a device already exists at the IP address or host name entered into the edit field, then a message appears and the change is not accepted.

"Port is being used by" combo box

Use this combo box to display a list of devices that has been already set up for the selected IP address or host name.

From "Baudrate" to "Handshake" combo boxes

These settings are for the serial port. The network device does not need this setting.

"InputBuf" edit field

The network device does not need this setting.

"OutputBuf" edit field

Use this edit field to display and set the output buffer size of the network.

The acceptable range is 32 to 1024, but restrictions apply to some devices depending on the device information file (INF).

For details, refer to the hardware manual of the device.

If the port selection is "None", then this edit field cannot be used to change the setting.

"Output Interval Time" edit field

Use this edit field for set the retry time to use when output fails. For details of output interval time, please refer to the "[5.8 Glossary](#)" of this manual. The acceptable range is 0 to 9999.

For details, refer to the hardware manual of the device.

If the port setting is "None", this field cannot be modified.

"Input SleepTime" edit field

This edit field supports only Version 1.xx devices.

Use this edit field to display and set the sleep time to use by the Sleep API in input.

The acceptable range is 0 to 99.

For details, refer to the hardware manual of the device.

If the port selection is "None", then this edit field cannot be used to change the setting.

[CheckHealth Interactive] button

Use this button to call the CheckHealth method for the selected device using Level = OPOS_CH_INTERACTIVE.

If the port setting is "None", then this button is disabled and afterwards interactive CheckHealth is not executed.

[Device Specific Settings] button

Use this button to display the "Device Specific Settings" dialog box and change

the device specific settings for the selected device.

If the port setting is "None" or the selected device does not support device specific settings, then this button becomes dimmed and disabled.

[Back] button

Use this button to go back to the device selection screen.

[Finish] button

Use this button to save the current settings to the registry and close the "Add New Device" wizard.

[Cancel] button

Use this button to cancel the settings and return to the main window.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

5.3.1.6. Communication Settings Screen (for TMPort device)

In this screen, set a port to connect as well as its details.

This screen appears if a device of the TMPort is selected in Select DeviceName Key screen.

"TMPORT Settings" combo box

Selects a port of TMPORT for connect.

[Make Port] button

Displays the [TMPORT making] dialog box.

[Update Port] button

Updates the TMPORT that is able to connected.

"Port is being used by" combo box

Use this combo box to display a list of devices that has been already set up for the selected port number. When Hydra connected device is set, confirm the displayed device.

Setup of Details of the port**"BaudRate" combo box**

Use this combo box to display or set the communication speed of the serial port. The values that can be used depend on the device information file (INF file). For information on the communication speed, refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, Bluetooth, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the speed is displayed but cannot be modified.

"BitLength" combo box

Use this combo box to display or set the data bit length of the serial port. The values that can be used depend on the device information file (INF file). For information on data bit length of the device, refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, Bluetooth, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the value is displayed but cannot be modified.

"Parity" combo box

Use this combo box to display or set the parity bit length of the serial port. The values that can be used depend on the device information file (INF file). For information on parity bit of the device, refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, Bluetooth, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the value is displayed but cannot be modified.

"StopBit" combo box

Use this combo box to display or set the stop bit of the serial port. The values that can be used depend on the device information file (INF file). For information on stop bit of the device, refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, Bluetooth, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the value is displayed but cannot be modified.

"Handshake" combo box

Use this combo box to display or set the handshake of the serial port. The values that can be used depend on the device information file (INF file). For information on handshake of the device, refer to the hardware manual of the device.

If the selected port is parallel, USB, network, wireless port, Bluetooth, or "None", then this setting cannot be set or changed. For a slave device in Hydra connection, the value is displayed but cannot be modified.

[CheckHealth Interactive] button

Use this button to execute the CheckHealth method using Level = OPOS_CH_INTERACTIVE for the device. If the port setting is "None", then this button is disabled. For details, refer to the manual of the device.

[Device Specific Settings] button

Use this button to display the "Device Specific Settings" dialog box and change the device specific settings for the selected device. If the selected device does not support device specific settings, then this button is disabled. For details, refer to the manual of the device.

[Back] button

Use this button to go back to the previous screen.

[Finish] button

Use this button to save the current settings to the registry and close the "Add New Device" wizard.

[Cancel] button

Use this button to cancel the changes and close the dialog box. Any changes in the settings will not be saved to the registry.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

5.3.1.7. TMPort Making

In this screen, creates a port of TMPort.

"Port Type" combo box

Displays a port type that can be created.

When the "COM", "LPT", or "Bluetooth" is selected:

Chooses a port in the "Physical Port (Only COM, LPT, and Bluetooth)" combo box.

When the "USB" is selected:

Chooses a device in the "Connect Device" combo box.

When the "NET" is selected:

Inputs an IP address in the "Physical Port (Only NET)" combo box.

"Physical Port (Only COM, LPT, and Bluetooth)" combo box

This setting only becomes effective when you select COM, LPT or Bluetooth in the "Port Type" combo box. Only the port recognized by the system is displayed.

You cannot create a new port at the physical port which has already registered in TMPort.

When using a Bluetooth device, select the COM port of the Bluetooth device.

Before registering a Bluetooth-connected device in SetupPOS, configure the COM port of the Bluetooth device. Refer to the "Product Manual" for instructions. When using the "TM Bluetooth connector Utility," select "COMx."

"Physical Port (Only NET)" edit field

This setting only becomes effective when you select NET in the "Port Type" combo box. As the default, "0.0.0.0" is displayed.

You cannot create a new port at the IP address which has already registered in TMPort.

You cannot specify the host name.

"Connect Device (Only USB)" combo box

This setting only becomes effective when you select USB in the "Port Type" combo box. The connectable ports are displayed.

"Make Port Name" edit field

Displays the name of port that is created.

[OK] button

Creates a port of TMPort according to the type of port selected.

[Cancel] button

Use this button to cancel the operation and return to the previous window.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

5.3.1.8. Communication Settings Screen (for key hook driver)

Use this screen to set the details of **Keyboard Hook Control**.

This screen appears if MSR/Keylock is selected in the add device screen.



"Start hooking keyboard input for this device" radio buttons

If the [Yes] button is selected, then the EPSON Keyboard Hook Control utility (hereafter referred to as "Hook Control") is placed in the Windows Startup group and when the system runs, monitoring of keyboard input to this device also starts. If the [No] button is selected, the hook control utility is not always loaded or placed in the Windows Startup group.

In this case, however, it is possible to use this device in the OPOS application.

If an input event occurs at this device when the keyboard interface device is not opened yet, then unanticipated keyboard input may result.

To avoid this problem, it is necessary to register the hook control utility as the target of this device.

If no keyboard-monitoring driver is used, then these buttons become dimmed and disabled.

"Show Keylock position adjustment dialog box at system start up" check box

If the [Yes] button is selected, when the system runs, a dialog box appears and prompts for change of the position.

Then follow the instructions and modify the position so that the key position can be correctly adjusted. Usually, select the [Yes] button.

If the [No] button is selected, then the dialog box will not appear.

However, to know the initial state of the Keylock correctly, when starting the system, the Keylock position must at least be modified once.

Therefore, unless there is a necessary, always select the [Yes] button.

This setting is available for the Keylock.

If the selected device is MSR, then these buttons become dimmed and disabled.

[CheckHealth Interactive] button

Use this button to call the CheckHealth method for the selected device using Level = OPOS_CH_INTERACTIVE.

[Device Specific Settings] button

Use this button to display the "Device Specific Settings" dialog box and change the device specific settings for the selected device.

If the selected device does not support device specific settings, then this button becomes dimmed and disabled.

[Back] button

Use this button to go back to the device selection screen.

[Finish] button

Use this button to save the current settings to the registry and close the "Add New Device" wizard.

[Cancel] button

Use this button to cancel the settings and return to the main window.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

Note that when version 1.xx and version 2.xx are installed, it is impossible to use MSR and Keylock devices of these versions at a time. Use the combination of MSR and Keylock of either version (only version 1.xx or only version 2.xx). MSR and Keylock use the keyboard interface and keyboard monitoring driver (EPSONHKO). However, this driver differs from version 1.xx and version 2.xx, so it is impossible to use MSR and Keylock devices of both versions at a time.

5.3.2. Using the USB AutoSetup Wizard

In the communication settings screen, if the [Auto Setting] button is selected, then a message appears.

If the [OK] button is selected, then the “USB AutoSetup” wizard starts and searches an appropriate USB port number to the selected devices.

5.3.2.1. Screen for the User to Select a Model

This screen appears if it is possible that the selected device is a slave device with Hydra connection (specifically, a device other than POSPrinter).

"Master Device" combo box (POSPrinter)

Use this combo box to select the master device of a device to be searched.

[Browse] button

Use this button to open the file open common dialog box and select a master device's INF file from a location other than the specified folder.

[OK] button

Use this button to start searching for the USB port number according to the information of the selected master device.

[Cancel] button

Use this button to cancel the operation, and then return to the device communication settings wizard of the “Add New Device” wizard or the “Communication settings” dialog box.

When the search is complete, one of the screens from (1) to (3) appears.

(1) If only one search target device is found:

"The number has been searched below." edit field

A port number in the search result appears.

[OK] button

Use this button to end the “USB AutoSetup” wizard, and set the port number (found in the search) to “USB Port Setting” of the communication setting.

[Search] button

Use this button to search the port number again.

[Cancel] button

Use this button to cancel the operation, then return to the device communication settings wizard of the “Add New Device” wizard or the “Communication settings” dialog box.

(2) If multiple search target devices are found:

"The number has been searched below." edit field

Multiple port numbers in the search result appear. To set the port number manually, click the [OK] button after selecting the port number.

To narrow the port number choice, press the [Search] button again.

[OK] button

Use this button to end the "USB AutoSetup" wizard, and set the port number (found in the search) to "USB Port Setting" of the communication setting.

[Search] button

Use this button to search the port number again. Usually, use this button to narrow the port number choice from the multiple port numbers. Before searching again, open the cover of the POSPrinter not to be included in the search target so that communication becomes not possible.

[Cancel] button

Use this button to cancel the operation, then return to the device communication settings wizard of the "Add New Device" wizard or the "Communication settings" dialog box.

(3) If no search target device is found:**[Search] button**

Use this button to search the port number again.

Before searching again, make sure the target devices are correctly stated and are connected.

[Cancel] button

Use this button to cancel the operation, then return to the device communication settings wizard of the "Add New Device" wizard or the "Communication settings" dialog box.

5.4. About the Warning Dialog Box**5.4.1. The Contradictory Warning of a Setup of Registry**

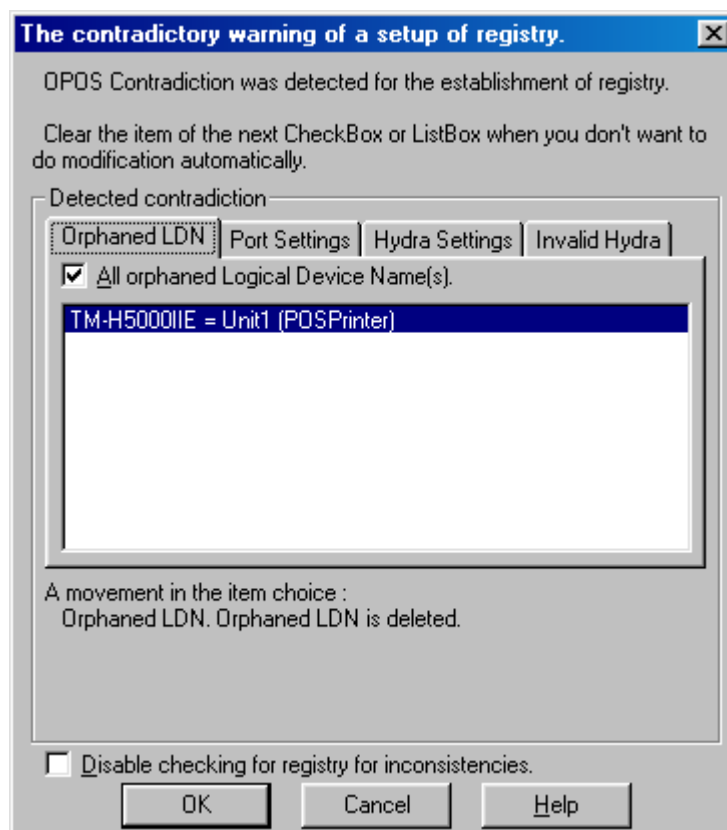
If inconsistency is found in the OPOS registry settings when the Setup program starts, or after devices are registered or settings modified, this dialog box appears. The detected inconsistency is listed in the list box of the appropriate tab of the dialog box. If multiple inconsistencies occur, they are listed in the list boxes of the respective tabs.

Four types of inconsistencies can be detected, as listed below:

- (1) Logical device inconsistency

- (2) Port settings inconsistency
- (3) Hydra settings inconsistency
- (4) Invalid Hydra inconsistency

It is possible to correct the registry inconsistency automatically for the selected item by clicking the [OK] button. Usually, use this method to correct the problem. However, this approach may result in losing all of the settings. To avoid some automatic corrections, clear the selected items in the list box, or click the [Cancel] button. By clicking the [Cancel] button, the registry inconsistencies listed in the list boxes are ignored and the operation continues. On the other hand, to perform the correction for each list, select the desired tabs and then select or clear items in the list boxes.



1. "Orphaned LDN" tab

The "Orphaned LDN" (logical device name) tab contains the following fields.

"All orphaned Logical Device Name(s)." check box

Select or clear multiple item in the list box, and this check box indicates the type of the selection, as follows:

- If all items are selected : this check box is selected.
- If only some items are selected : this check box is in an undetermined state.
- If all items are cleared : this check box is cleared.

"Orphaned LDN" list box

This list box displays a list of orphaned logical device names.

An orphaned logical device name is a logical device name, which has been used, but its device has been deleted.

Logical device names selected in this list box will be deleted.

Not to delete them, clear their selections from the list box.

2. "Port Settings" tab

The "Port Settings" tab contains the following fields.

"All inconsistent port setting for hydra device(s)." check box

Select or clear multiple item in the list box, and this check box indicates the type of the selection, as follows:

- If all items are selected : this check box is selected.
- If only some items are selected : this check box is in an undetermined state.
- If all items are cleared : this check box is cleared.

"Port Settings" list box

All devices with Hydra connection must have the same communication settings.

If there are slave devices with different communication settings, they are listed in the list box, and they will be corrected so as to have the same communication settings as their master devices.

To continue using different communication settings despite the warning, just clear the selections from the list box.

3. "Hydra Settings" tab

The "Hydra Settings" tab contains the following fields.

"All hydra device settings." check box

Select or clear multiple item in the list box, and this check box indicates

the type of the selection, as follows:

- If all items are selected : this check box is selected.
- If only some items are selected : this check box is in an undetermined state.
- If all items are cleared : this check box is cleared.

"Incorrect Hydra mark(s):" list box

This list box indicates inconsistency in the Hydra information set in the registry.

It first shows the correct setting (ON or OFF).

This problem only occurs in Hydra settings for version 1.xx of the application.

It occurs when there is inconsistency in the Hydra or Shydra registry setting, which indicates Hydra.

"Bad uses:" list box

This list box indicates inconsistency in Hydra's master device information set in the registry.

The registry of each Hydra-related device must have information of the master device (Uses).

If this information is illegal, the inconsistency in the master device is listed, and then the correct master device is installed.

Hydra settings selected here will be automatically corrected.

To continue using the illegal Hydra settings despite the warning, just clear the selections from the list box.

Usually, the SetupPOS program automatically registers normal Hydras to the registry.

The Hydra settings inconsistency may occur when the OPOS registry is modified directly.

Therefore, do not use a utility such as REGEDIT to change the OPOS registry directly.

4."Invalid Hydra" tab

The "Invalid Hydra" tab contains the following fields.

"All invalid Hydra settings." check box

Select or clear multiple item in the list box, and this check box indicates the type of the selection, as follows:

- If all items are selected : this check box is selected.
- If only some items are selected : this check box is in an

undetermined state.

· If all items are cleared : this check box is cleared.

" Invalid Hydra" list box

This list box displays a list of invalid Hydras.

Depending on the capability of the device, the Hydra connection's range is determined.

If a device on a port exceeds its capability, a list of its related devices (inconsistency) appears in the list box.

Inconsistent devices selected in the list are automatically corrected.

In other words, settings of ports such as COM1 become undefined ("None").

Not to have the port setting become undefined, clear the selection of the device from the list box.

"Disable checking for registry for inconsistencies." check box

To stop validating Hydra inconsistency from now on, select this check box.

This check box setting is reflected in the "Validate" check box in the main window.

To restart registry validation, select the "Validate" check box again in the main window.

We recommend clearing this check box. When automatic validation is not required, clear this check box.

In this case, the user must make sure the OPOS registry is correctly set.

[OK] button

Use this button to enable automatic correction on the selected items and return to the main window.

[Cancel] button

Use this button to cancel the settings and return to the main window.

[Help] button

Use this button to display the help on the dialog box by clicking this button.

5.5. Messages

When performing operations or settings from a dialog box, some messages may appear.

This section explains the various messages.

5.5.1. Messages Explanations

1. Adding a logical device name

- (1) When specifying a LDN which has been used (logical device name for a deleted device), the following message appears:

"Logical Device Name "LogicalDeviceName" was used to refer to "DeviceNameKey". Do you want to update it ?"

This operation will change an old device name key for an existing logical device name to a device name key to be added.

2. Deleting a logical device name

- (1) When deleting a logical device name, the following message appears:

"Logical Device Name "Unit 1" is used to refer to "TM-T88IVM". Do you want to remove it ?"

This operation will delete the selected logical device name.

3. Deleting a device

- (1) When deleting a device name key, the following message appears:

"Are you sure you want to delete all setting information about device (TM-T88IVM)?"

This operation will delete the selected logical device.

- (2) When the device to be deleted is a master device with Hydra connection, the following message appears:

"Hydra connection with the device. Delete the device simultaneously."

4. Modifying logical device name

- (1) When changing the name of a logical device from "Unit1" to "Unit2", the following message appears:

"If you change the LDN from "Unit1" to "Unit2", it may affect some application settings. Do you want to rename it ?"

- (2) When trying to change the name of an existing logical device, the following message appears:

"If you change the LDN from "Unit1" to "Unit2", it may affect some application settings. Do you want to rename it ?"

If selecting the [Yes] button, then the following message appears:

"Logical Device Name "Unit 1" is used to refer to "TM-T88IVM". Do you want to replace it ?"

5. When changing a port causes an inconsistent Hydra state, the following message appears:

"Device will not be set as COMn."

6. When trying to connect a device (slave device), which requires a master device, to a port by itself only, the following message appears:

"No master device detected. You must set/add a proper master device for this device."

7. When changing the port of a master device with Hydra setting, the following message appears:

"This device has been set as a master device. Hydra settings must be changed accordingly."

8. When trying to connect a new device to a port with Hydra configuration or which is connected to devices, if the new device is neither alone nor Hydra, then the following message appears:

"Hydra connection detected for 'device name'. If the device is to be stand alone, select [No]. Do you want to set it to be a slave device ?"

9. When trying to connect a new device to a port connected to Hydra-configurable devices, if the devices which has limit of device which can connect and the device to fall outside the limit exist, then the following message appears:

"The master device and the slave device combination is not proper. You must use this device under proper device combination. The changes to the serial port setting will be canceled."

10. The range of Hydra connection depends on the capability of the master device. When trying to add a new device to a port so that this operation exceeds the capability of the port, the following message appears:

"Improper Hydra device(s) detected: Device will not be set as COMn."

Note: The location of COMn, depending on the specified port, becomes LPTn, USBn or IP address/Host name.

<CheckHealth Interactive>

- (1) When running the interactive CheckHealth,
Its execution result appears.

<Export registry dialog box>

- (1) When exporting the registry, the following message appears:
 " OPOS registry is kept as registry data file. To keep registry data file, the backup copy of "OPOS ADK Disk-1" is made, it, drive A : Insert it. The same OPOS environment can be reproduced by using "OPOS ADK Disk-1" that it is renewed on this more than one target machine. "

All : Register all information.
 Selecting : Register selected items' information.
 Hydra : Register all information about Hydra connection.

- (2) When overwriting a file, the following message appears:
 " (A file name) already exists! Do you want to replace them? ([Yes]=Replace. [No]=Append.) "

<Port range dialog box>

- (1) When specifying a number small than the maximum number of COM ports used by devices, the following message appears:
 "Currently using a COM port that is higher than the specified maximum value. Cannot set COM Port maximum."
- (2) When specifying a number small than the maximum number of LPT ports used by devices, the following message appears:
 "Currently using a LPT port that is higher than the specified maximum value. Cannot set LPT Port maximum."

<Trace settings dialog box>

- (1) When changing the trace settings for TM-T88IVM and selecting an another device, the following message appears:
 "Trace settings for POSPrinter\TM-T88IVM have been changed. Do you want to save the change(s) ?"

<Message handling dialog box>

- (1) During synchronization, if message handling is selected, the following message appears:
 " Do not execute OPOS methods/properties when the OPOS method is executed during synchronous. Otherwise, it is possible that the program may unexpectedly hang. Do you want to change it ?"

<Sleep time settings dialog box>

- (1) When changing the sleep time settings for TM-T88IVM and selecting an another device, the following message appears:
 "POSPrinter\TM-T88IVM has been changed. Do you want to save the

change?"

<Change of component dialog box>

- (1) If the installer does not exist in the specified folder, the following message appears:
"Fails to start up Setup.exe."

<Add New Device wizard>

- (1) For browsing INF files, when specifying an illegal INF file, the following message appears:
"Select a correct device class INF file."
- (2) When the combination of master and slave devices is illegal, the following message appears:
"The master device and the slave device combination is not proper.
You must use this device under proper device combination. The port setting for this device will be UNDEFINED."
- (3) When the combination with Hydra is improper, the following message appears:
"Improper Hydra device(s) detected: Device will not be set as COMn. It will leave the port setting UNDEFINED."
- (4) When specifying an illegal Hydra connection, the following message appears:
"This setting may cause a port conflict if the device is not setup properly. "
- (5) When trying to connect a device, which requires a master device, to a port by itself, and the master device cannot be found, then the following message appears:
"No master device and the slave device combination is not proper. You must use this device under proper device combination. The changes to the serial port setting will be canceled. "
- (6) When addition of a device causes its combination with the master device to become improper, the following message appears:
"The master device and the slave device combination is not proper. You must use this device under proper device combination. The port setting for this device will be UNDEFINED."

Note: The location of COMn, depending on the specified port, becomes LPT, USB, Ethernet, or IEEE802.11.

<CCO registration>

When a file does not exist at the specified location, the following message appears:

"Specified file cannot be found."

<Sharing with non-EPSON CO and SO>

(1) Sharing with non-EPSON CO

When accessing CO where non-EPSON CO is registered, the following message appears:

"Another company's CO registered. Cannot use this function."

(2) Sharing with non-EPSON SO

When accessing SO where non-EPSON SO is registered, the following message appears:

"Another company's SO registered. Cannot use this function."

5.6. Using the Keyboard Hook Control Utility

The keyboard hook control utility is a DLL driver to process Keylock and MSR data properly.

5.6.1. Functions

The keyboard hook control utility contains the following function.

Keyboard data monitoring

Because MSR and Keylock are devices of the keyboard interface, data of each of the devices is sent to the PC as keyboard data.

The keyboard hook control utility separates their data for each of the devices.

During the device (MSR or Keylock) operation (from Open to Close), the hook driver is called to separate the keyboard data automatically.

In other times, when keyboard data separation is required, by loading the keyboard hook control always, it is possible to have the hook driver perform keyboard data separation.

This is because at the communication settings of MSR/Keylock, Keyboard Hook Control has been set to be always loaded. When the keyboard hook control is always loaded, then data of the keyboard, MSR, and Keylock devices are always separated.

This prevents incorrect operation or data from being displayed in the editor or input application if card or Keylock information is loaded during operations of the editor or an application allowing input.

If the Keylock device is used, after startup, until the key position is changed, the current key position cannot be determined correctly.

When the keyboard hook control utility runs, a dialog box appears and prompts for change of the position.

Following the instructions and modifying the position adjust the key position correctly.

5.6.2. Activation

During setup, when the MSR or Keylock is chosen and if the device is set to always load at the communication settings, then the keyboard hook control utility is placed in the Windows Startup group or the keyboard hook driver will automatically load each time Windows is started.

Also during setup, a choice can be made of whether or not to display a dialog box that asks the user whether to confirm the Keylock position or not.

If the choice is made to confirm the position and to display the dialog box, each time the keyboard hook control utility is started, a message appears asking that the Keylock be turned to certain position when Keylock device is registered.

"Please change the key position to [L2]"

Note: Depending on the devices, characters shown with L2 may differ.

If the key is placed in the [L2] position, the default value is set and afterwards the initial Keylock utility will be loaded correctly.

If the key is set to a position other than [L2] then the following message appears.

"The key was not turned to [L2]. Please continue."

"The key position has not changed. Please continue."

Until the Keylock is turned once, the Keylock device position may not be recognized correctly.

5.6.3. Device Dependency

The keyboard hook control utility separates data, but the way data is separated depends on the device being used.

Because of this, it is necessary to know what device is being used when the keyboard hook utility is started.

During operation of the keyboard hook control utility, the devices with keyboard interface other than that for keyboard hook control cannot execute the Open method.

The device being used can be specified during the hook control utility setup.

When the keyboard hook control utility is registered at the setup, the device with

the available keyboard interface is only the device for keyboard hook control utility.

5.7. Using Devices

This section explains how to set up or delete a device, using POSPrinter TM-T88IVM as an example.

5.7.1. Setting up a Device

To set up POSPrinter TM-T88IVM as a device, use the "Add New Device" wizard of SetupPOS.

Start the wizard either by selecting "Add New Device" from the Edit view, or clicking the  [Add New Device] button on the tool bar.

Or, select the POSPrinter in the left pane of the main window, and then right-click to bring up a pop-up menu and select Add New Device. (In this case the step starts from **<Select device screen>**.)

Following the instructions in each screen of the wizard, set up the device as described in the following procedure.

(Operations in each screen)

[Next] button : Click this button to accept the changes in the current screen and then proceed to the next screen.

[Back] button : Click this button to clear the current settings and go back to the previous screen.

[Cancel] button : Click this button to cancel the operation.

[Help] button : Click this button to display appropriate help.

Perform these operations as required in each screen.

<Device class selection screen>

For device class: Select POSPrinter and then click the [Next] button.

<Device selection screen>

Select Device Name : Select "TM-T88IVM."

Select detailed model : Select "TM-T88IVM."

Used Port : "COM" is displayed.

Device Description : "EPSON TM-T88IVM POS Printer" is displayed.

INF File Name : INF file name of TM-T88IVM is displayed.

- [Browse] button : Not used for this time.
(Use this button when selecting a different INF file of TM-T88IVM which exists other than the folder where EPSON OPOS ADK is installed.)
- Add New LDN : If necessary, enter "Unit1."
(It is possible to enter this information after the wizard ends. From the Edit menu, select "Add New LDN." Another method is to select a device name key and right-click it. From the pop-up menu, select "Add New LDN.")

Click the [Next] button.

<Communication settings screen>

"COM Port Setting" * Select the following items.

Port setting : "COM1"

[Auto Setting] button : In this example, it is not displayed.
(It appears when setting a device using the USB port.)

"Port is being used by":

It does not appear if there is no device in use. If the port is already used in another device, then the name of the device appears here.

"Setup of details of the port." * Set the following items.

BaudRate : 19200
Bit length : 8 bits
Parity : NONE
Stop bit : 1 bit
Handshake : DTR/DSR
Output buffer length : 1024
Output interval time interval : 1000

[CheckHealth Interactive] button:

Click this button and make sure this setting allows for normal communication and printing.

[Device Specific Settings] button:

If necessary, click this button to change the device specific settings.

Click the [Finish] button

The window displays the settings of TM-T88IVM.

5.7.2. Removing a Device

Delete a device using one of the following three methods.

- (1) In the main window, select the device name key to be deleted (in this case TM-T88IVM). From the Edit menu, select Delete. When a confirmation message box appears, select Yes to delete the device.
- (2) Select the device name key to be deleted (in this case TM-T88IVM). Right-click to bring up a pop-up menu and select Delete. When a confirmation message box appears, select Yes to delete the device.
- (3) In the main window, select the key of the device name to be deleted (in this case TM-T88IVM). Then click the [Delete]  icon on the tool bar. When a confirmation message box appears, select Yes to delete the device.

5.8. Glossary

INF file	Each model comes with one information file. During installation, the information files are copied to the same folder where EPSON OPOS ADK is installed. These files are used to configure devices.
Device Class	In the Device Class view of the main window, device class is the device type displayed below Device. Device classes used in EPSON OPOS ADK are as follows: CashDrawer, HardTotals, Keylock, LineDisplay, MICR, MSR, CheckScanner, POSPrinter and ElectronicJournal. For detailed explanations of the devices available, refer to the "OPOS Registry Usage" section of the "OPOS Application Programmer's Guide".
Device Name Key	When a device is added, it appears in the Device Class view of the main window below device class. If no device is set for the selected device class, then this information is not displayed. For example, the device name key of POSPrinter's TM-295 is "POSPrinter¥TM295."

Logical Device Name (LDN)	An another name used to identify a device. Each device can have a logical device name. However, a logical device name which is already being used by the same device class cannot be specified. For example, if a logical device name "Unit1" is already assigned to a device called TM-295 of POSPrinter, then "Unit1" cannot be used again as a logical device name for TM-88IV in the same device class (POSPrinter). Using the logical device names does not need to specify the device name key directly and so makes application development versatile.
Registry	OPOS uses the system registry to store and reference device information. Although it is possible to view the information using the REGEDIT utility, do not change the SetupPOS program. For details, refer to the "OPOS Registry Usage" section of the "OPOS Application Programmer's Guide".
Input buffer length	It sets the size of the input buffer (unit in bytes; a area in the PC's memory for temporarily storing input data). Because this preset value has been optimized, usually it is not necessary to modify it. If this value is changed smaller than the preset one, the input response may deteriorate. On the other hand, if the new value is larger than the preset one, input response for large amount of data input would improve. Therefore, adjust this value if the input response is bad.
Output buffer length	It sets the size of the output buffer (unit in bytes; a area in the PC's memory for temporarily storing output data). Because this preset value has been optimized, usually it is not necessary to modify it. If this value is changed larger than the preset one, data loss may occur. On the other hand, if the new value is smaller than the preset one, the output performance may deteriorate. Therefore, unless there is a necessary, do not change the preset value.
Output Interval Time	Time interval for retry in the case output fails. If not even 1 byte is output for the set time, the Timeout error occurs. While this preset value has been optimized, modify it (usually by increasing it rather than decreasing it) if the Timeout error frequently occurs during printing at your environment. On the other hand, timeout errors which occur when communication is not possible due to physical problems such as cover being opened are normal. Change this value for cases where timeout error occurs during printing even though printing is possible.
Input Sleep Time	During the waiting for input, EPSON OPOS ADK can call Sleep (a WIN32 API). The input sleep time is the value (msec) to pass to the Sleep function. This preset value has been optimized and usually it does not need to be changed. Input processing (ReadFile) uses the CPU exclusively. When input process is performed while there is no input data in particular, because data event does not occur, the application looks like it is not running (hangs). To solve this problem, use the Sleep function between input processes so that control (of CPU) can be passed to other applications as much as possible. (Executing the Sleep function passes to other processes.) Try to use the preset input sleep time value, because if changing the value decreases the input response. To change the input sleep time value by all means, change it in a way that strikes a balance between input response and application performance.
Port	EPSON OPOS ADK uses the following types of ports: serial(COM), parallel(LPT), USB, Ethernet, IEEE802.11., and Bluetooth

Trace	The trace (log) function provides support for application development. Usually this function is used only for debugging OPOS applications; it is not used for other purposes. To change the trace settings, use the "OPOS API trace settings" dialog box. Note that for version 1.xx, the trace function can be used only if an EPSON OPOS ADK supporting trace is installed.
Message handling	Depending on the application, some methods may require a lot of time to process. If the application needs to extract a message during the method processing duration, the message handling function is used. Do not use this function.
Sleep time	In the programs for waiting for input from device inside the driver or long programs which keep the CPU time use exclusively, EPSON OPOS ADK can call Sleep (a WIN32 API) to pass control to other programs. The input sleep time is the value (msec) to pass to the Sleep function. Changing the sleep time changes the ratio of CPU time used by a particular device. To change the sleep time for input wait programs, use the port detail settings of each device.
Log file	A file for recording the execution result of the trace.
Master Device	The device for Hydra connection, the host device will be referred to as "Master Device". When the Hydra connection: POSPrinter When the Pass-through connection: LineDisplay
Slave Device	The device for Hydra connection, the destination device will be referred to as "Slave Device". When the Hydra connection: MICR, CheckScanner, ElectronicJournal, LineDisplay, CashDrawer When the Pass-through connection: POSPrinter, MICR, CheckScanner, ElectronicJournal, CashDrawer
TMPORT	Ports created by OPOS and virtual ports are called TMPORT.

Section 6. EPSON TM USB Driver

The methods for installing the USB driver are explained below.

6.1. Installation

The installation can be done only by authorized users of the PC installation. Please request the authorized users for help.

6.1.1. Windows 2000

- (1) The “Add New Hardware” Wizard is displayed when the EPSON UB-U01II/02II is detected.
- (2) Select Search for the best driver for your device (recommended), then click on the [Next] button.
- (3) Select “Specify a location”.
- (4) Specify the location in which the driver is located on the hard disk.
- (5) When the driver is found, the conformation screen is displayed.
- (6) After confirming as EPSON UB-U01II/02II, click on the [Next] button. Driver is installed.
- (7) After completed the installation of the driver, click on the [Finish] button to complete the installation.

6.1.2. Windows XP/ WEPOS/ POSReady/ Embedded Standard

- (1) The “Found New Hardware” Wizard is displayed when the EPSON UB-U01II/02II is detected.
- (2) Click the [Next] button with Install from a list or specific (Advanced) selected.
- (3) Select Search for the best driver in these locations for Search and installation options. Check Include this location in the search, and specify the location, and then click the [Next] button. (The driver files are copied to the OPOS destination folder).
- (4) When the driver is found, the “has not passed Windows Logo testing to verify its compatibility with Windows XP” message appears; click the [Continue] button.
- (5) After completed the installation of the driver, click on the [Finish] button to

complete the installation.

6.1.3. Windows Vista/ Windows 7/ Windows 8/ Windows 8.1/ Windows Server 2008/ Windows Server 2012/ POSReady 7

- (1) The "Found New Hardware" Wizard is displayed when the EPSON UB-U01II/02II is detected.
- (2) Select "Locate and install driver software (recommended) ", then click on the [Next] button.
- (3) Select "Browse my computer for driver software (advanced)" Check "Include subfolders", and specify the location, and then click the [Next] button. (The driver files are copied to the OPOS destination folder).
- (4) After completed the installation of the driver, click on the [Finish] button to complete the installation.

6.1.4. Precautions

- When UB-U01II/02II is being used, it is indicated with EPSON UB-U01II/02II.
- Ensure that USB equipment is installed in accordance with USB standards. Certain parts of these standards are listed below. There are several conditions in effect when creating a tree consisting of hubs that can accept multiple connections of USB mice, keyboards and other devices (known as nodes.)
 - A maximum of 127 nodes and hubs can be connected.
 - Only a single tree for USB hosts and route hubs exists.
 - A maximum of six branch levels are possible with the hubs. As this includes the PC route hub, only five will appear to actually exist.
 - The maximum length of the cable connecting hubs with hubs or nodes must be within five meters (under normal conditions.)
- If the following drivers are installed later when the EPSON TMFLogo Utility is being installed, the version of TMUSB driver is affected by those drivers. In such case, please update the TMUSB driver. For details of update ways, please refer to the "[6.2 Updating an old version of the USB device driver](#)" of this section.
 - - EPSON Advance Printer Driver
 - - EPSON TM-J9000/J9100 Driver
 - - EPSON OPOS ADK for .NET
 - - EPSON JavaPOS ADK

6.2. Updating an old version of the USB device driver

The method for updating the USB driver is explained below. It takes Windows 2000 for instance. The images of the windows are different in each operating system, but the updating procedures are the same as those of Windows 2000.

6.2.1. Windows 2000

- (1) Connect the printer and turn on power to the computer and click the [Device Manager] in "Hardware" tab on the "System Properties". Select "EPSON USB Controller for TM/BA/EU printers" under "Universal Serial Bus controllers". Click the [Properties] button to open the Properties screen.
- (2) Select the "Driver" tab and click the [Update Driver] button.
- (3) "Update Device Driver" wizard will appear. Click the [Next] button.
- (4) Select "Display a list of the known drivers so that I can choose a specific driver" and click the [Next] button.
- (5) Click [Have Disk...] and specify the location in which the new driver is located on the hard disk. The driver files are copied to a "TMUSB" folder at the EPSON OPOS ADK destination folder.
- (6) Click the [Next] button.
- (7) Check that "EPSON USB Controller for TM/BA/EU Printers" is displayed. Click the [Next] button to begin updating the driver.
- (8) After the driver has been updated, click the [Finish] button.

6.3. Naming Devices

6.3.1. Device Names in the Registry

A "device name" is required for OPOS and other applications to access devices such as the TM printer. The device name for the USB to be used by the EPSON OPOS ADK is automatically added by the setup utility (hereafter SetupPOS) and the TM Flash log setup (hereafter TMFlogo) utility based on the information generated by the USB device driver. This device name is retained in the registry.

The following describes how to name USB devices.

When the port numbers are displayed by SetupPOS and TMFlogo, the device names to be used by EPSON OPOS ADK are assigned to each of the USB interface ports (hereafter "port") following reference to the registry information generated by the USB device driver. These device names are assigned in the order of "TMUSB000", "TMUSB001", "TMUSB002", etc., and assigned according to the following method. Once a device name has been assigned to a port, the

same device name will always be used even if the device is disconnected from the PC and then reconnected again (same when the PC is reset). This is owing to the fact that the device names are managed as the ID numbers of ports. Accordingly, the device name assigned to the ID number of a given port is not assigned to a port having another ID number.

An example is shown below.

Operations when a TM printer has never been connected to the PC.

1. TM printer "A" connected (Port ID number "1") => TMUSB000
2. TM printer "B" connected (Port ID number "2") => TMUSB001
3. TM printer "C" connected (Port ID number "3") => TMUSB002
4. TM printer "A" disconnected (Port ID number "1")
5. TM printer "A" connected (Port ID number "1") => TMUSB000
(because TMUSB000 has been assigned to Port ID number "1")
6. TM printer "B" disconnected (Port ID number "2")
7. TM printer "D" connected (Port ID number "2") => TMUSB001
(because TMUSB001 has been assigned to the port ID number "2", regardless of the TM printer)

*As device names are assigned to the port ID numbers, the USB tree structure for connected printers can be located anywhere.

6.4. Port Names on OPOS

The USB port names used with EPSON OPOS ADK will be assigned as follows in accordance with the device names set in the registry.

Registry	Port name (SetupPOS, TMFlogo, etc.)
TMUSB000	=> USB1
TMUSB001	=> USB2
TMUSB002	=> USB3
TMUSB003	=> USB4
...	
TMUSB010	=> USB11
TMUSB100	=> USB101

Section 7. Warnings

7.1. Precautions for Connecting with USB I/F

- When connecting a new device, turn on the device before running the “Add New Device” wizard. The disconnected or power-off devices are not recognized.
- If the port number of the device is not indicated on the Port Connection View screen during the connection, select the corresponding device on the Main View, or select [Auto Setting] button on the “Communication Settings” dialog box. The connected devices are recognized.
- When the same type of devices are registered already and some of the devices are turned off, there is a possibility that the turned off device's USB port number is replaced with the new one. Therefore, beware of registering multiple numbers of the same devices.
- Please note the following points when re-assigning USB ports:
 - Turn on the applicable device when doing automatic USB port setting.
 - If using several devices of the same model on different USB ports, only do automatic setting for one device.
 - Editing the registry during automatic USB port setting is not supported.

7.2. Other precautions

- If an installation is not completed successfully and reinstallation also fails, please do the following steps:
 - (1) Uninstall OPOS.
 - (2) Install OPOS version which is originally installed prior to installing this version.
 - (3) Install this version.
- When using Windows Vista or newer, Version 2.60 or older cannot overwrite Version 2.60 or newer.
- When installing this version over an existing install of version 2.67 or older and the following devices are registered, the devices must be re-registered after installation is complete. Re-register the devices according to the specifications in "Section 5 SetupPOS Utility" of the "EPSON OPOS ADK MANUAL User's Manual (Installer/ SetupPOS/ TMUSB)" manual.

TM-H6000II(M), TM-H6000III(M), TM-T88III(M), TM-T88IV(M), TM-T70(M),
 TM-T90(M), TM-L90(M), TM-U220A(M), TM-U220AM_BM, TM-U220B(M),

TM-U220BM_BM, TM-U220D(M), TM-U220DM_BM